University of Dhaka
Department of Robotics & Mechatronics Engineering

Development of an Origami-Inspired Inchworm Robot with Minimal Actuation for Proprioceptive Pipeline Corrosion Detection and Localization

RME 4200: Project

Proposed By

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What Happens When We Can't See Inside a Pipeline?

Corrosion, cracks, and structural fatigue develop inside pipelines, where routine inspection is often impossible.

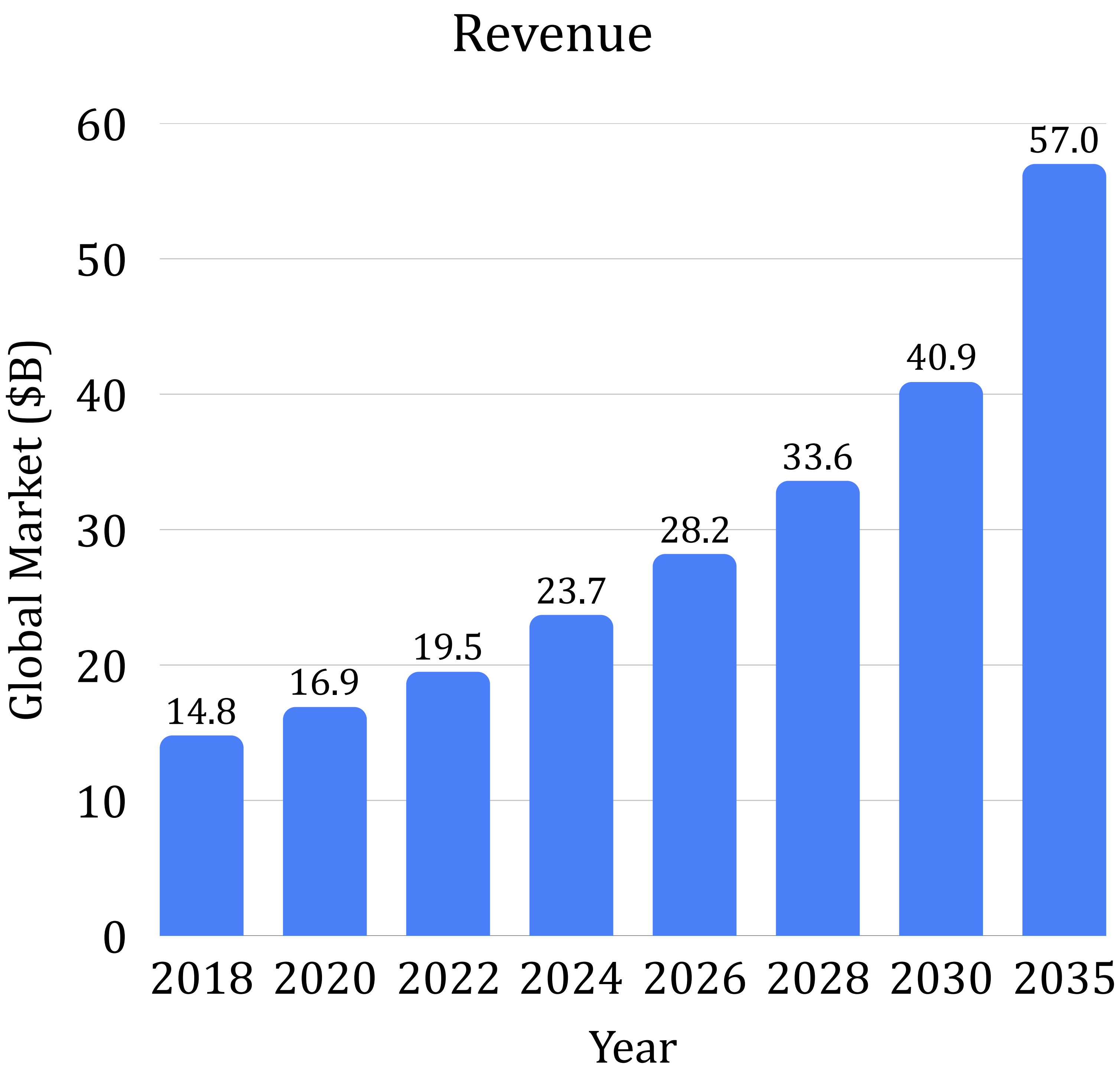
Existing inspection robots face difficulty

- Zero visibility

Dark, sealed environments where cameras become unreliable.

- Harsh operating conditions

High vibration, moisture, corrosion, and temperature variation damage conventional sensing systems and pipe sections with small diameter remain largely inaccessible.



Statistics taken from [1]

Potential Deployment Sectors in Bangladesh

Bangladesh Chemical Industries Corporation (BCIC)

Statistics taken from [2]

What are the existing solutions?

Images taken from [3]

Pipeline Inspection Gauges (PIGs)

Uses magnetic flux leakage to detect corrosion in steel pipes of large diameter

- passive and so rely entirely on line flow to propel them
- large, rigid body geometry means they wedge or jam at bends
- no housing small enough (<6") to carry the magnetic sensing hardware at all

Images taken from [4]

Wheeled Robotic Platforms

Offers real time video feedback with precise operator control

- Inspection quality is entirely dependent on camera
- Defect assessment is not automated

Images taken from [5]

UAVs

Drones excel in spaces with large volume without physical contact with the structure

- Needs open flight volume to operate
- Can't confirm wall condition without a camera
- Offers no clearance inside an enclosed interior.

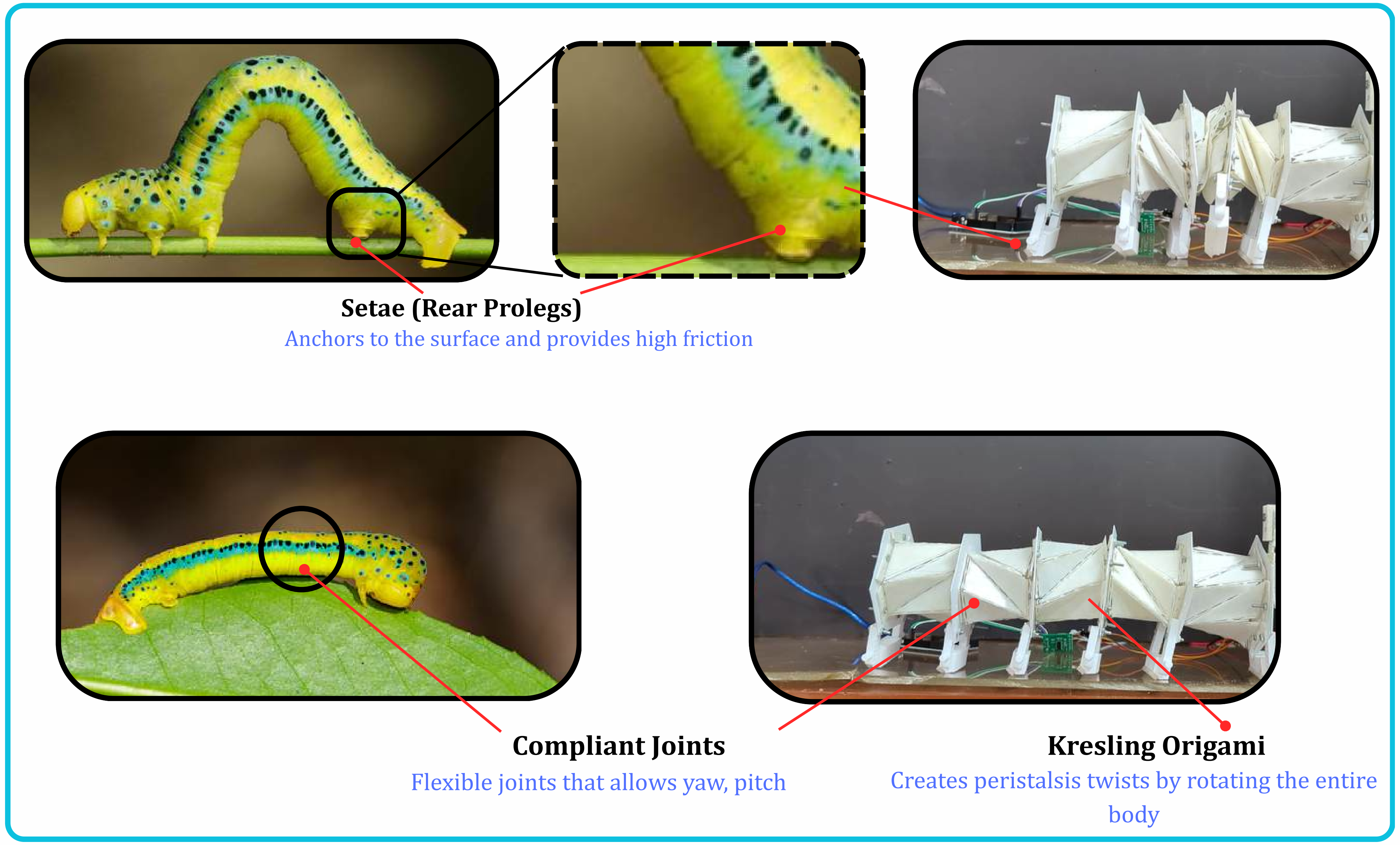
None of them solves automated sensing of wall condition in a small bore, sealed, corroded pipe

Is there any solution to this problem?

- Natural solution: inchworm.
- This bio organism moves through constrained spaces with ease, using a simple looping gait

How can we implement this locomotion of into a mechanism?

- Relying on "mechanical intelligence" instead of electronics"
- ***Origami*** allows a structure with a ***large functional volume*** to be compressed into a ***tiny space*** through simple folding.
- ***Doesn't require actuator*** for every degree of freedom



Images taken from [6]

Contributions

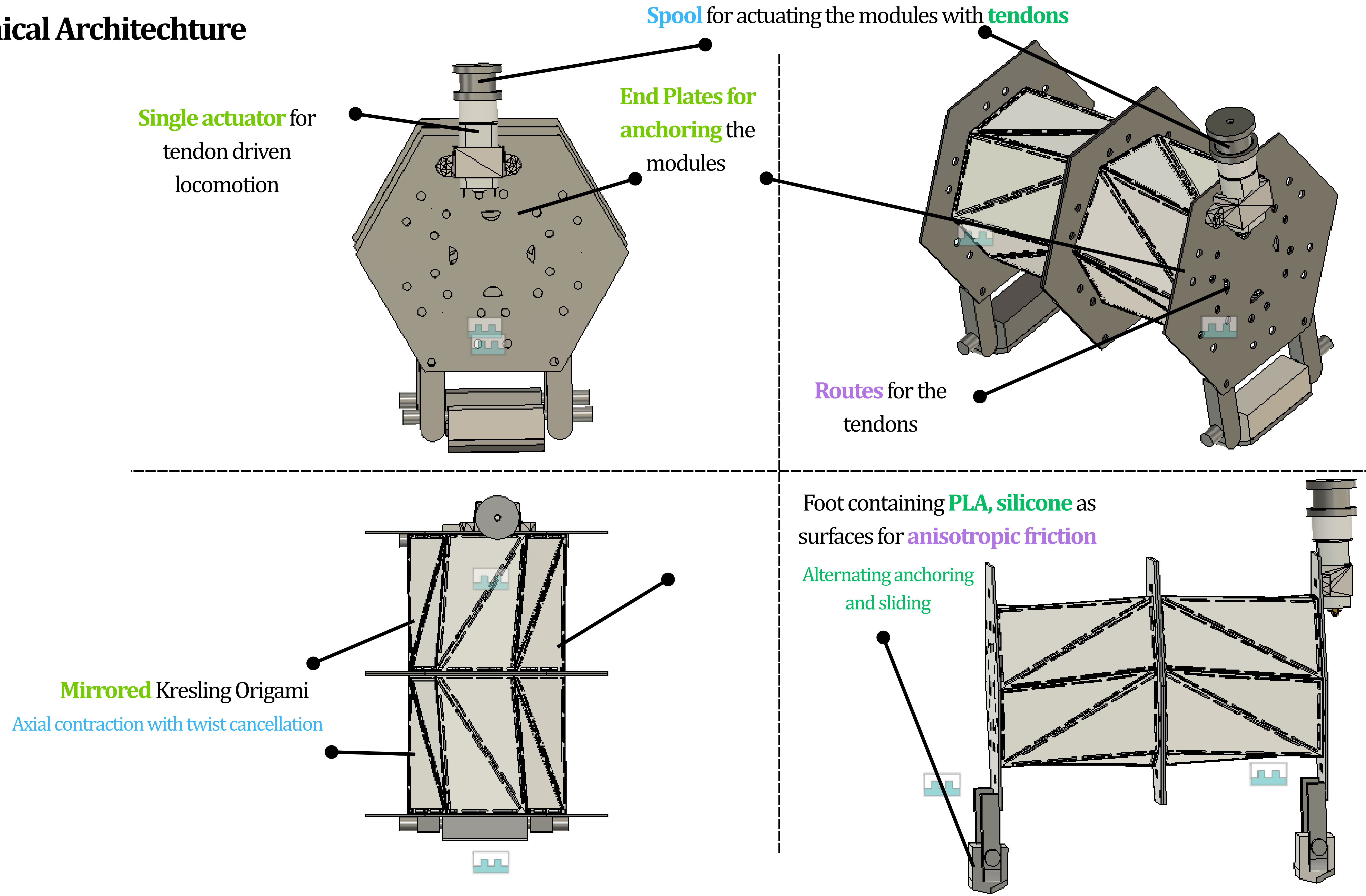
- Bio -Inspired pipeline inspection robot combining compliant structures of Kresling origami , with tendon actuation

- Corrosion detection and localization using IMU , Rotary Encoder and Force Sensor , without visual perception

- Training a lighweight algorithm for onboard detection of defects

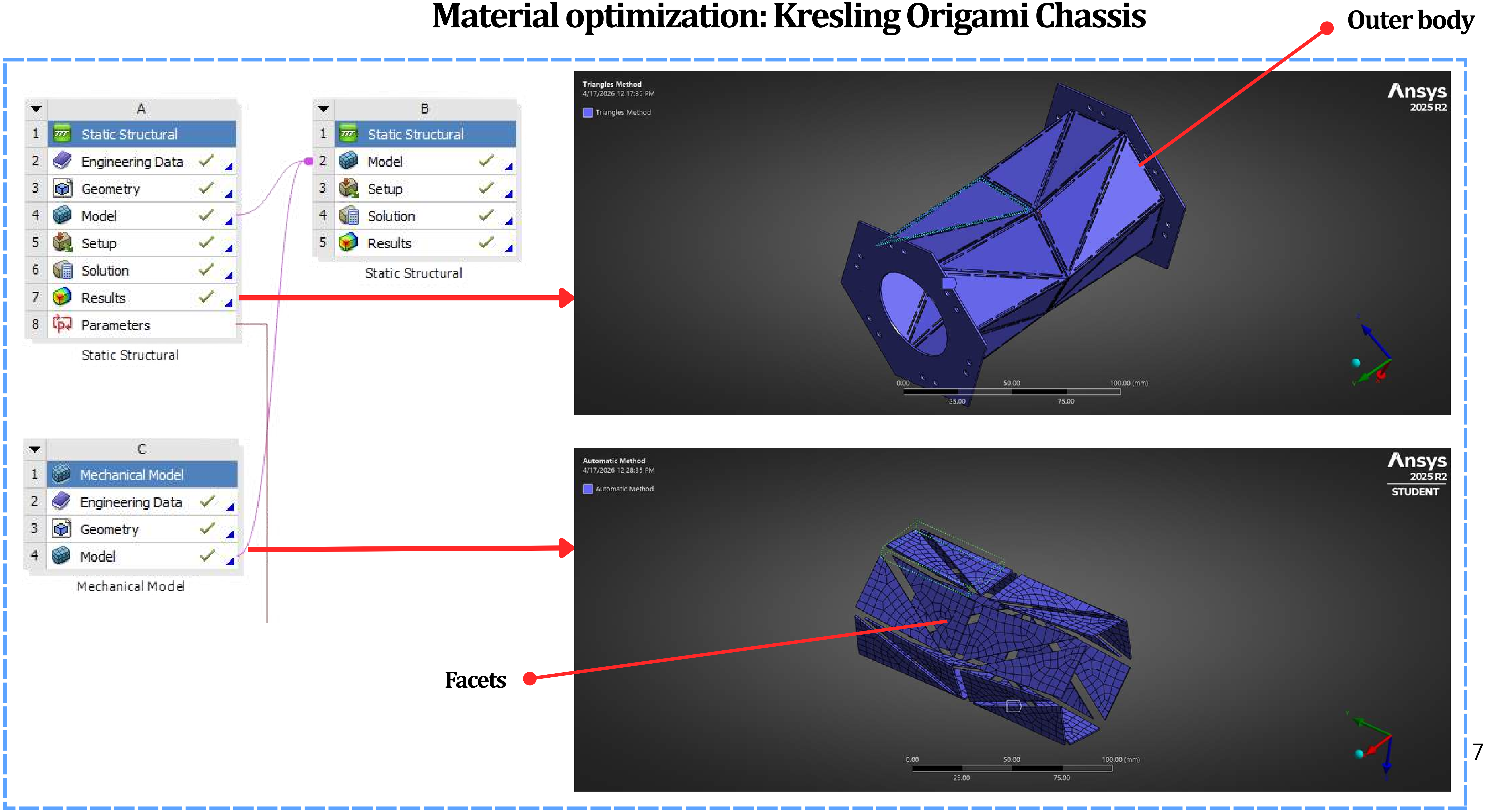
- Transmission of data containing information about the estimated defect severity and location

Mechanical Architechture

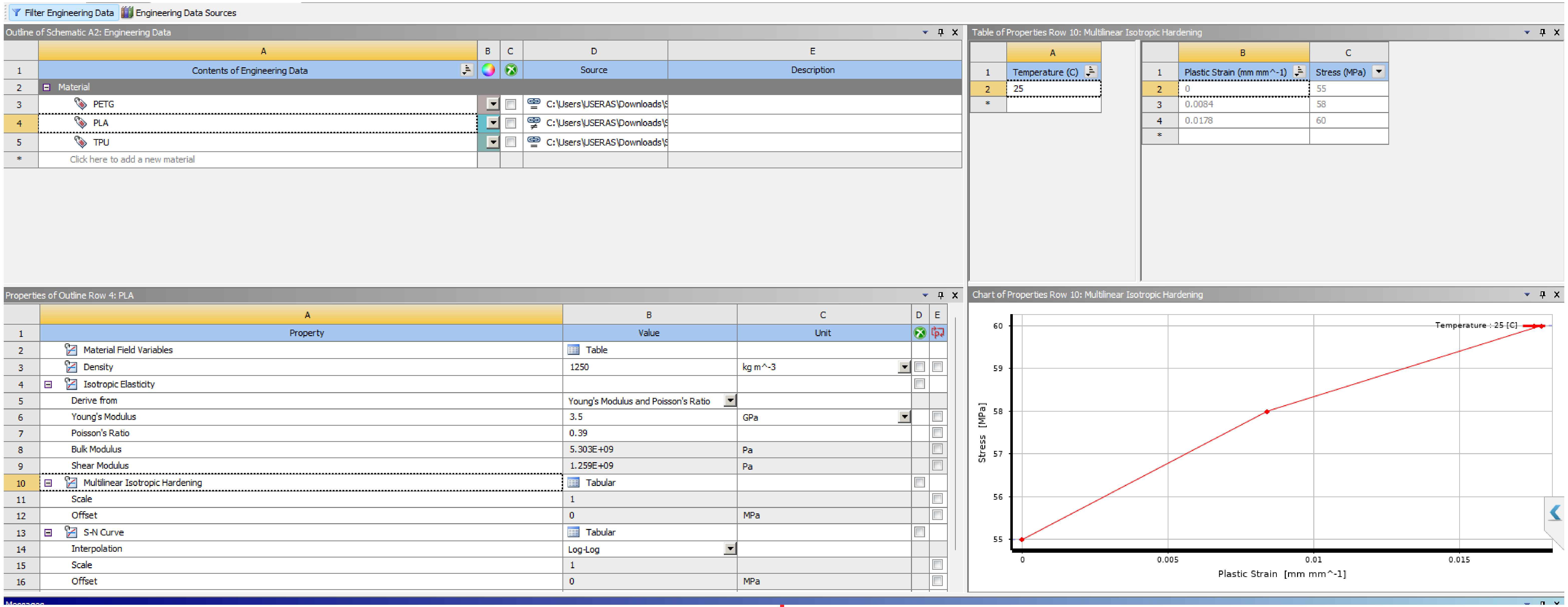


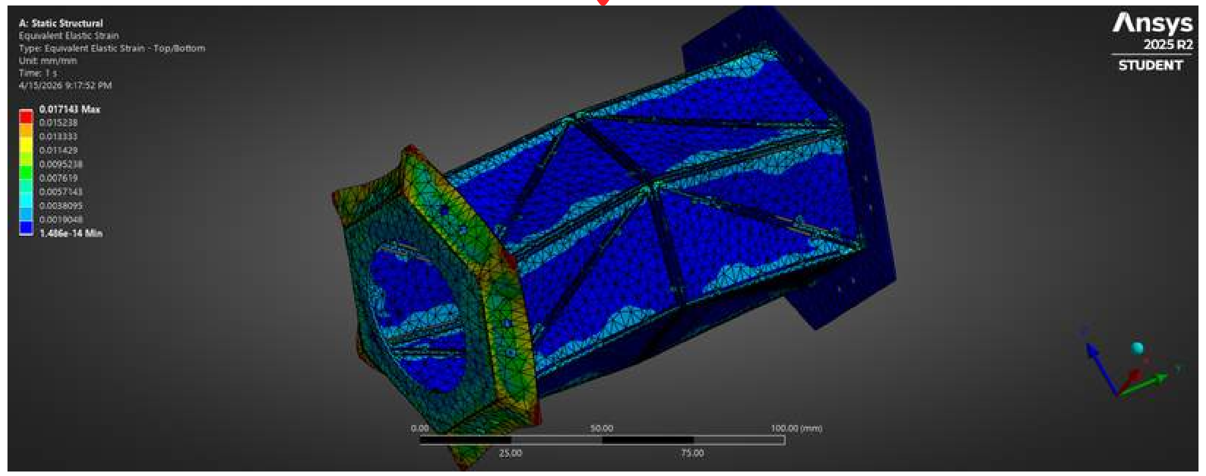
How was this architecture developed and optimized?

Material optimization: Kresling Origami Chassis

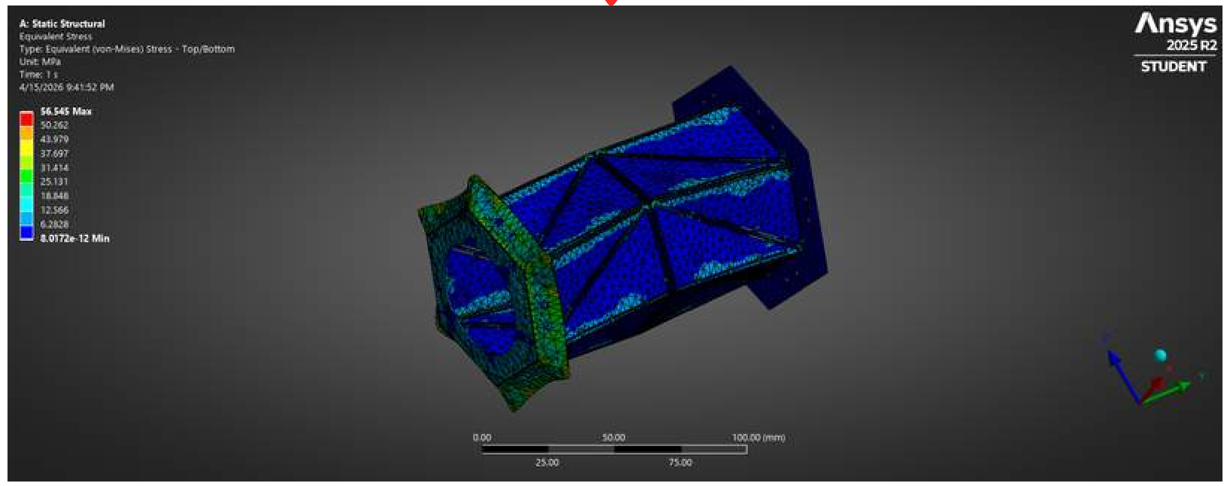


Which material will be the most suitable for the chassis?

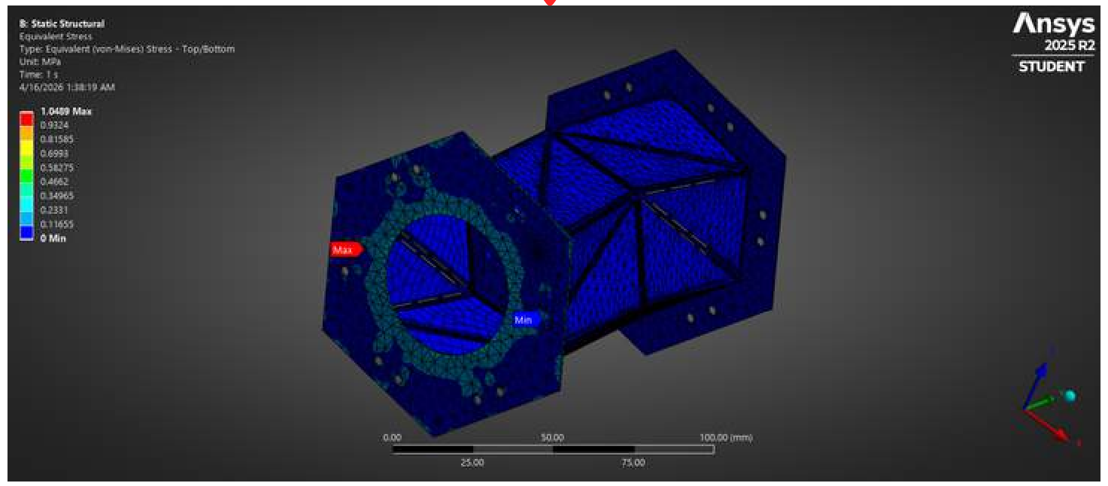




PLA-PETG

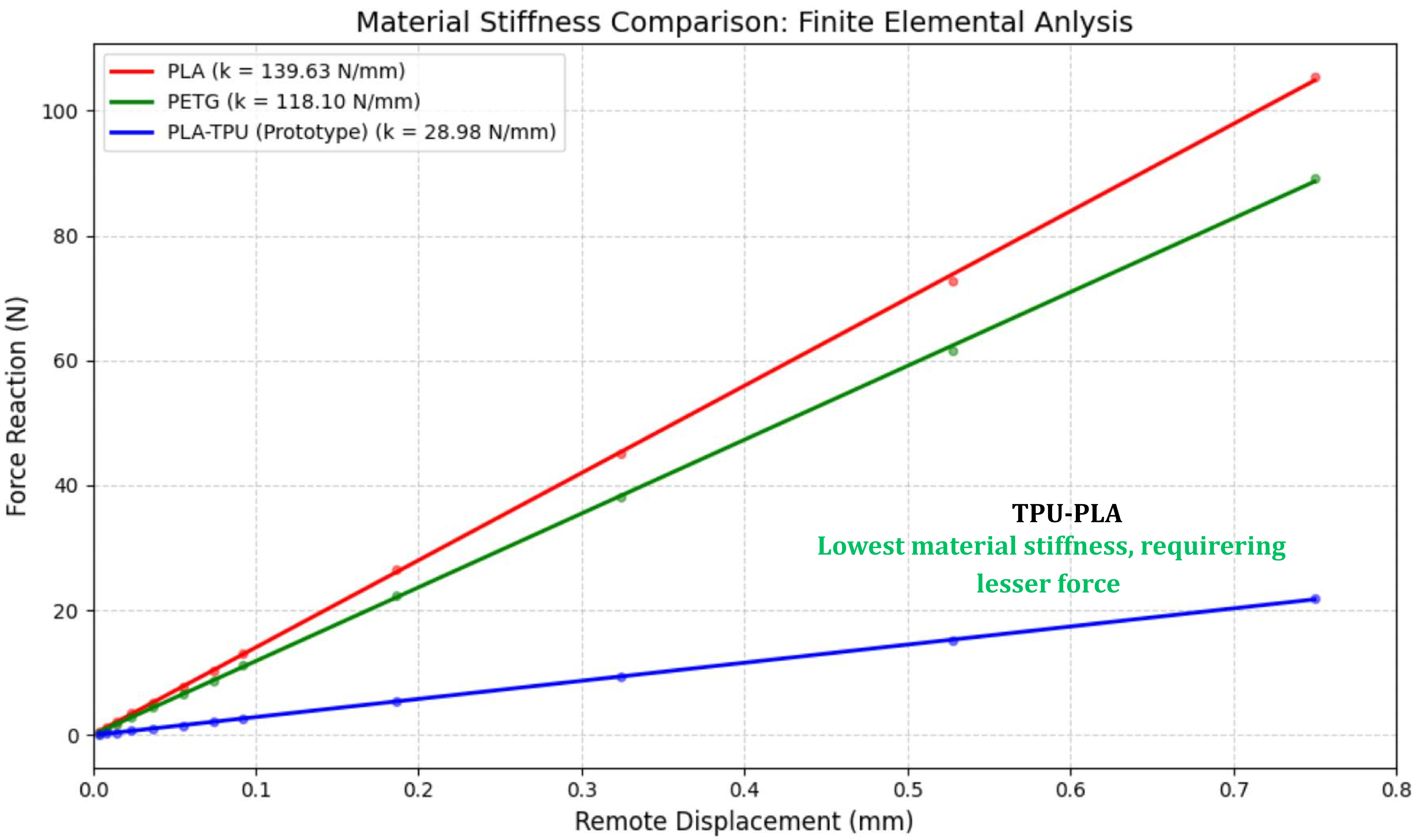


TPU-PETG



TPU-PLA

Trade-off between flexibility and structural integrity

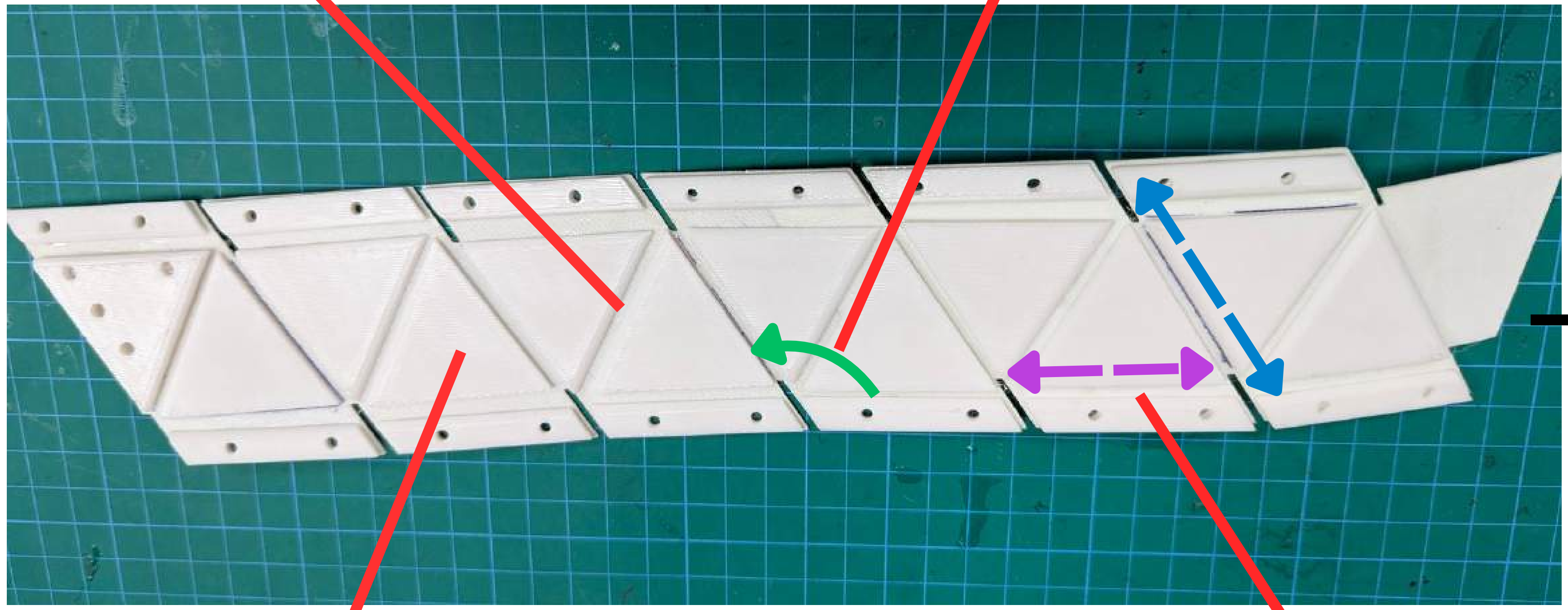


Dimension optimization: Kresling Origami Chassis

Iteration 1: ***PLA-TPU Sheets***

2mm Tolerance for living hinges

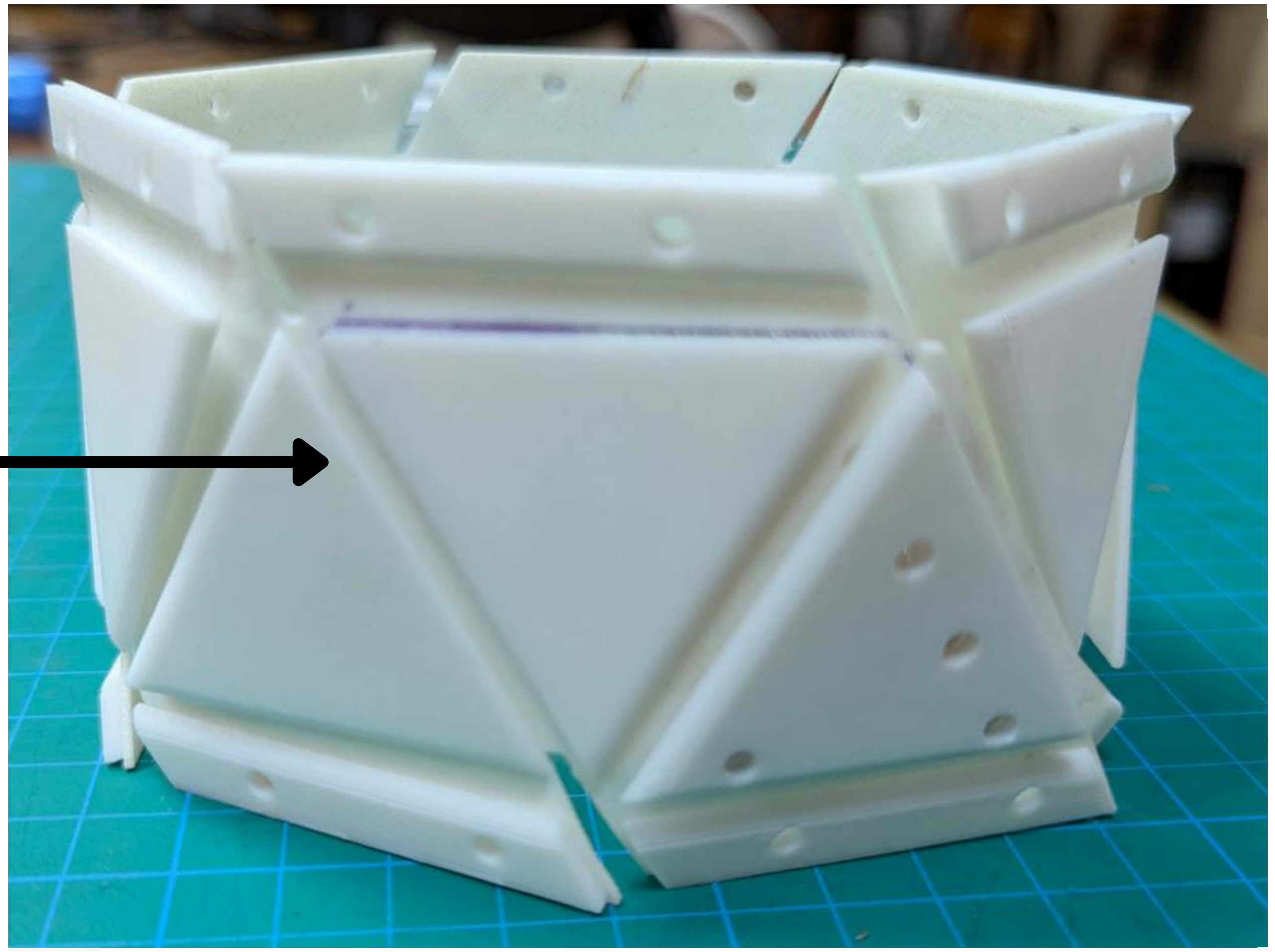
Fold Angle(60^0)



TPU Sheet (4mm)

PLA Facets (2mm)

Flattened PLA-TPU Sheet



Folded PLA-TPU Unit Module

Dimension optimization: Kresling Origami Chassis

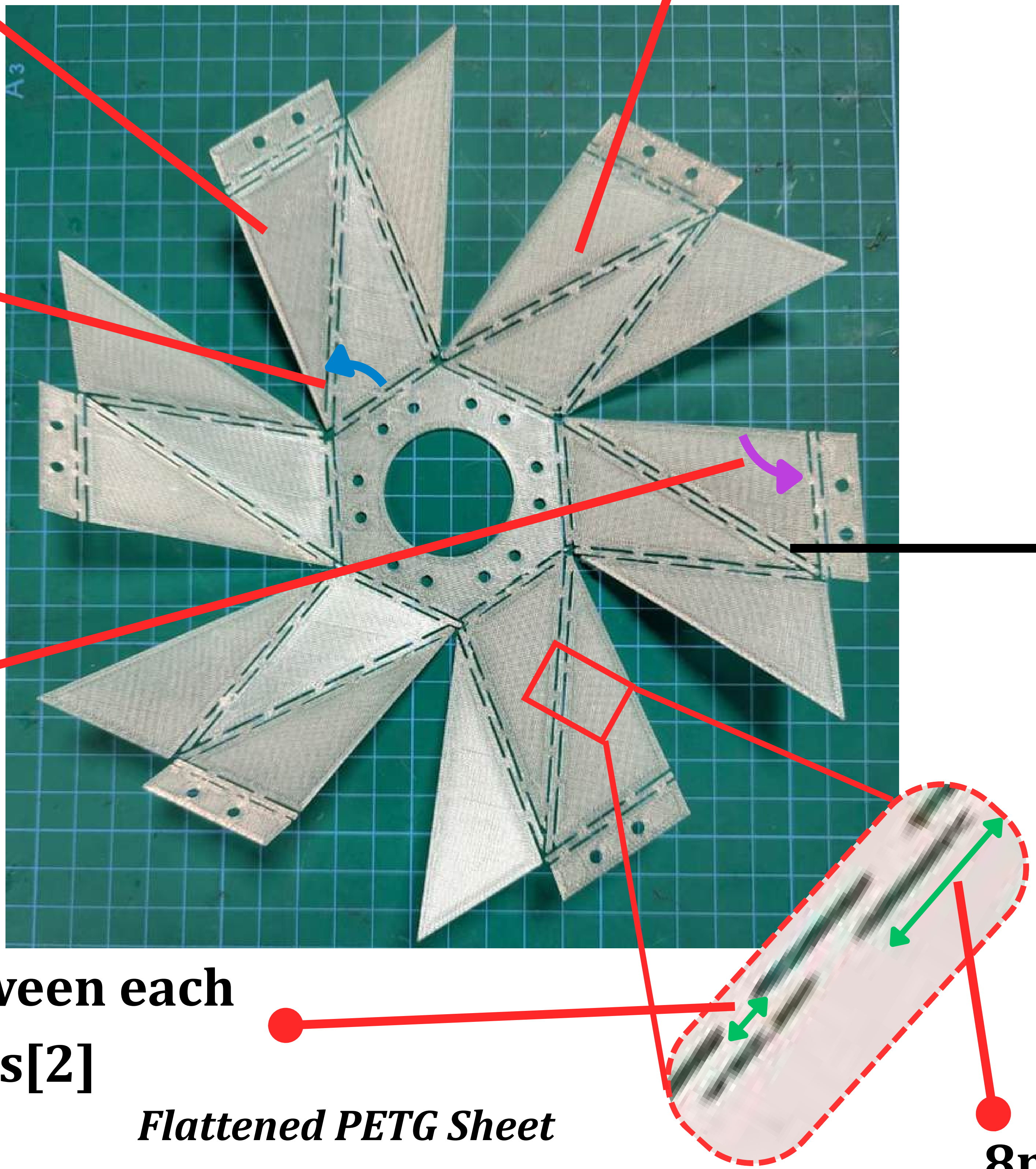
Iteration 2: *PETG Sheets*

Height (64.4mm)

.75mm petg

Fold Angle(45^0)

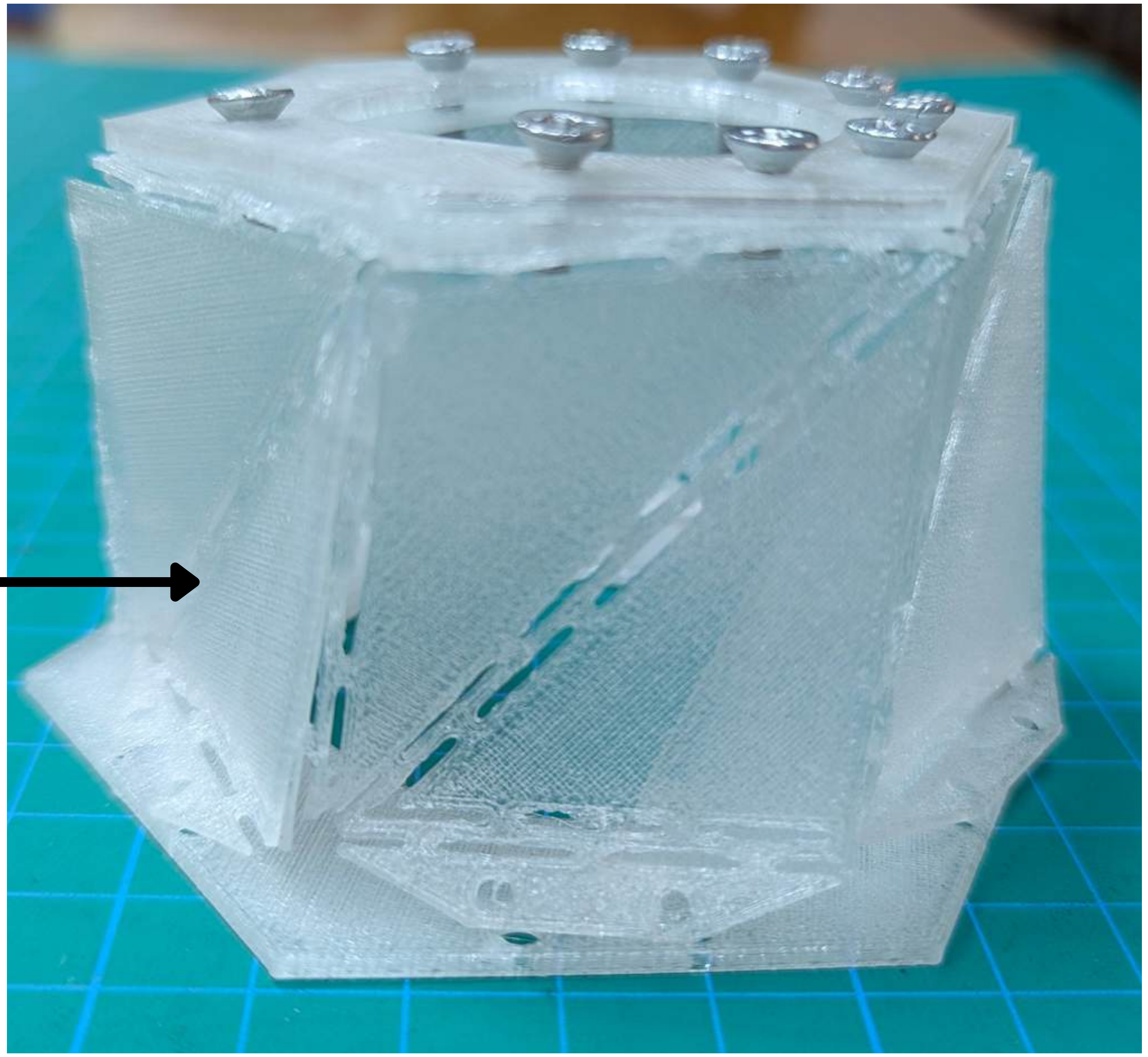
Valley Angle(22.5^0)



1.6mm gap between each pair of slits[2]

Flattened PETG Sheet

8mm Elliptical Slits to reduce tension



Folded PETG Unit Module

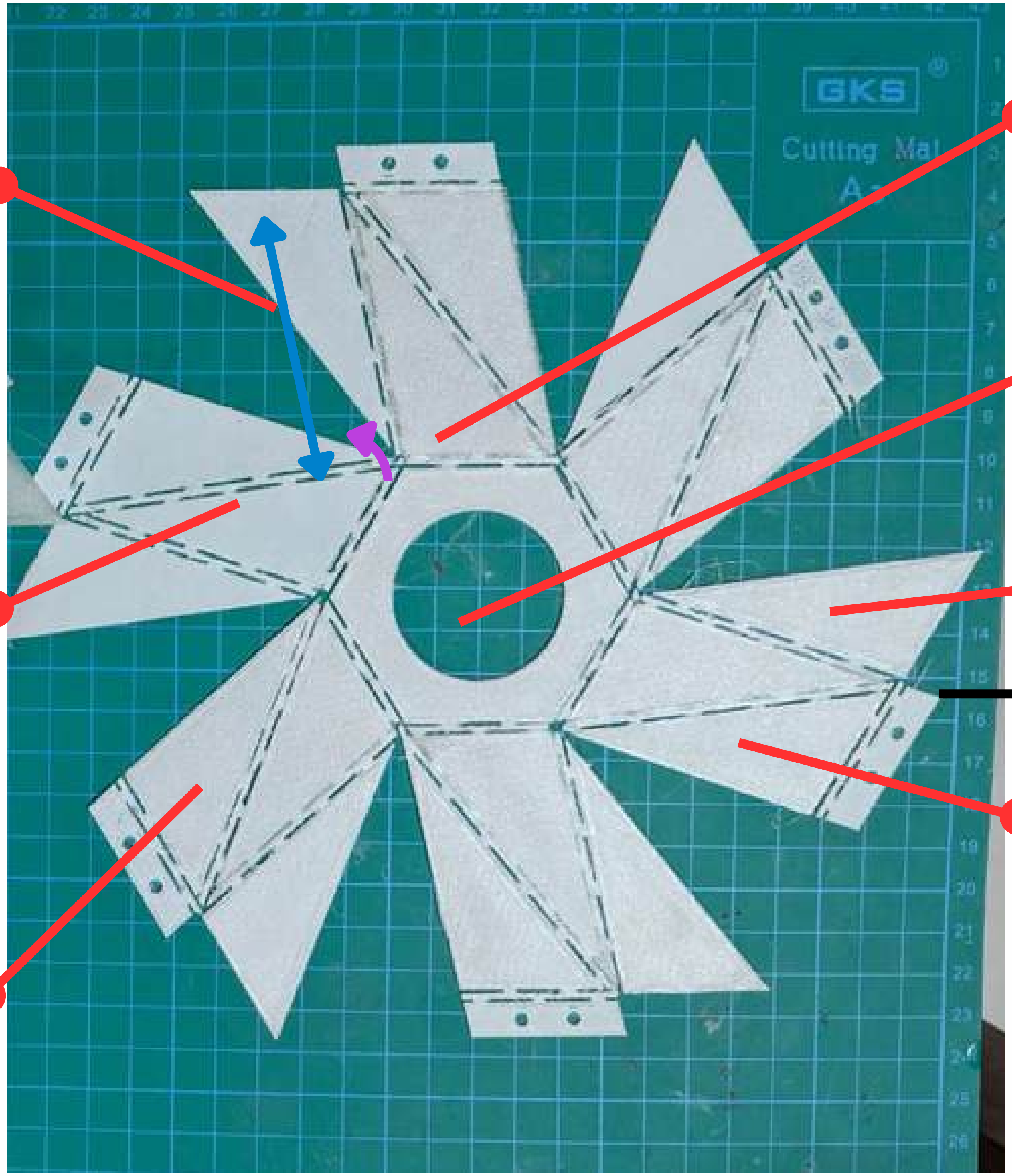
Dimension optimization: Kresling Origami Chassis

Iteration 3: *TPU-PETG Sheets*

64.44mm

TPU Sheet (0.41mm)

12mm Elliptical Slits to reduce tension

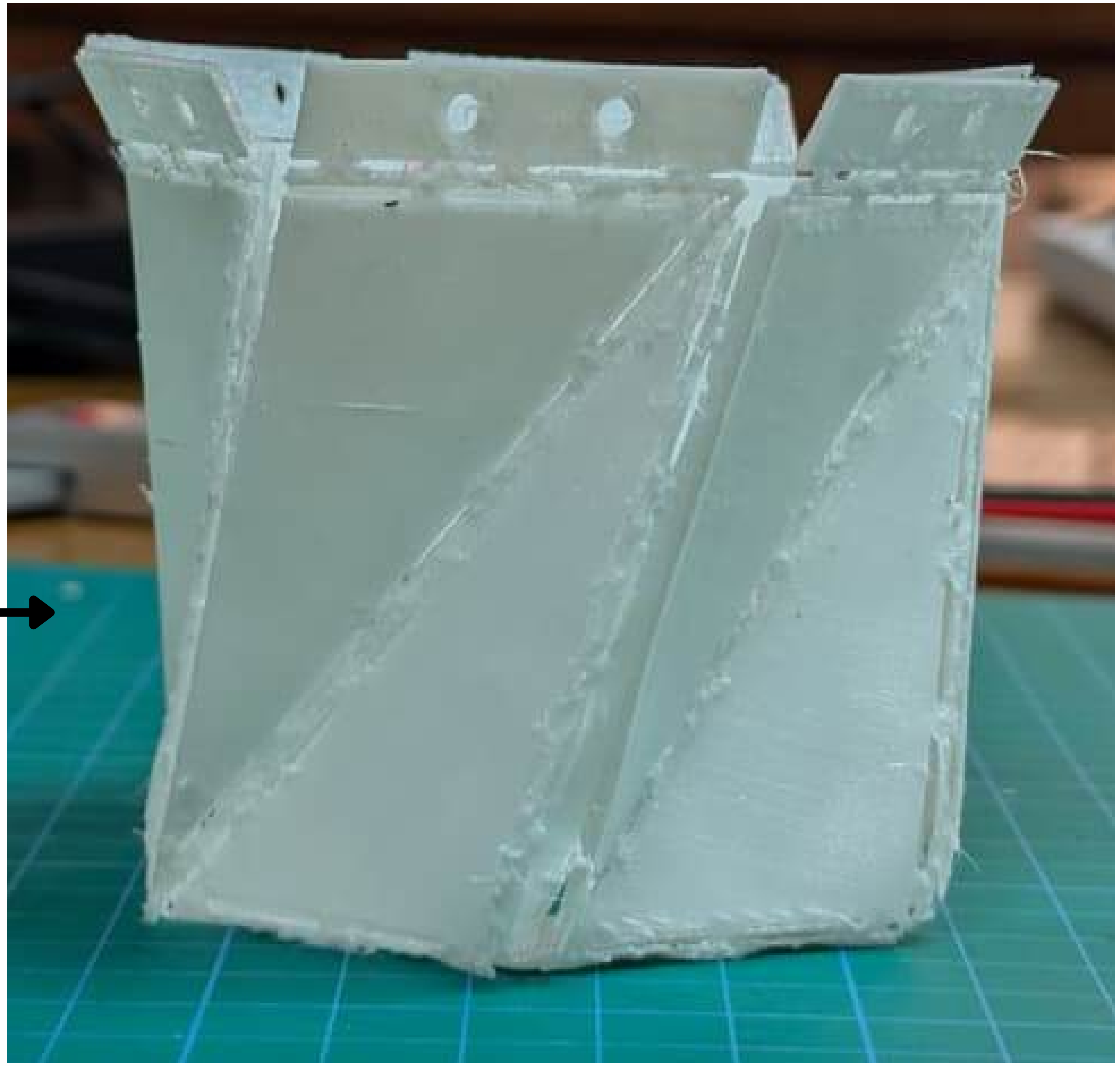


74.84^0

20mm Diameter Circle

PETG Facets (0.75mm)

2mm gap between each pair of slits[3]

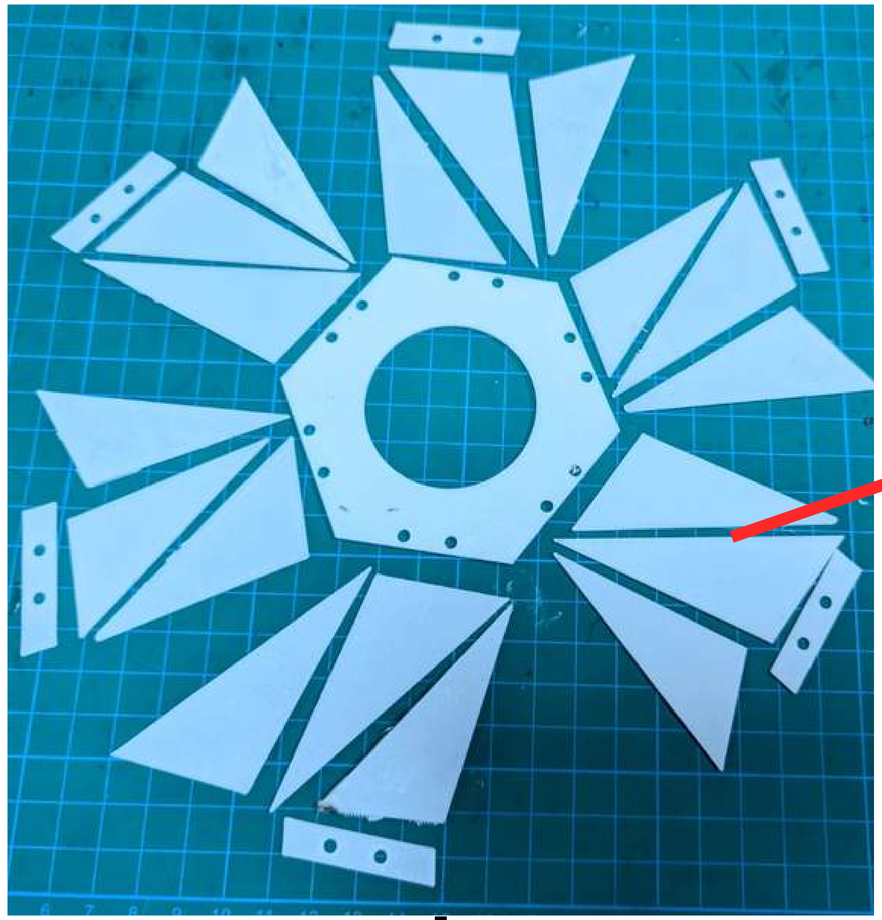


Flattened TPU-PETG Sheet

Folded TPU-PETG Unit Module

Dimension optimization: Kresling Origami Chassis

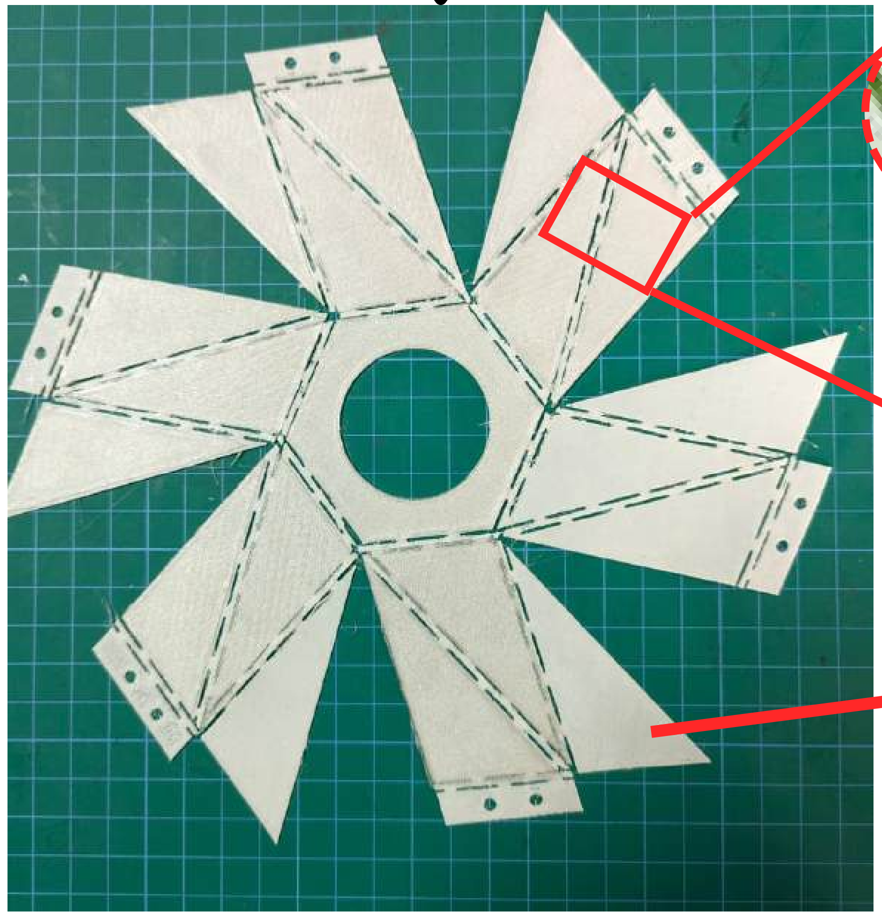
Iteration 4: *TPU-PLA Sheets*

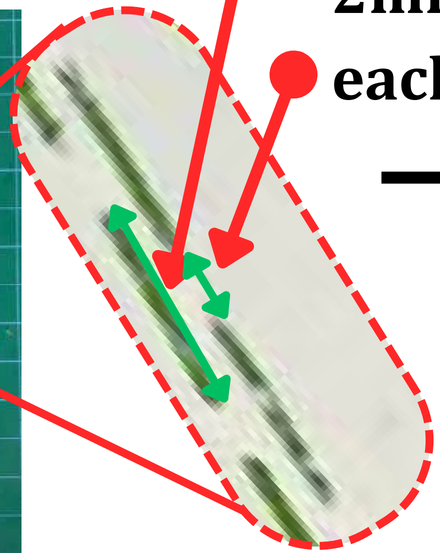


PLA Facets (width 1mm)

12 mm Elliptical Slits to reduce tension

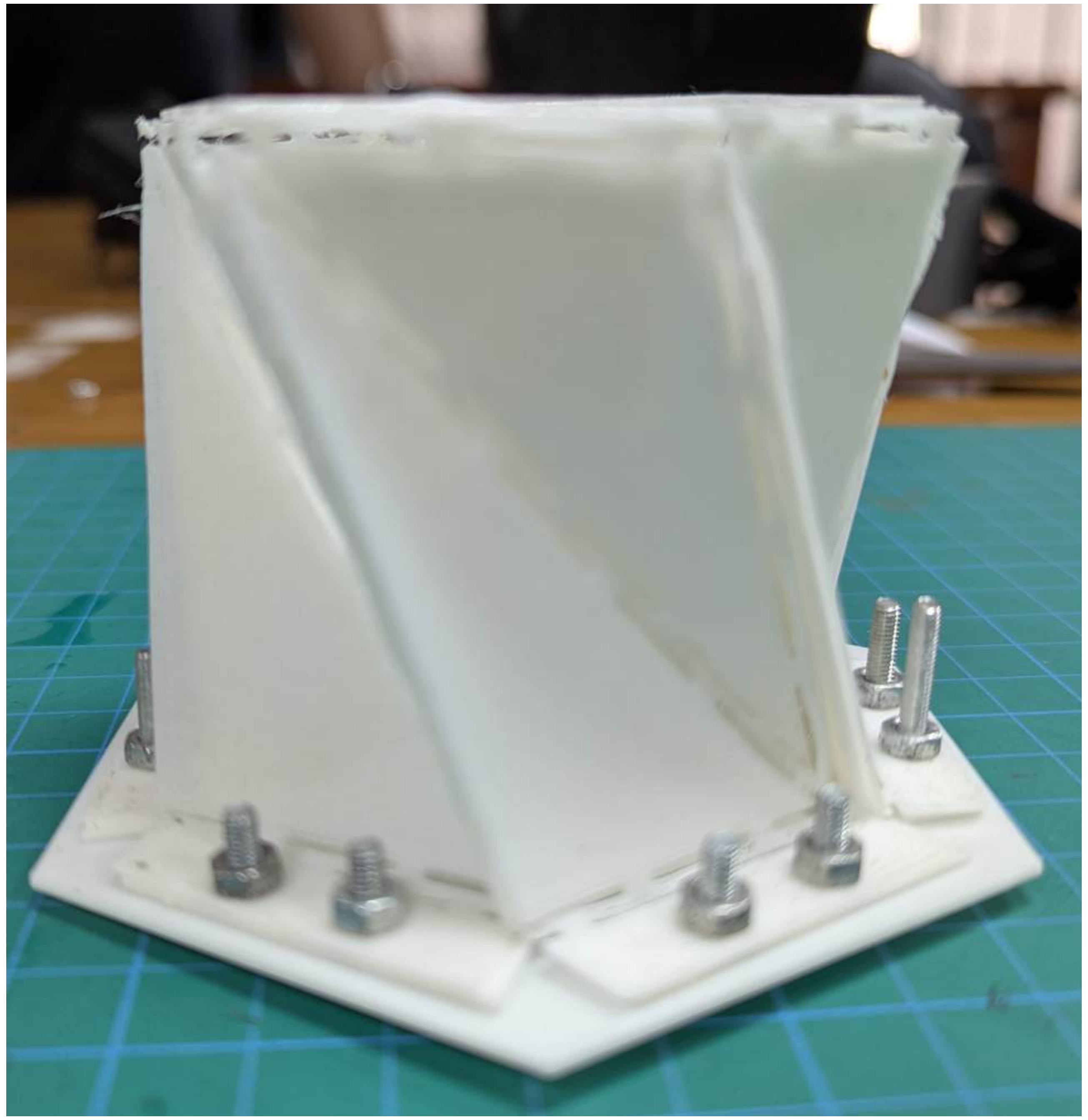
2mm gap between each pair of slits





TPU Sheet (width .41mm)

Flattened TPU-PLA Sheet



Folded TPU-PLA Unit Module

How is the module actuated?

Kinematic Modeling via Denavit-Hartenberg (D-H) Convention

$$\mathbf{T}_0^n = \prod_{i=1}^{n} \mathbf{H}_i^{i-1}, \qquad \mathbf{H}_i^{i-1} = \begin{bmatrix} \mathbf{R}_i & \mathbf{p}_i \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

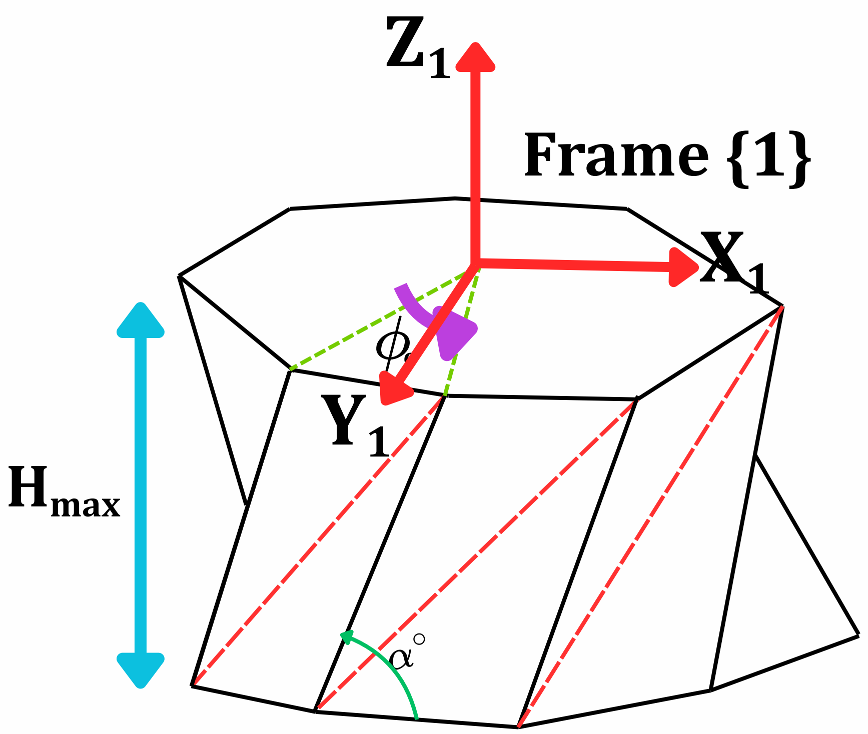
$$\mathbf{H}_{\text{Kresling}} = \begin{bmatrix} \cos\varphi & -\sin\varphi & 0 & 0 \\ \sin\varphi & \cos\varphi & 0 & 0 \\ 0 & 0 & 1 & H \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Two states

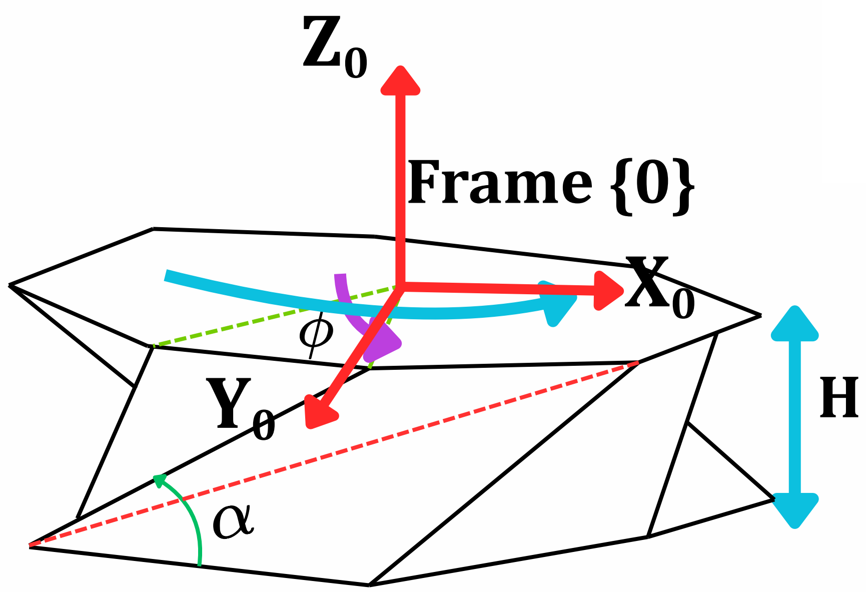
- **State 1** — Extended (Stiff Link).
- **State 0** — Contracted (Screw Joint).

$$\mathbf{H}_{(1)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & H_{\max} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad H_{\max} = 63.1\text{ mm}$$

$$\mathbf{H}_{(0)} = \begin{bmatrix} \cos\varphi & -\sin\varphi & 0 & 0 \\ \sin\varphi & \cos\varphi & 0 & 0 \\ 0 & 0 & 1 & H(t) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



State 1: Rigid Pure Translation



State 0: Screw Transformation

How is the module assembled?

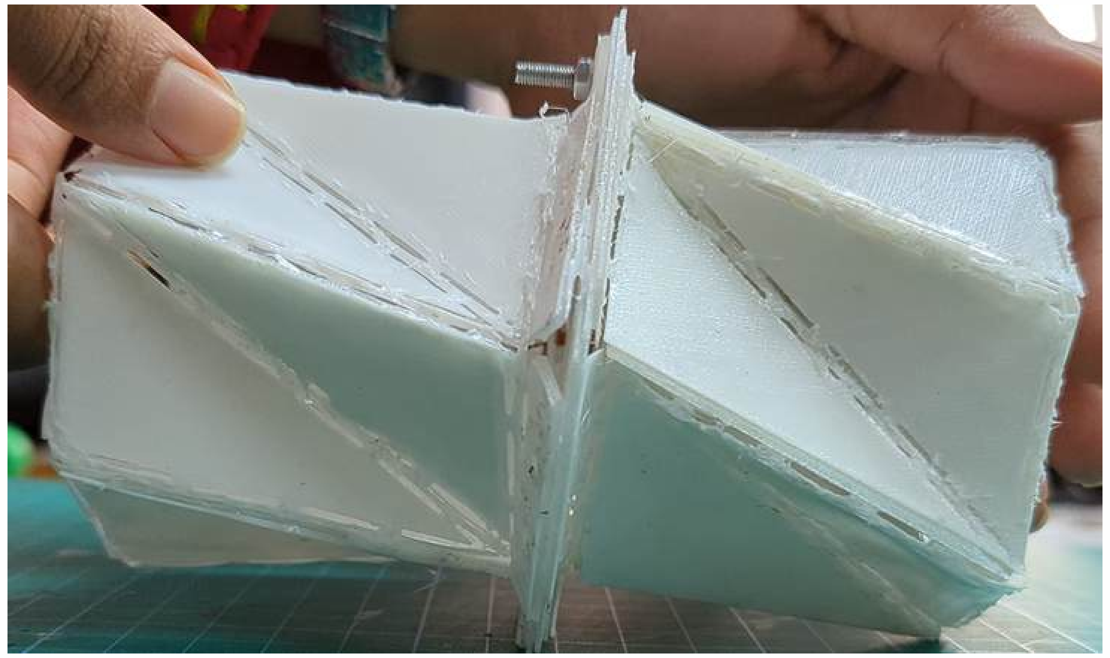
Twist Cancellation: The Kresling Dipole

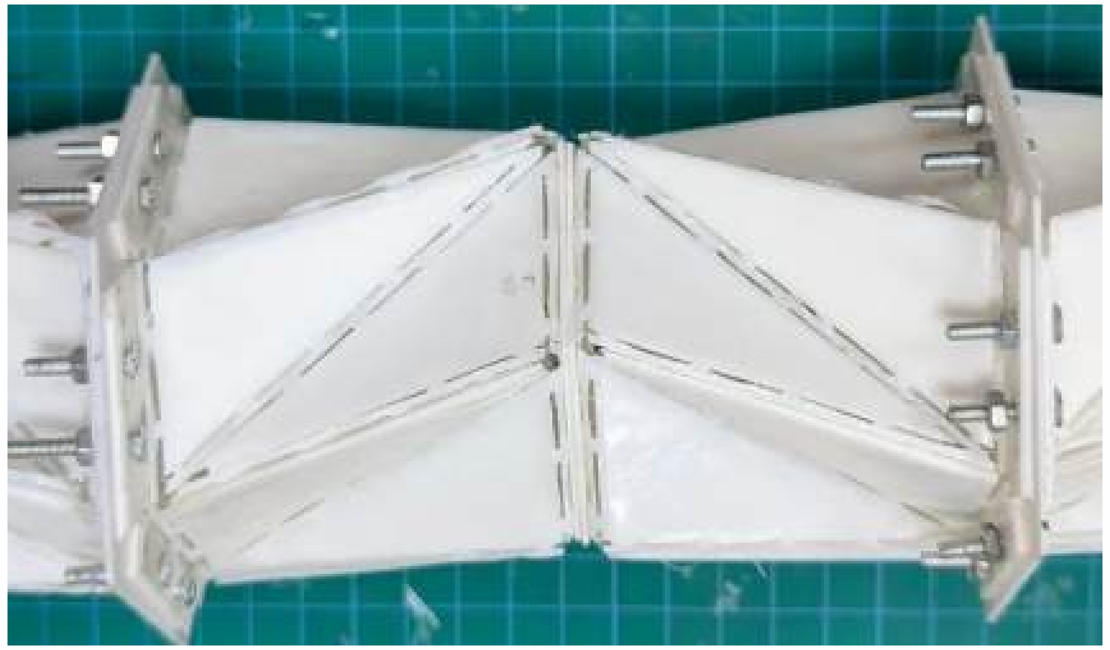
$$\mathbf{T}_{\text{dipole}} = \mathbf{H}_{(0)}^{\text{CW}}(+\varphi) \cdot \mathbf{H}_{(0)}^{\text{CCW}}(-\varphi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2H(t) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}_{\text{total}} = \mathbf{T}_{\text{dip1}} \cdot \mathbf{T}_{\text{dip2}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4H(t) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link No.	Unit	d_i (mm)	θ_i
1	U1 (CW)	0	$+\varphi(t),\ [0^\circ \to 50^\circ]$
2	U1 (CW)	$H(t)$	0
3	U2 (CCW)	0	$-\varphi(t),\ [0^\circ \to -50^\circ]$
4	U2 (CCW)	$H(t)$	0

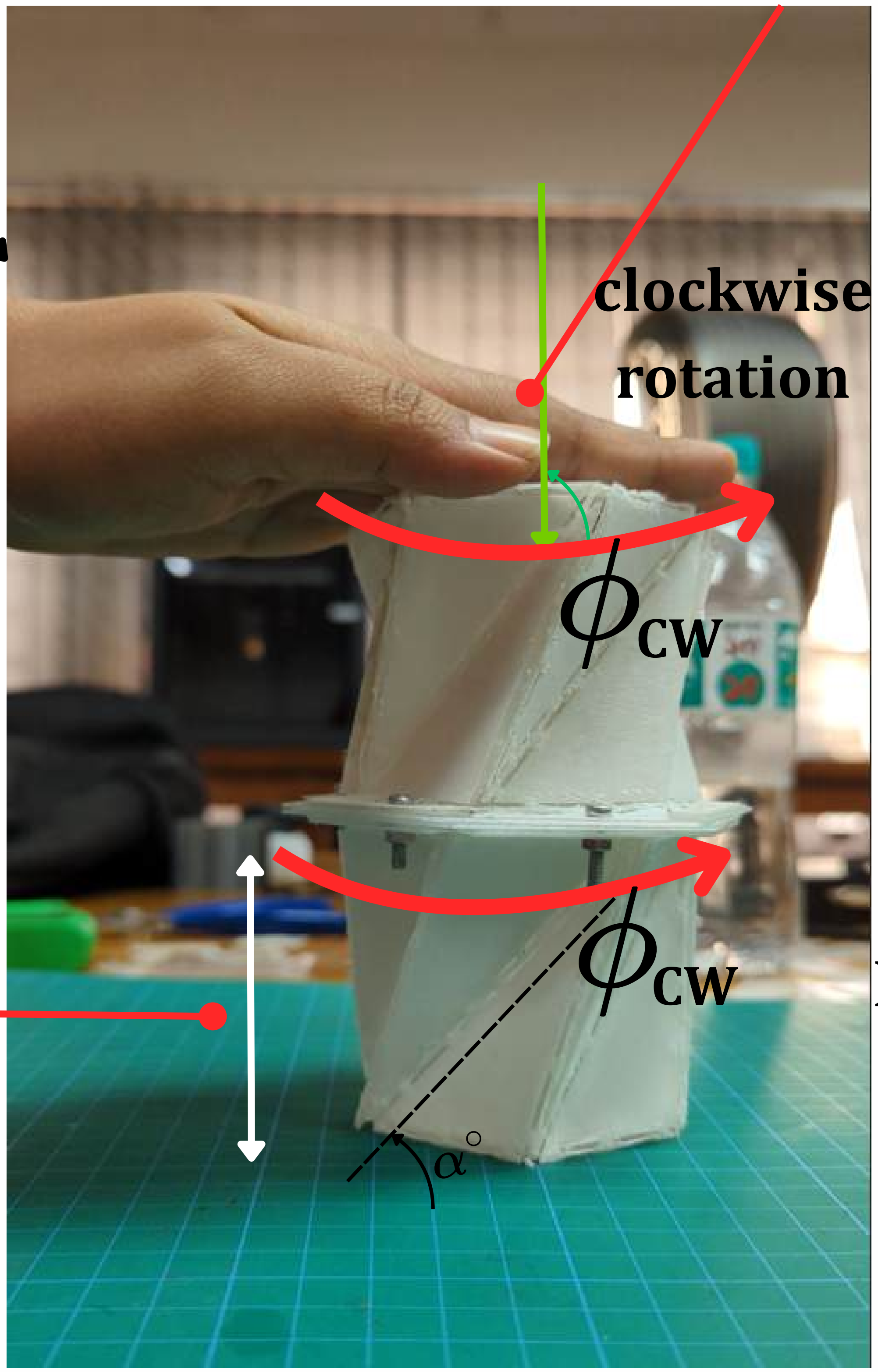
D-H Parameter Table [Odd links: Revolute; Even links: Prismatic. ai = αi = 0 for all links[5]



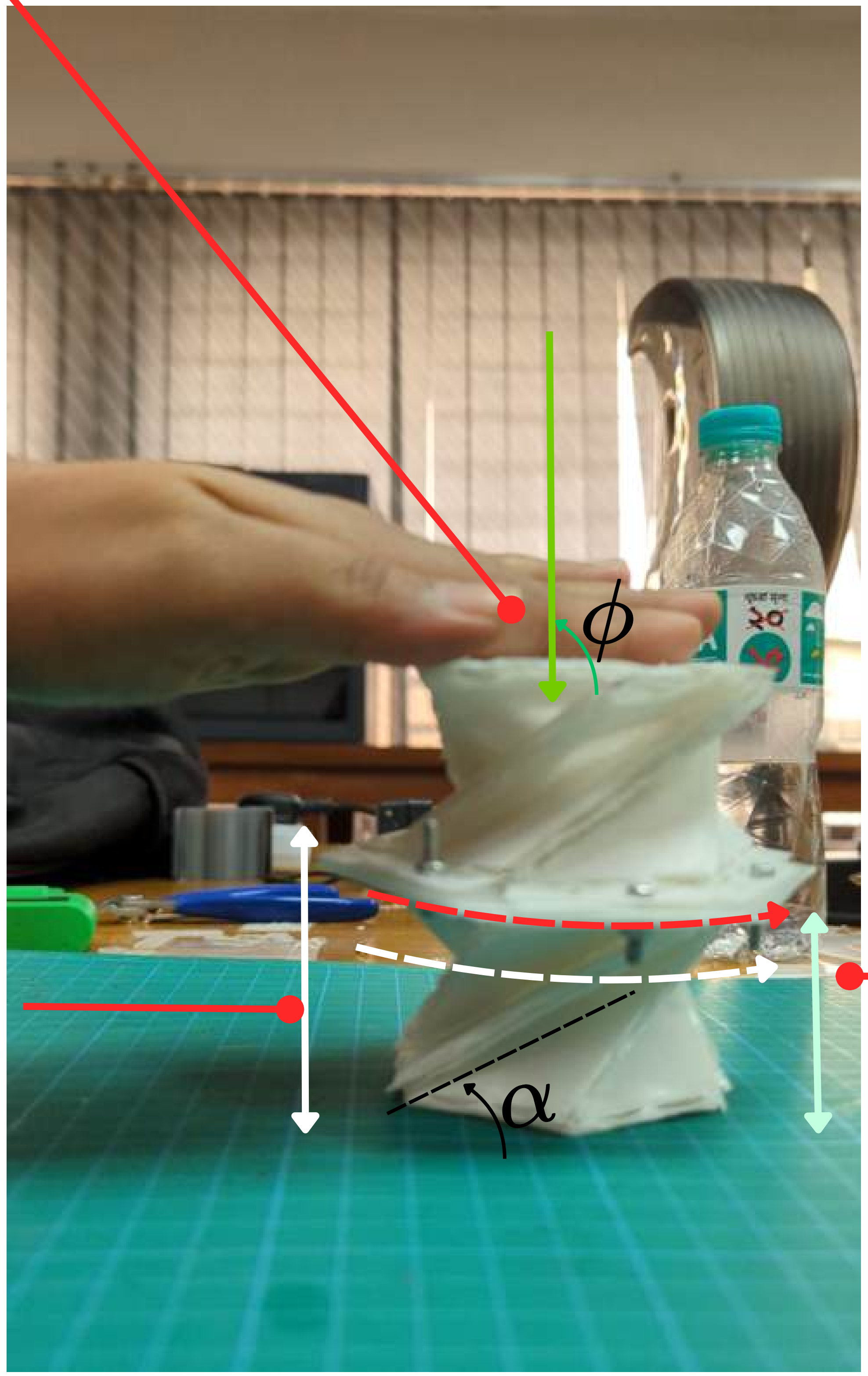


Asymmetric Twist(coupled motion): Coupled clockwise rotation and compression

Instantaneous twisting angle

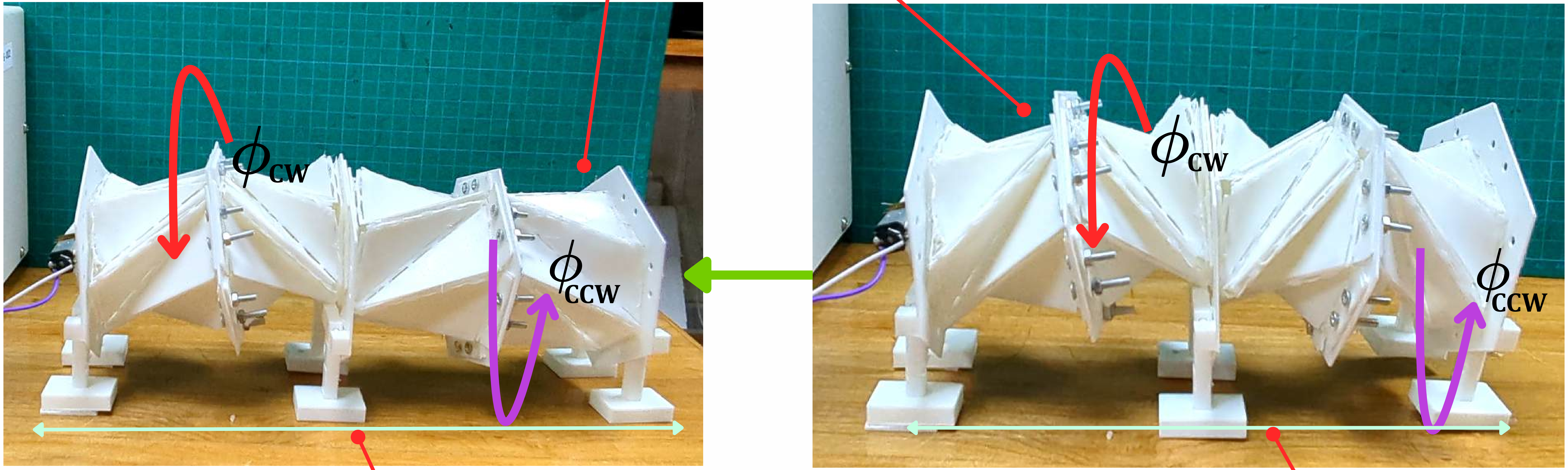


Relaxed State



Contracted State

Symmetric Twist (Twist Cancellation)

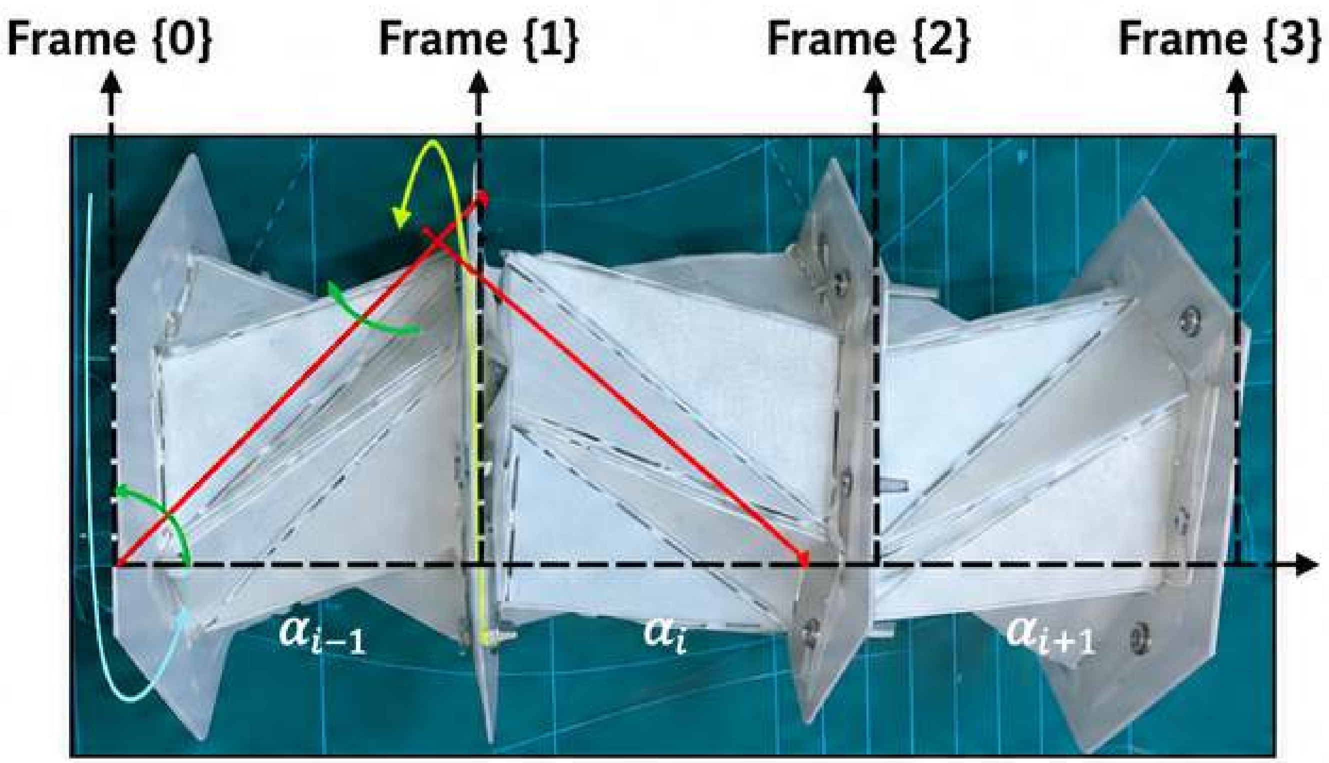


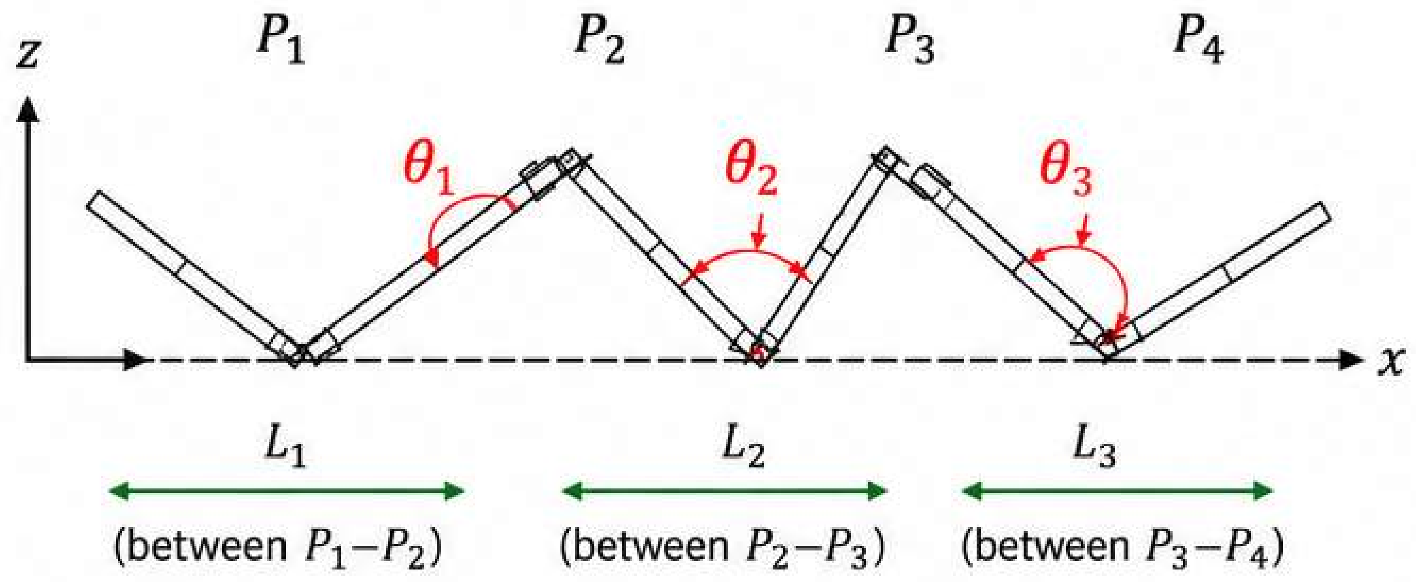
H_0 = 300mm

H = 221mm

Pure Vertical Translation, Zero Net Twist $\Delta\phi = \phi_{cw} - \phi_{ccw} = 0^0$

How is the assembled chassis actuated?





Valley angles: $\theta_1, \theta_2, \theta_3$

Plate rotations (about local z at hinges): ϕ_1, ϕ_2, ϕ_3

D–H PARAMETERS (initial)

Joint i	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i (rad)
1	0	L_1	0	θ_1
2	0	L_2	0	θ_2
3	0	L_3	0	θ_3

Assuming the length to be the only variable

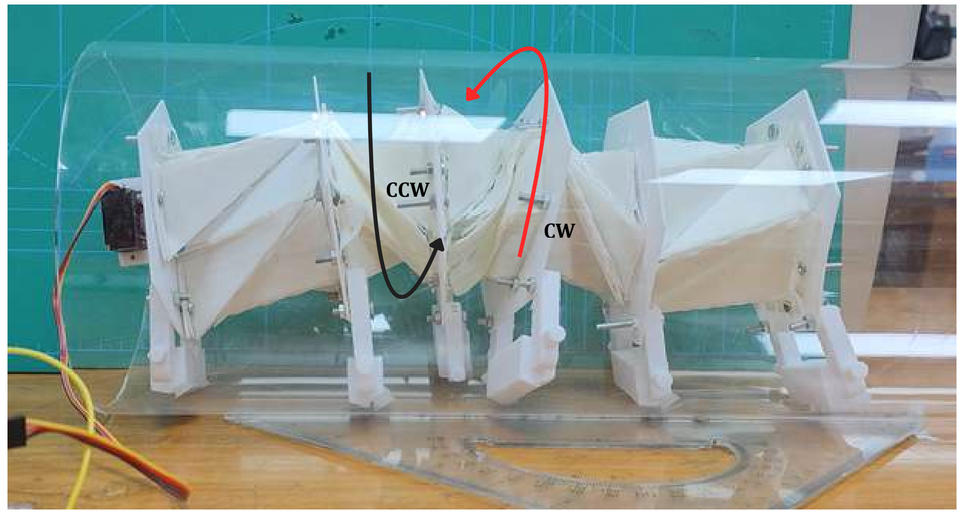
Initial Homogeneous Transformation

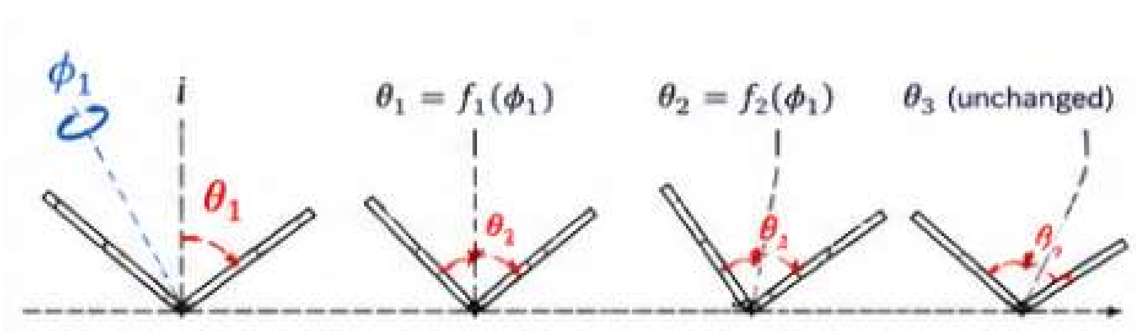
$$ {}^{0}\boldsymbol{T}_1 = \begin{bmatrix} c_1 & -s_1 & 0 & L_1 c_1 \\ s_1 & c_1 & 0 & L_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^{1}\boldsymbol{T}_2 = \begin{bmatrix} c_2 & -s_2 & 0 & L_2 c_2 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{2}\boldsymbol{T}_3 = \begin{bmatrix} c_3 & -s_3 & 0 & L_3 c_3 \\ s_3 & c_3 & 0 & L_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ c_i = \cos\theta_i, \quad s_i = \sin\theta_i $$

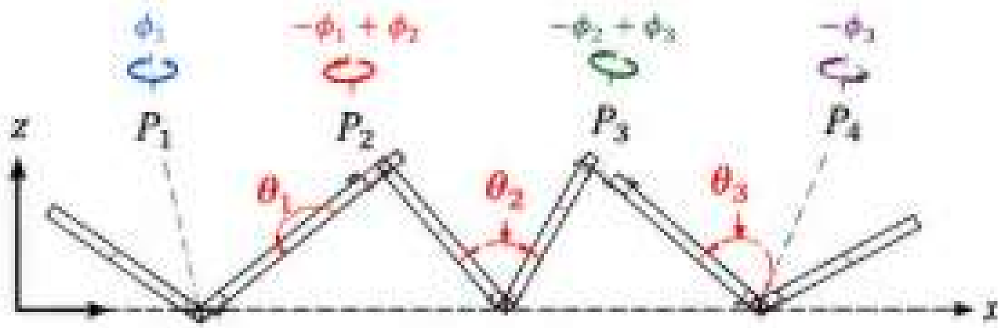
Changes due to valley angle due to plate rotation





Transformation due to rotation of ϕ

*



Rotation due to plate rotation ϕ

D–H PARAMETERS (after ϕ_1)

Joint i	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i (rad)
1	0	L_1	0	$\theta_1(\phi_1)$
2	0	L_2	0	$\theta_2(\phi_1)$
3	0	L_3	0	θ_3

θ_1 **and** θ_2 **change, and the distance is +x**

$$^{0}T_1(\phi_1) = \begin{bmatrix} c_1 & -s_1 & 0 & L_1c_1 \\ s_1 & c_1 & 0 & L_1s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{1}T_2(\phi_1) = \begin{bmatrix} c_2 & -s_2 & 0 & L_2c_2 \\ s_2 & c_2 & 0 & L_2s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^{2}T_3(\phi_1) = \begin{bmatrix} c_3 & -s_3 & 0 & L_3c_3 \\ s_3 & c_3 & 0 & L_3s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_z(\phi_1) = \begin{bmatrix} c_{\phi_1} & -s_{\phi_1} & 0 & 0 \\ s_{\phi_1} & c_{\phi_1} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R_z(-\phi_1+\phi_2) = \begin{bmatrix} c_{-\phi_1+\phi_2} & -s_{-\phi_1+\phi_2} & 0 & 0 \\ s_{-\phi_1+\phi_2} & c_{-\phi_1+\phi_2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_z(-\phi_2+\phi_3) = \begin{bmatrix} c_{-\phi_2+\phi_3} & -s_{-\phi_2+\phi_3} & 0 & 0 \\ s_{-\phi_2+\phi_3} & c_{-\phi_2+\phi_3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R_z(-\phi_3) = \begin{bmatrix} c_{-\phi_3} & -s_{-\phi_3} & 0 & 0 \\ s_{-\phi_3} & c_{-\phi_3} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$c_x = \cos x, \quad s_x = \sin x$

Total Transformation

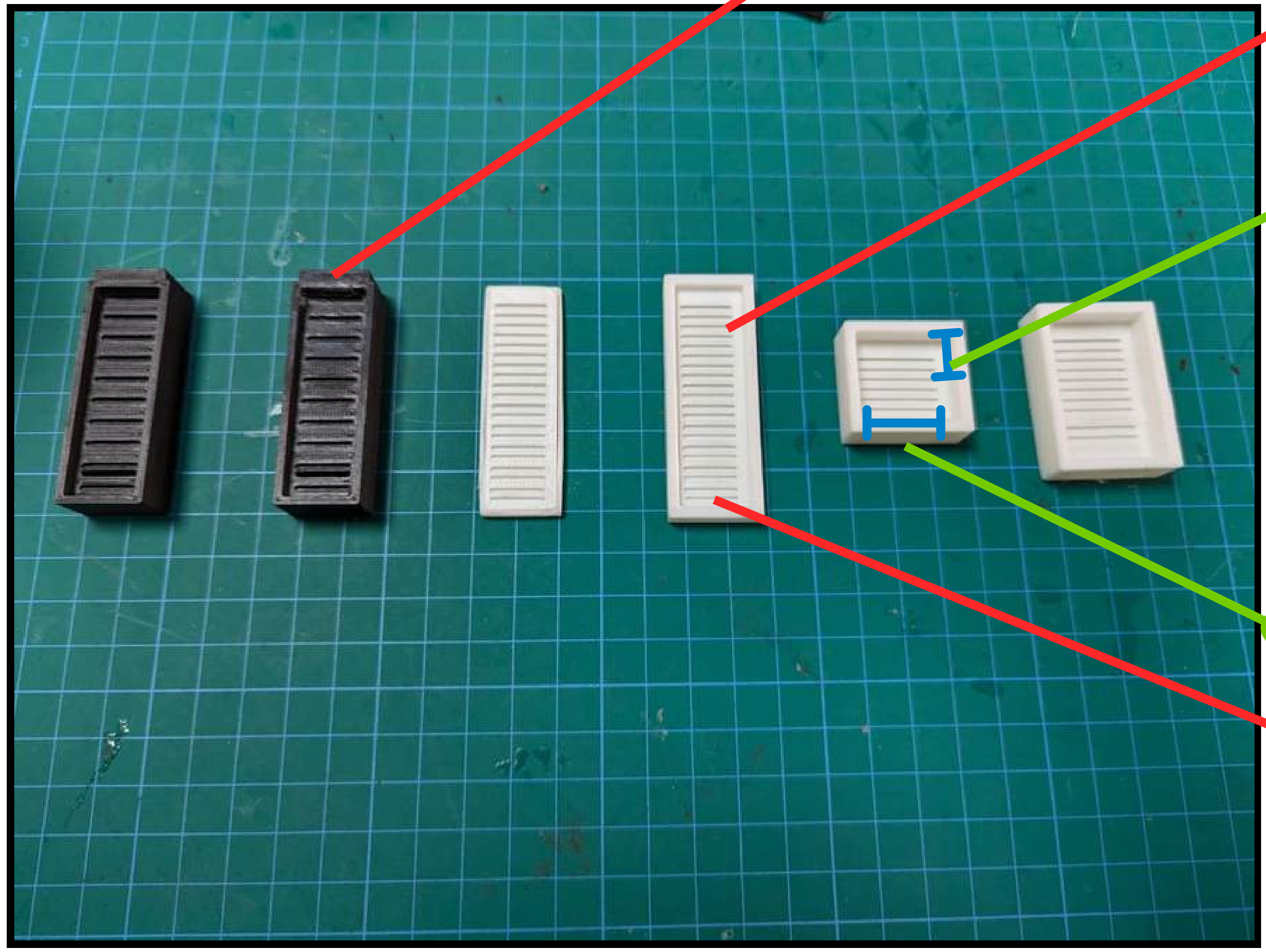
$$^{0}T_3^{total} = \underbrace{\left(R_z(\phi_1)\ \ {}^{0}T_1\right)}_{\text{Plate } P_1} \underbrace{\left(R_z(-\phi_1+\phi_2)\ \ {}^{1}T_2\right)}_{\text{Plate } P_2} \underbrace{\left(R_z(-\phi_2+\phi_3)\ \ {}^{2}T_3\right)}_{\text{Plate } P_3} \underbrace{R_z(-\phi_3)}_{\text{Plate } P_4}$$

Optimized feet for aiding locomotion: Iterations

Optimized feet for aiding locomotion: Iterations

Iteration 1: **PLA-Silicone alternataively placed with equal spacing**

PLA-matte mold



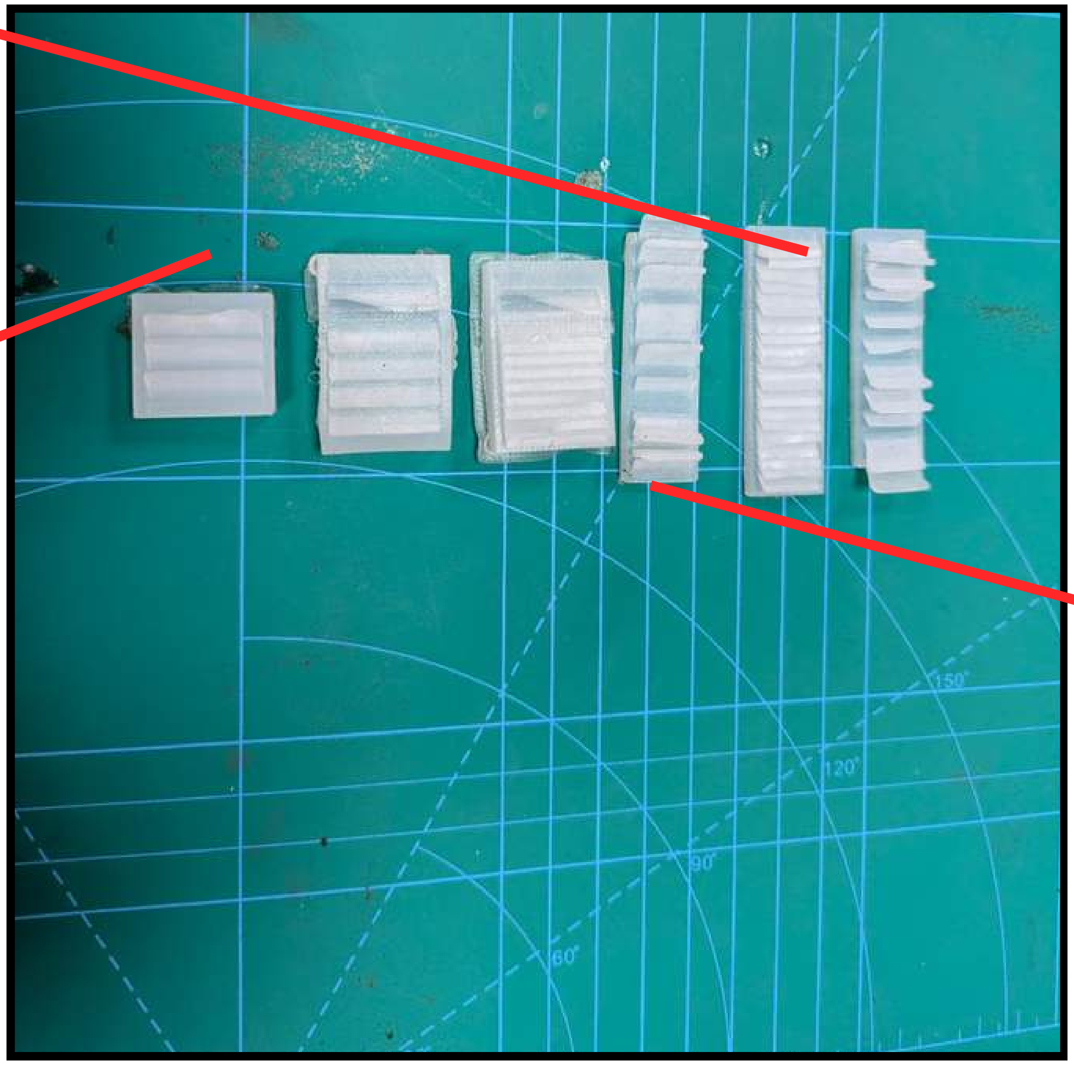
45^0 groove

1mm

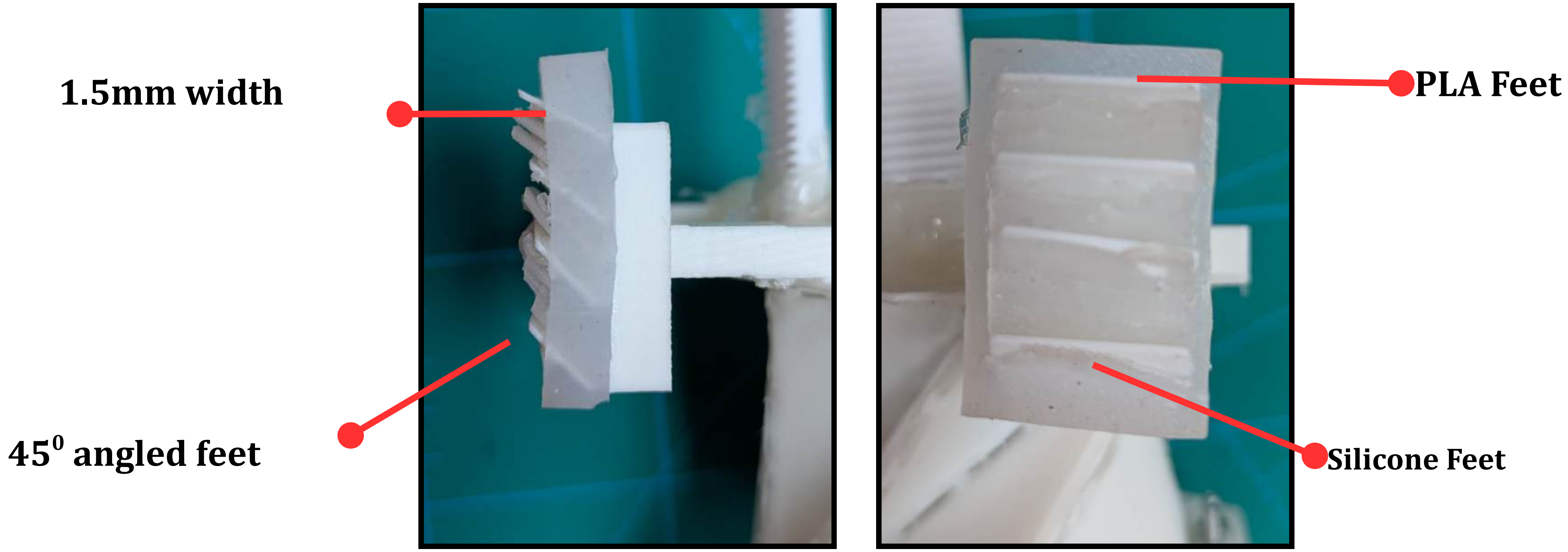
anisotropic feet

1.5mm

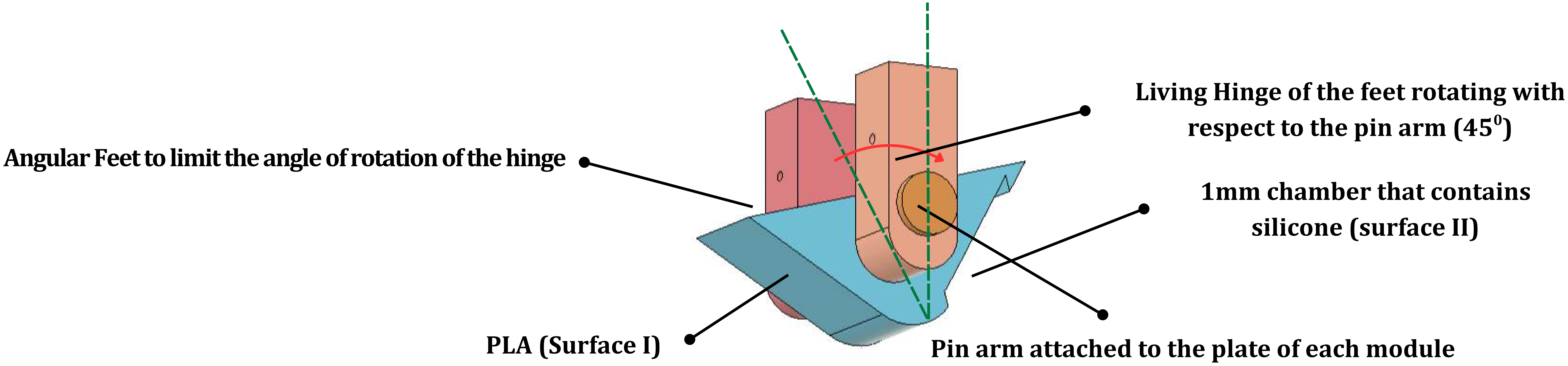
PLA mold



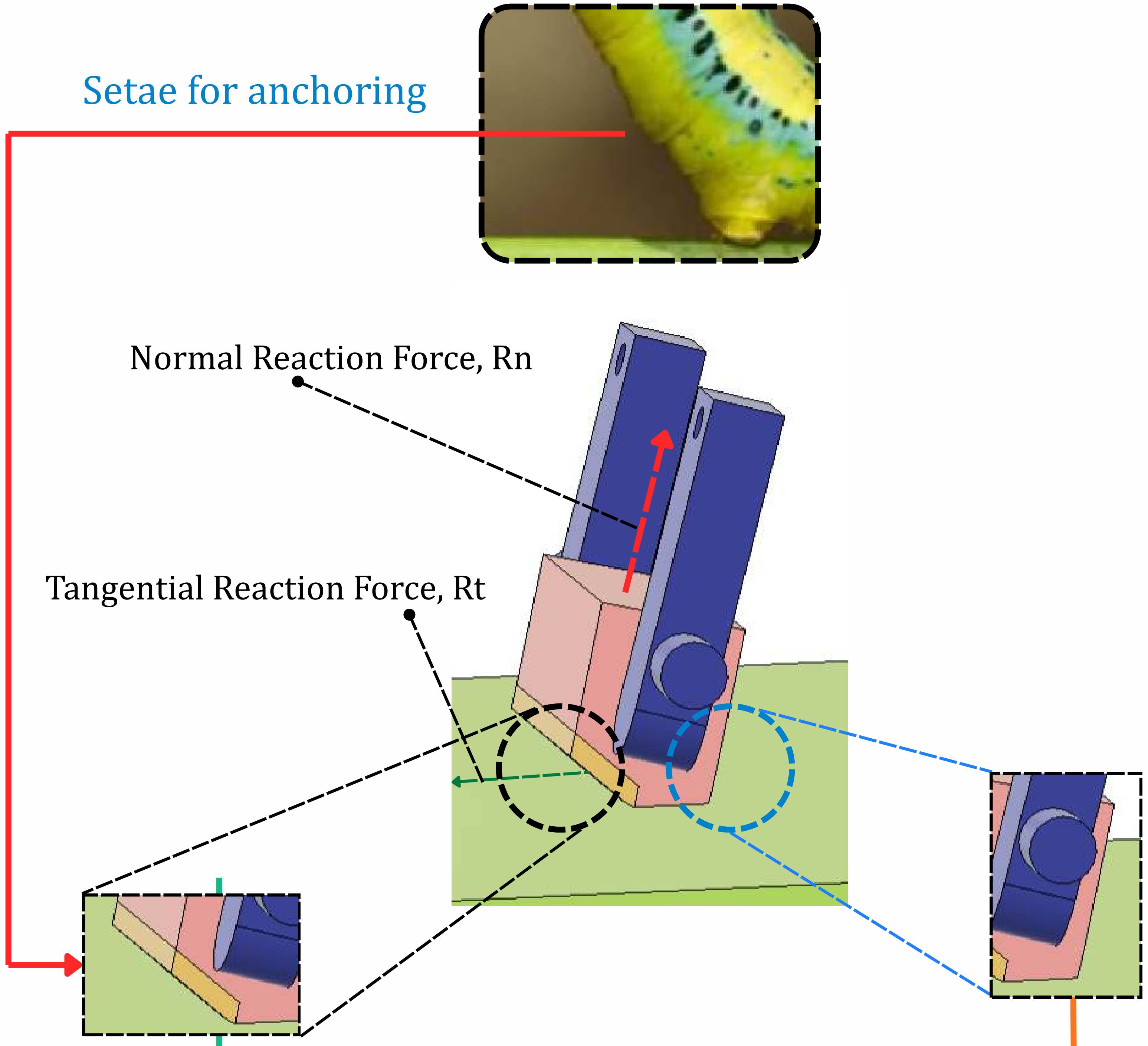
Silicone Ecoflex 00-30

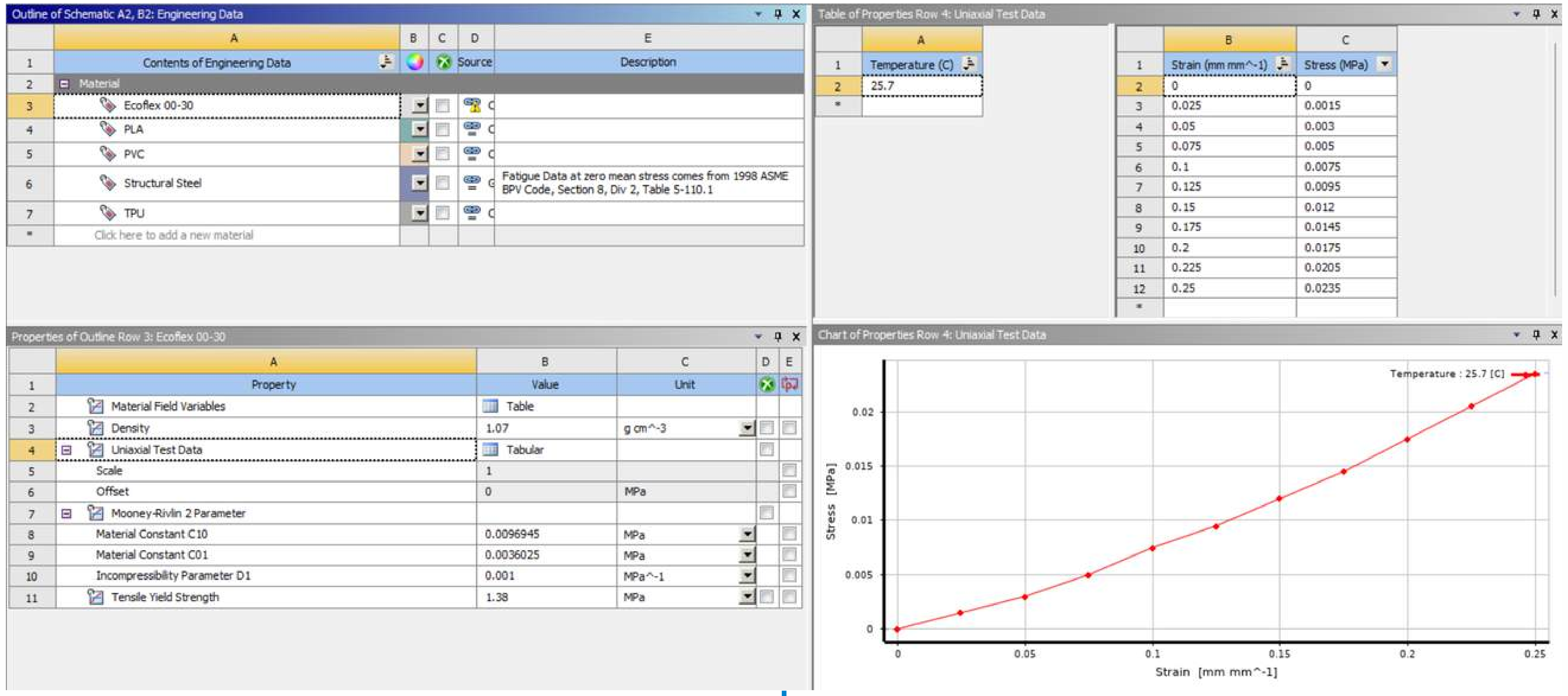


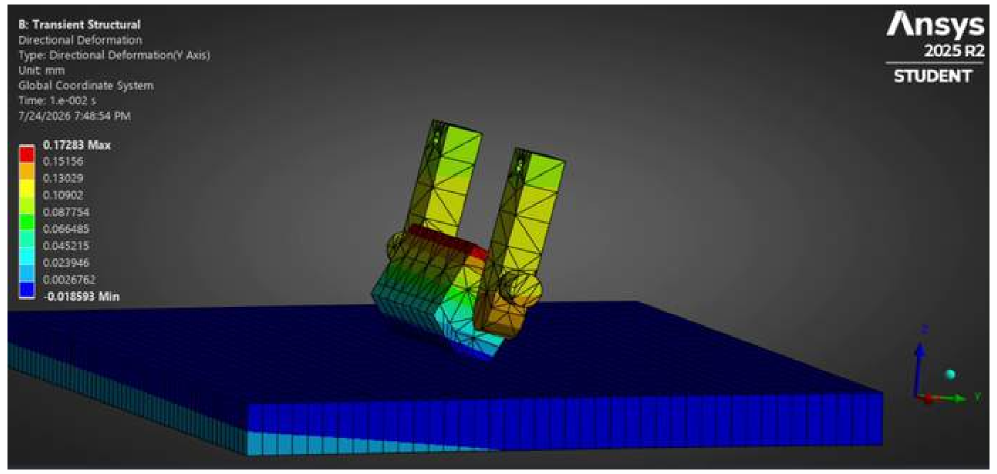
Iteration 2: PLA-Silicone in alternative contact area with a living hinge



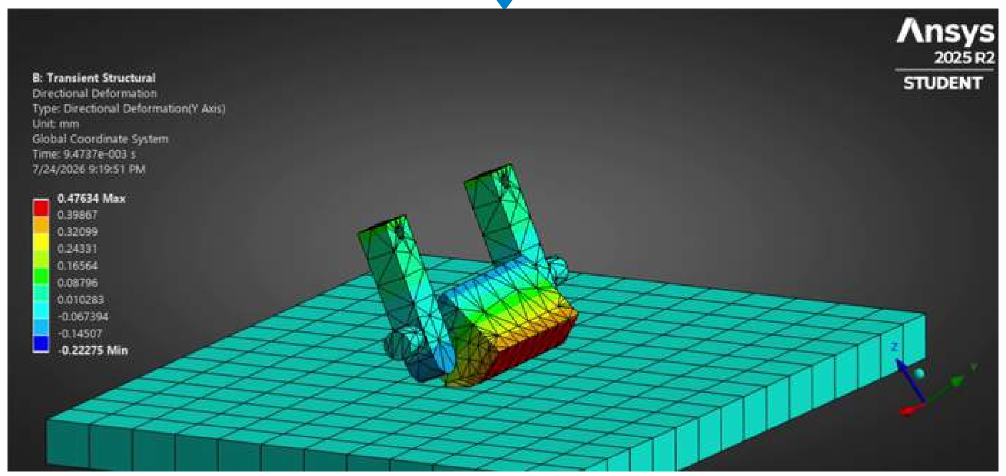
How is the characteristics of an inchworm setae implemented for this robot?





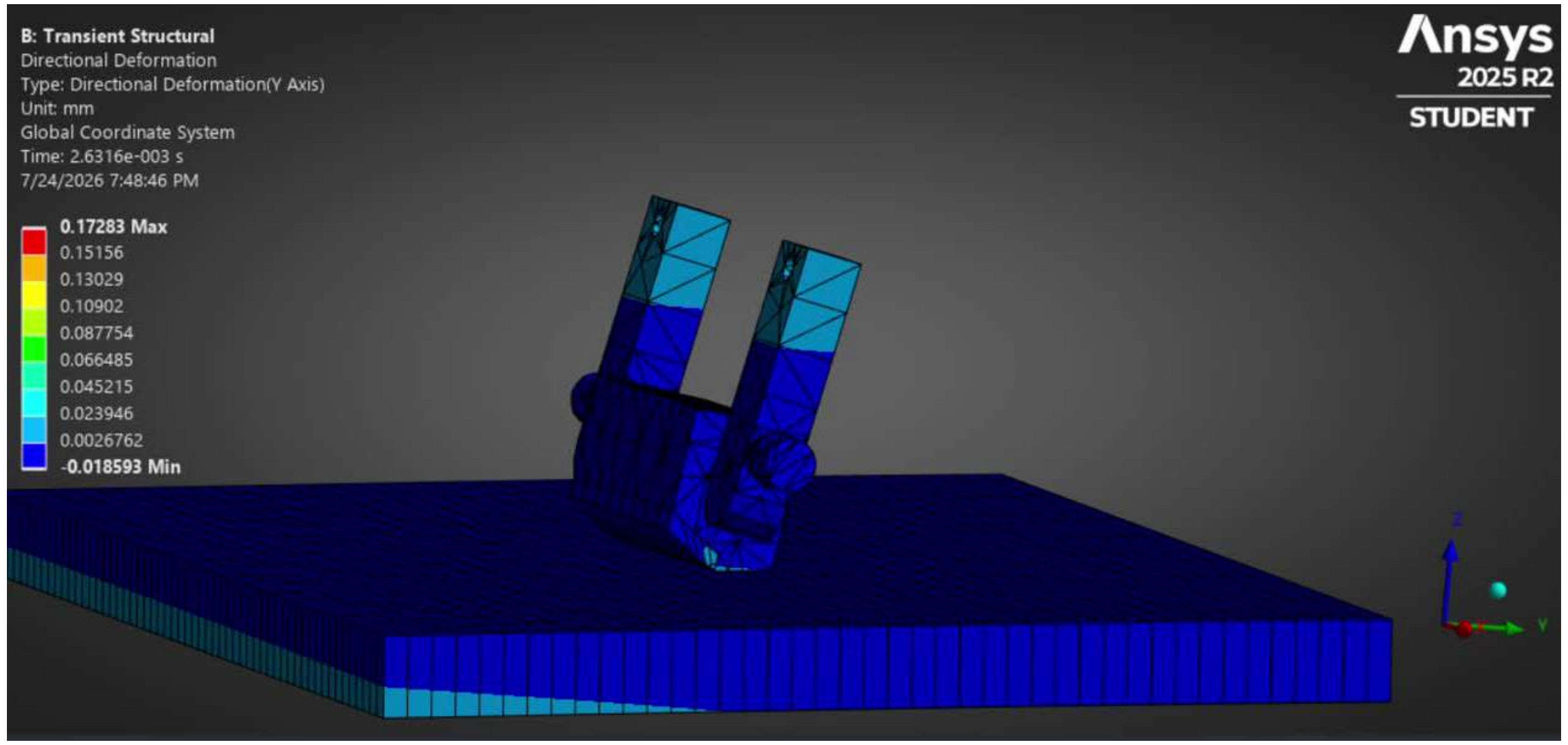


Silicone-Acrylic surface friction analysis

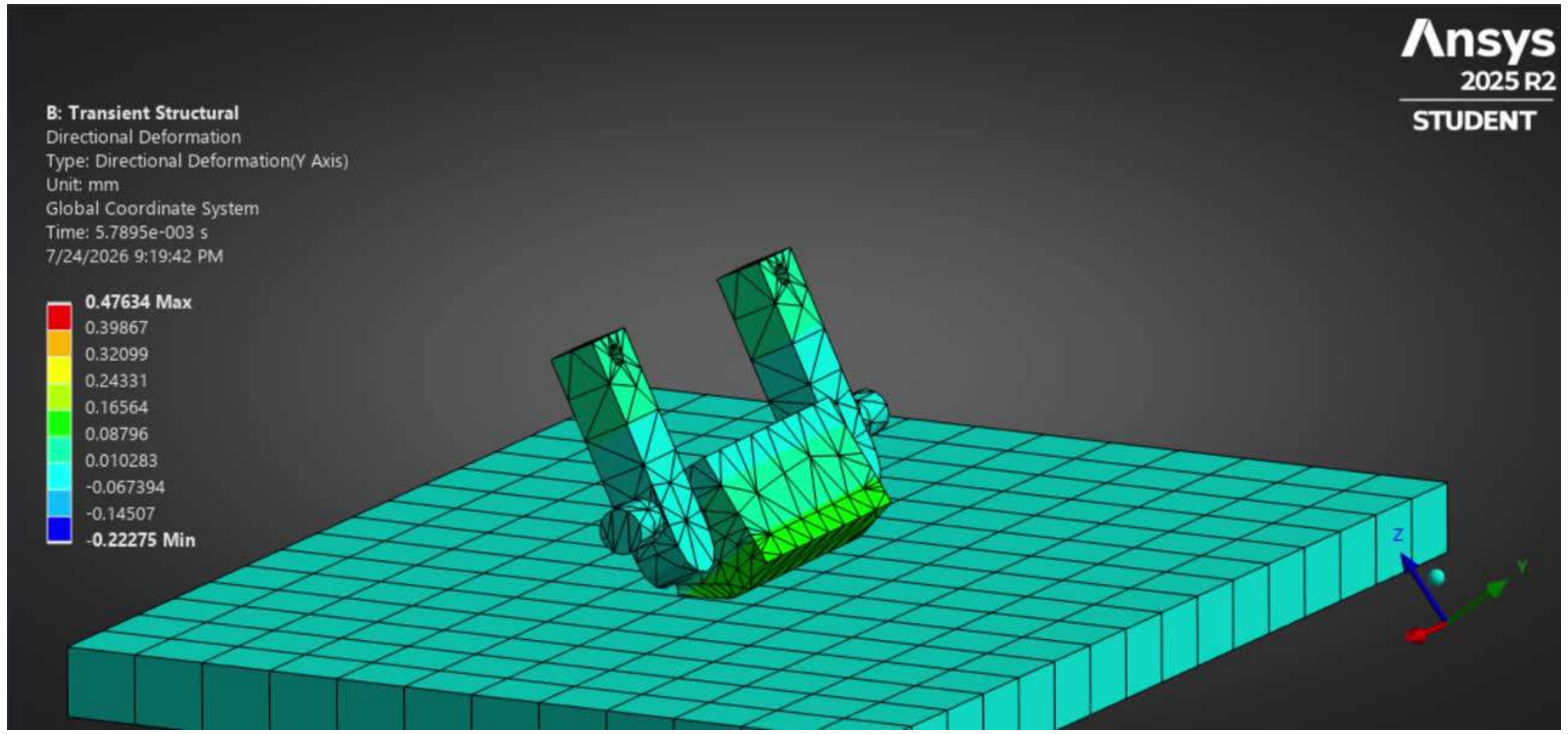


PLA-acrylic surface friction analysis

Passive Anchoring Mechanism:Anisotropic Friction Validation



PLA-acrylic surface friction analysis



Silicone-Acrylic surface friction analysis

Parameter	Silicone–Acrylic	PLA–Acrylic
Static Friction coeffcients	0.6	0.1
Kinemtic friction coefficients	0.1061	0.02356
Mobilization ratio	17.70%	23.60%
Sliding distance (mm)	5.998 E-03	5.606 E-03
Frictional work (N·mm)	0.002218	0.0006492

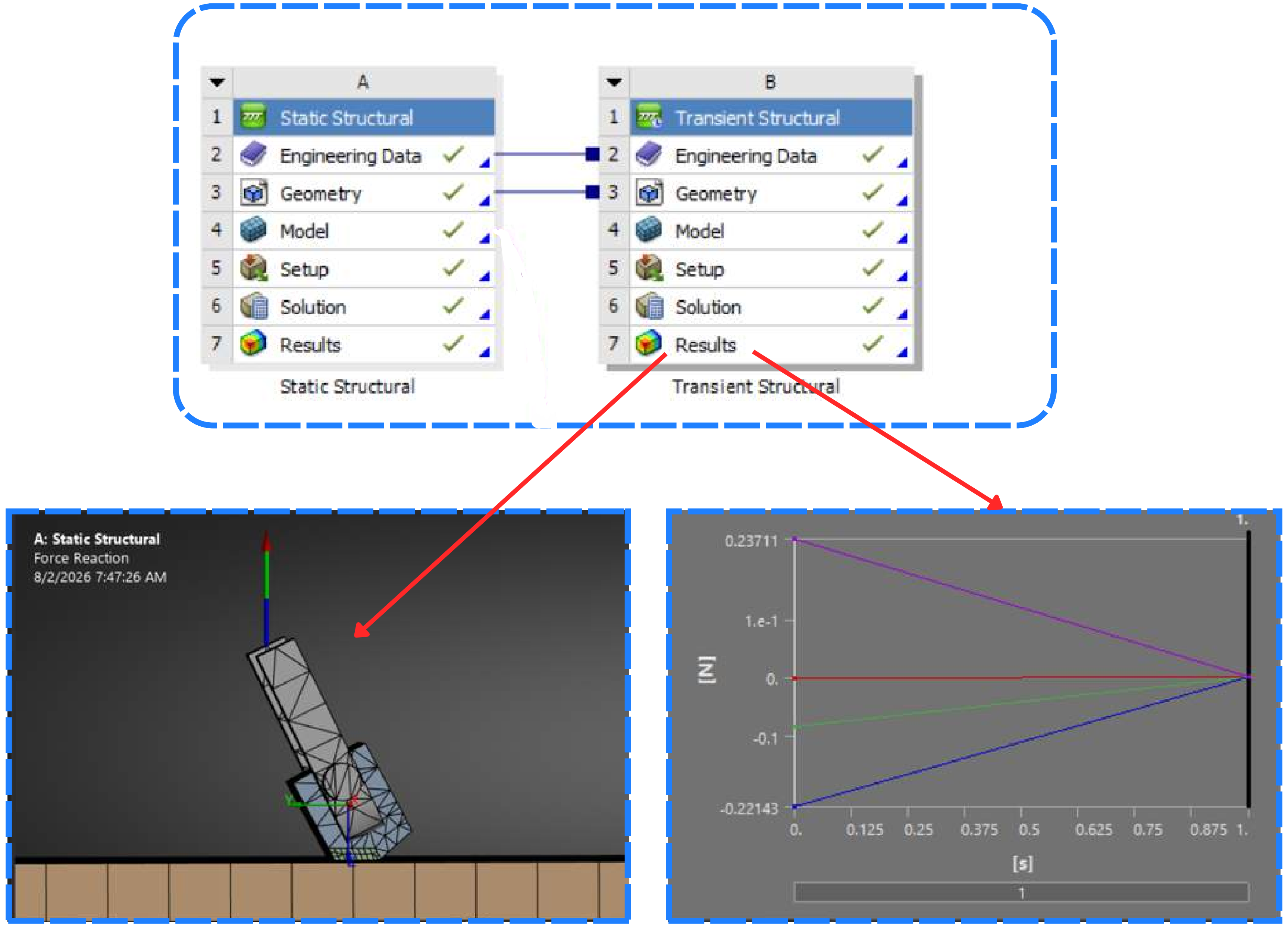
$$W_{\text{silicone-acrylic}} = 3.42 * W_{\text{PLA-acrylic}}$$

Stopper Angle Optimization and Stability Window

θc =(35°)(±5°)
Optimal Operating Stability Window

η = 7.97
Peak Directional Traction Ratio

8.72× Margin
Anchoring Safety Factor (SF_anchor)

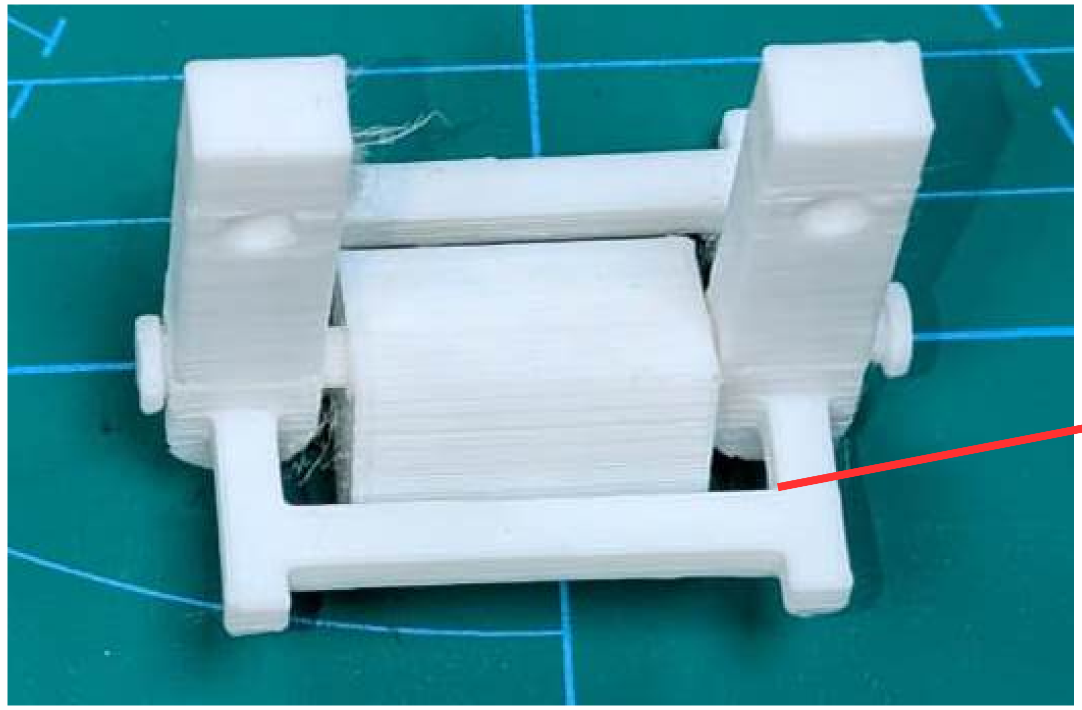


Stability Envelope: Within **[30°, 40°]**, anisotropic feet anchor reliably without **tipping (θc ≤ 15°)** or local **deformation slip (θc ≥ 45°)**.

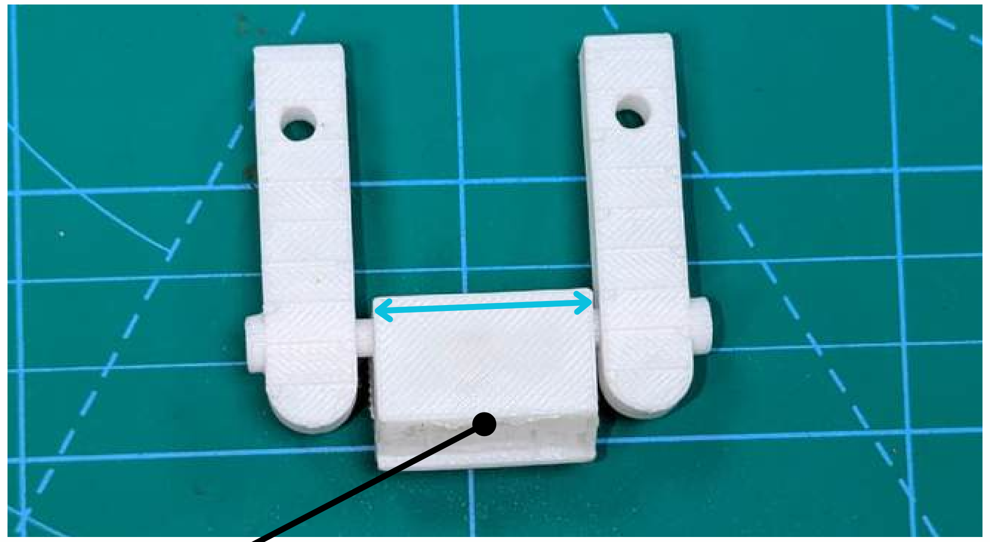
Parametric Evaluation (d_s = 3.0 mm)

Contact Angle θc (°)	Glide Friction Ff,glide (N)	Anchor Friction Ff,anchor (N)	Traction Ratio η	Stability Status
15	Unstable	Unstable	Unstable	Tipping / Premature Slip
30	0.412	2.48	6.02	Stable (θc - 5° boundary)
35	**0.385**	**3.0681**	**7.97**	**Optimal Stable Configuration**
40	0.401	2.71	6.76	Stable (θc + 5° boundary)
45	Unstable	Unstable	Unstable	Slip
60	Unstable	Unstable	Unstable	Interfacial Failure

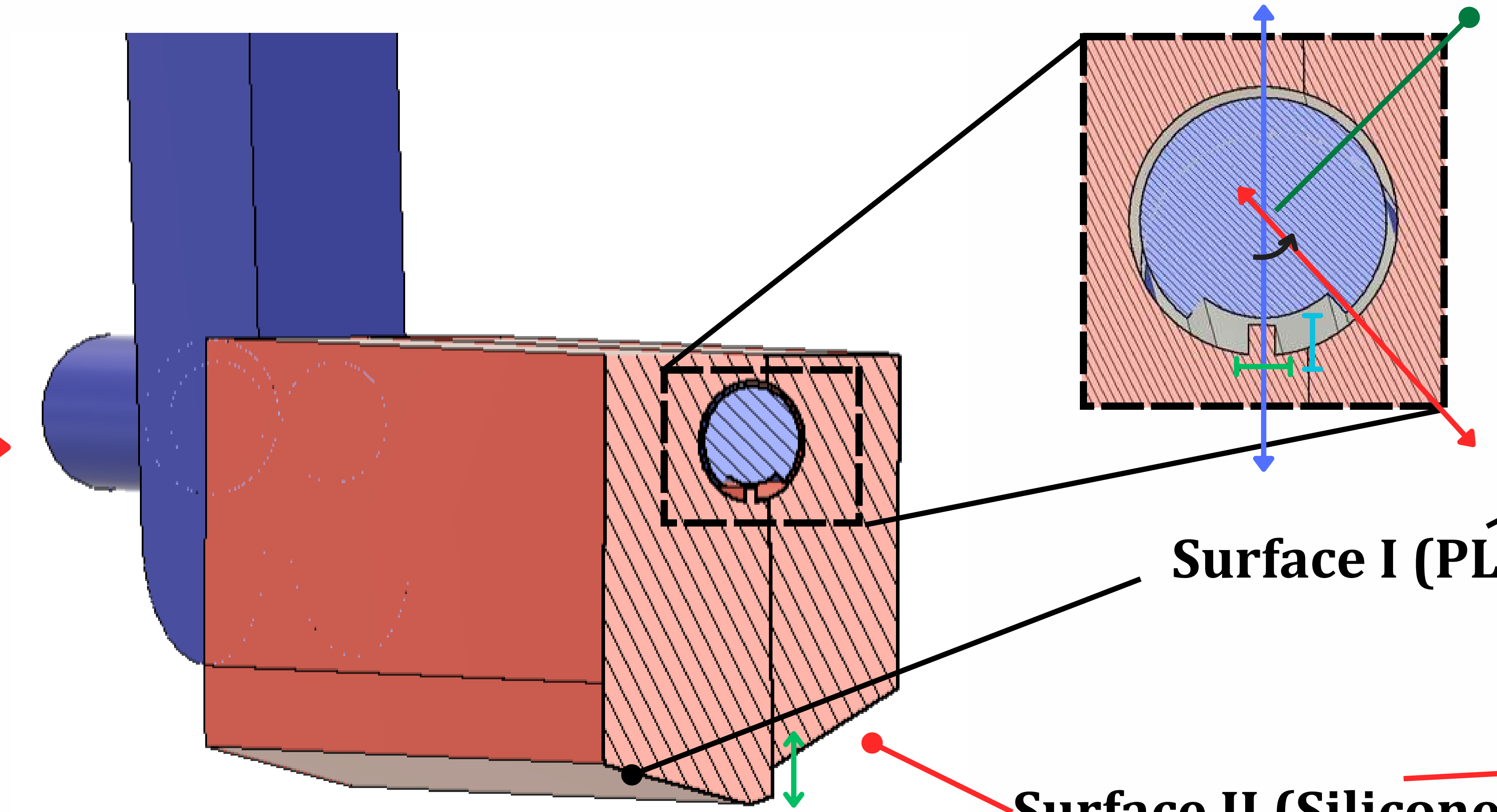
Iteration 3: PLA-Silicone in alternative contact area with a living hinge



Stopper to limit the angle of rotation of the living hinge (60^0)

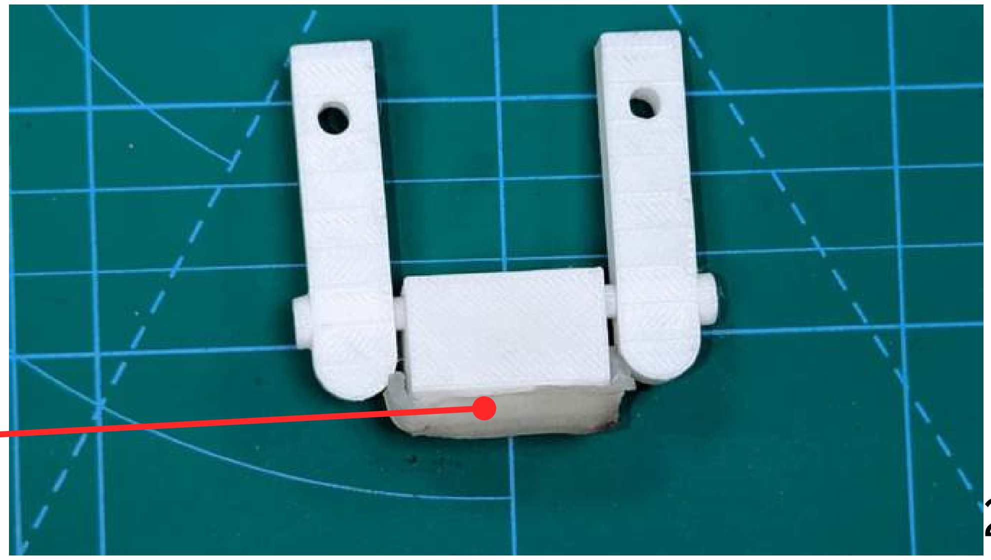


35^0 angled stopper



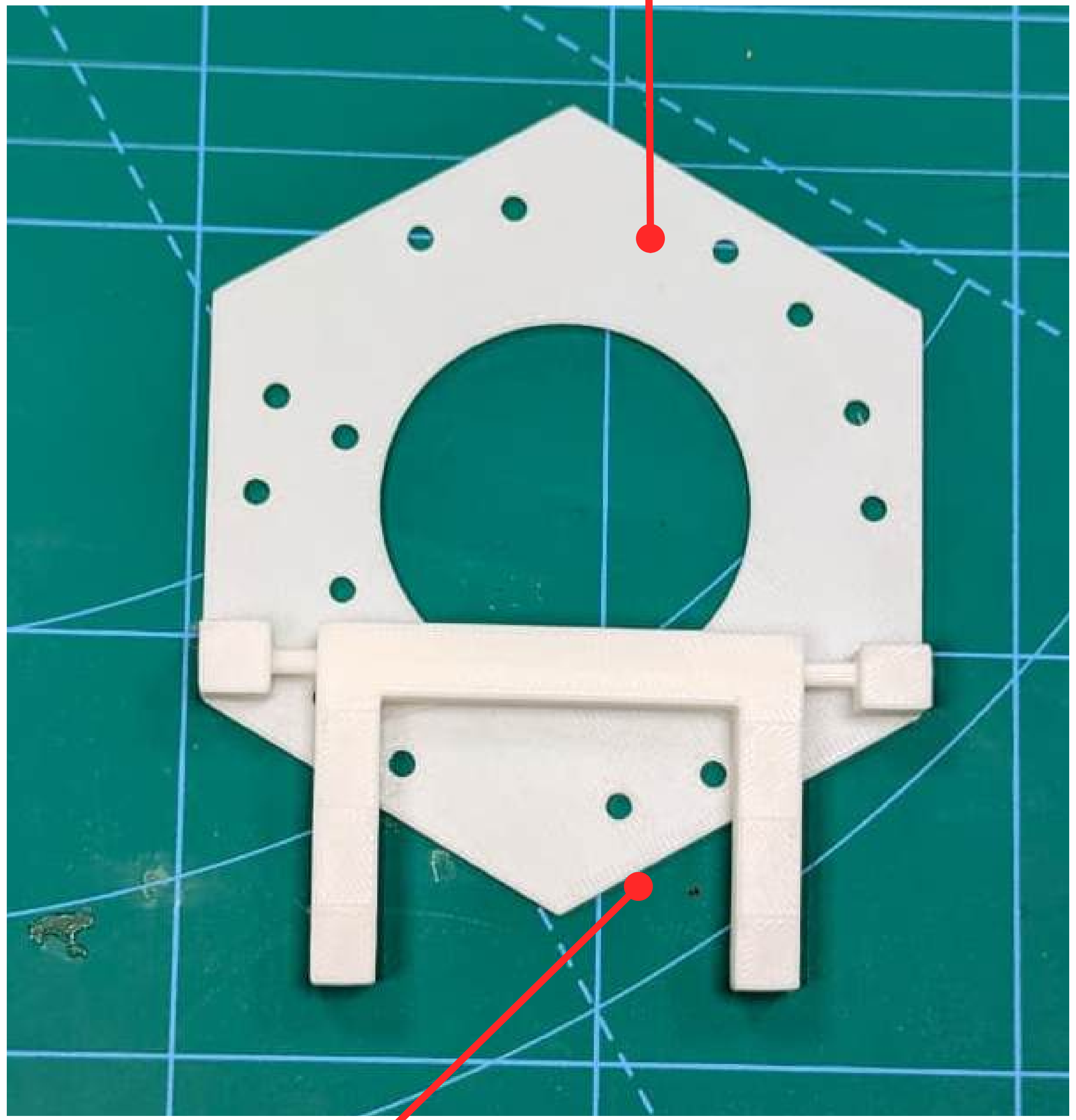
Surface I (PLA)

Surface II (Silicone)



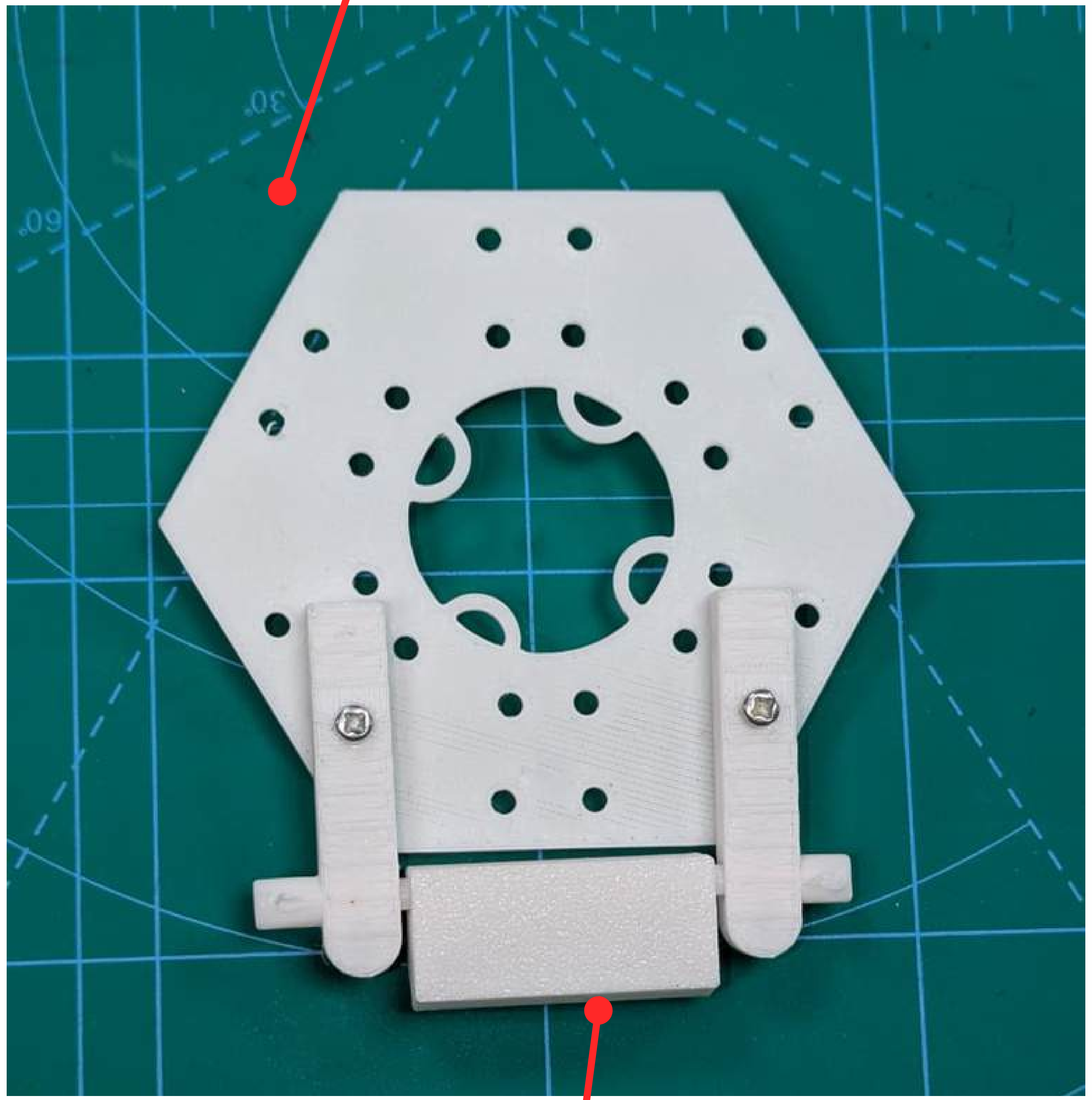
How is the chassis with the feet assembled?

Allows only one module to connect



Feet for balancing the module consisting of a single surface

Allows each pair of modules to connect while maintaining proper weight distribution

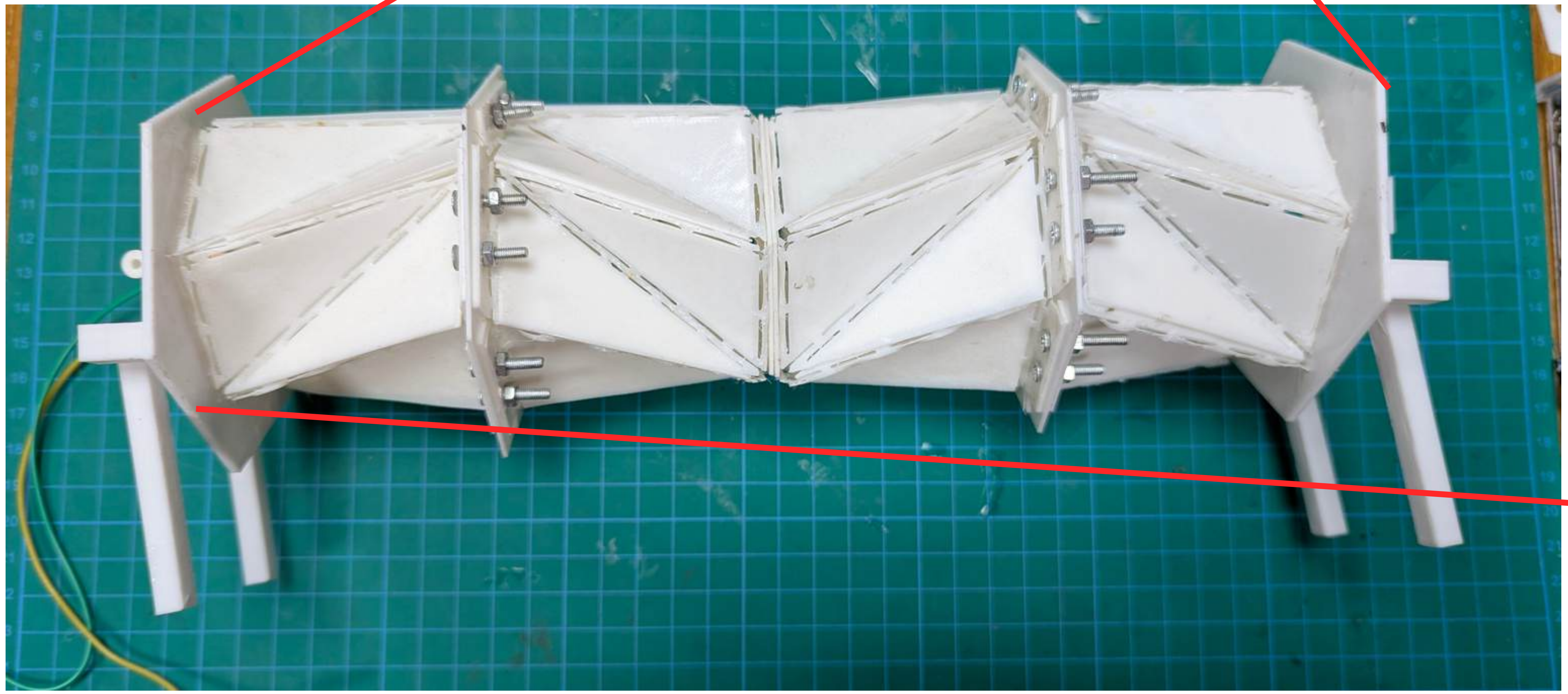


Feet consisting of two surfaces for anchoring and Sliding

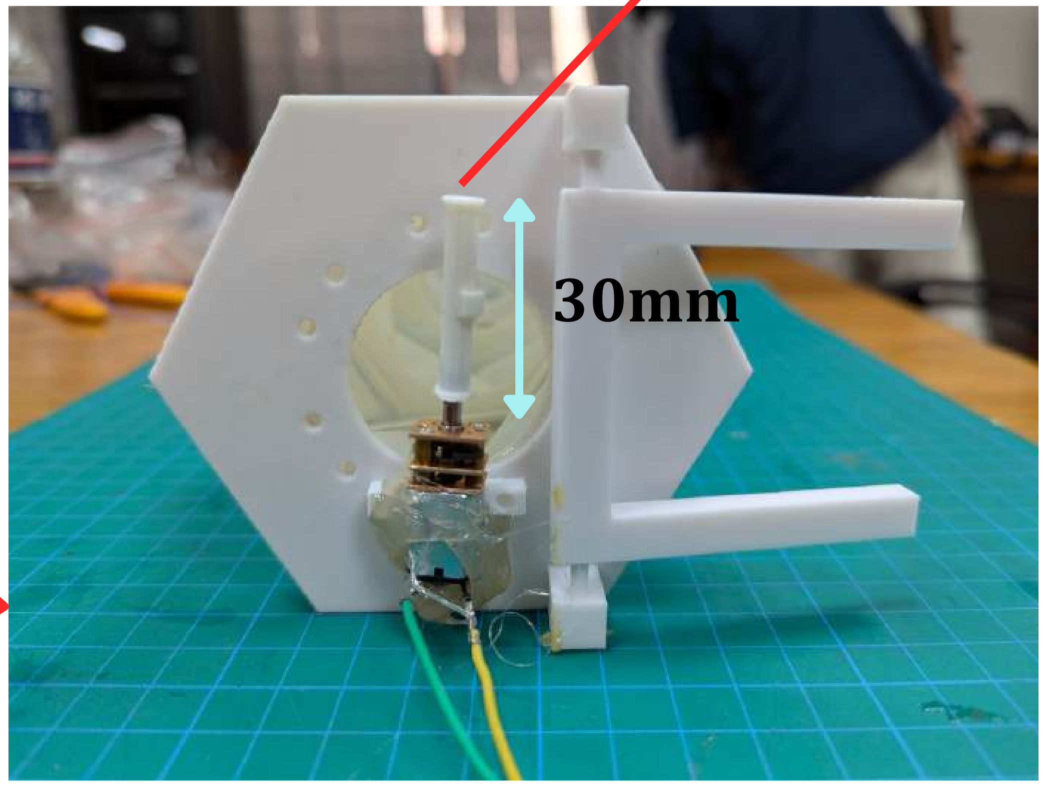
Optimization of the interated system

Prototype 1

2 plates at the end of modules

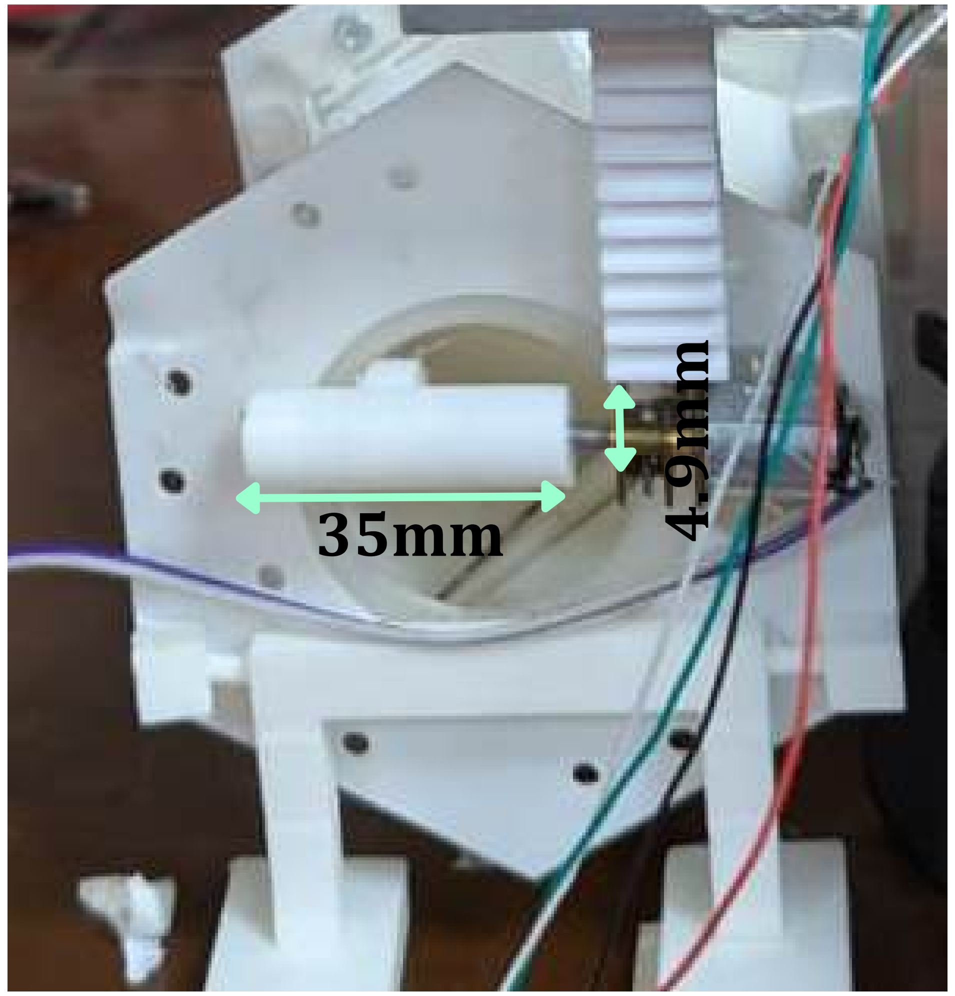


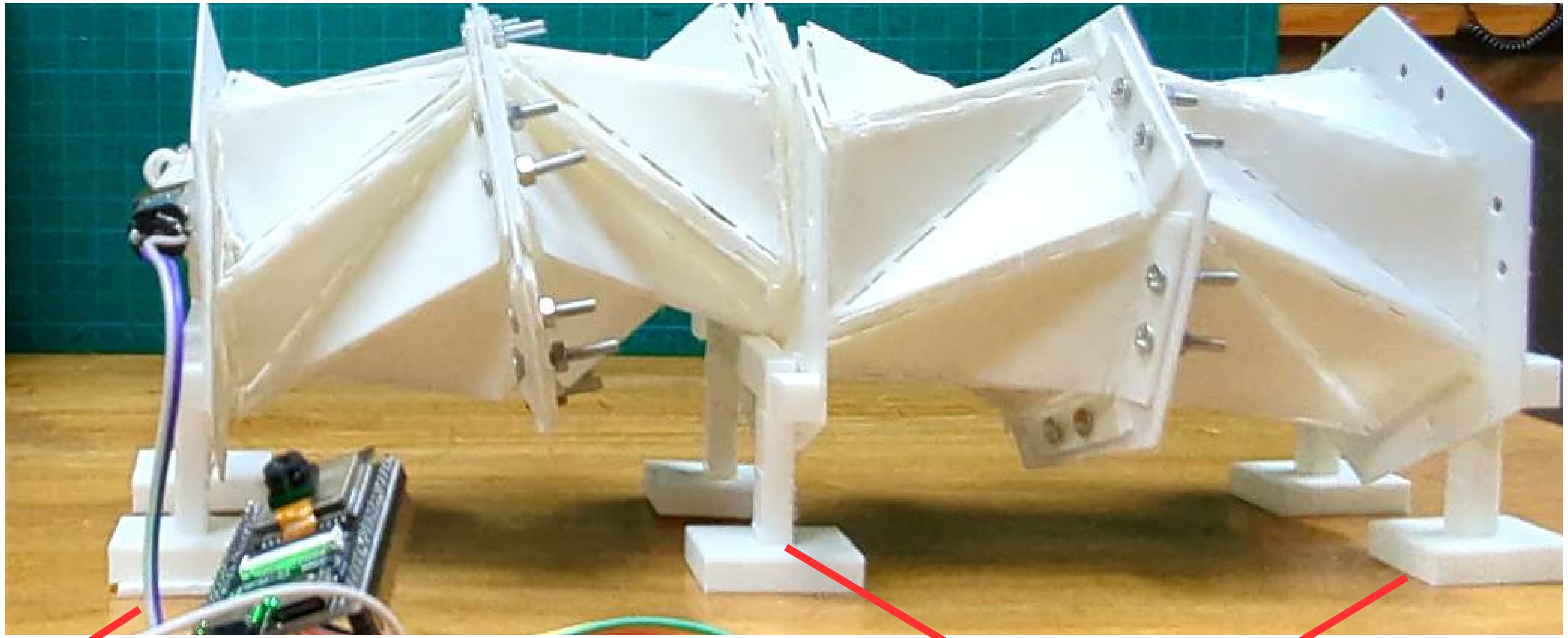
4.9mm diameter



Imbalance of the robot's **center of mass**

Prototype 2





6 feet

Each plate with 2 feet

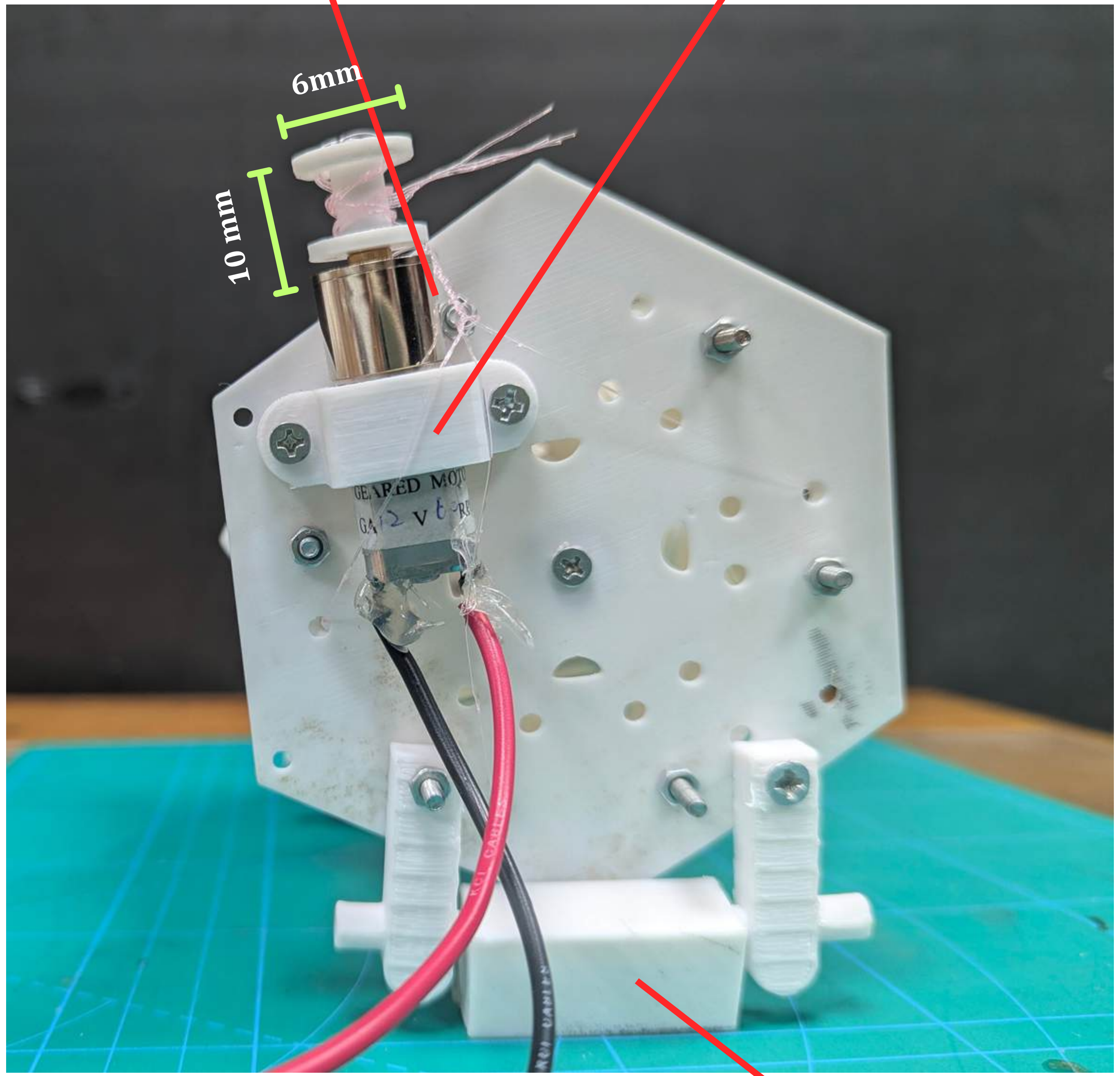
Lack of traction , couldn’t move **forward**

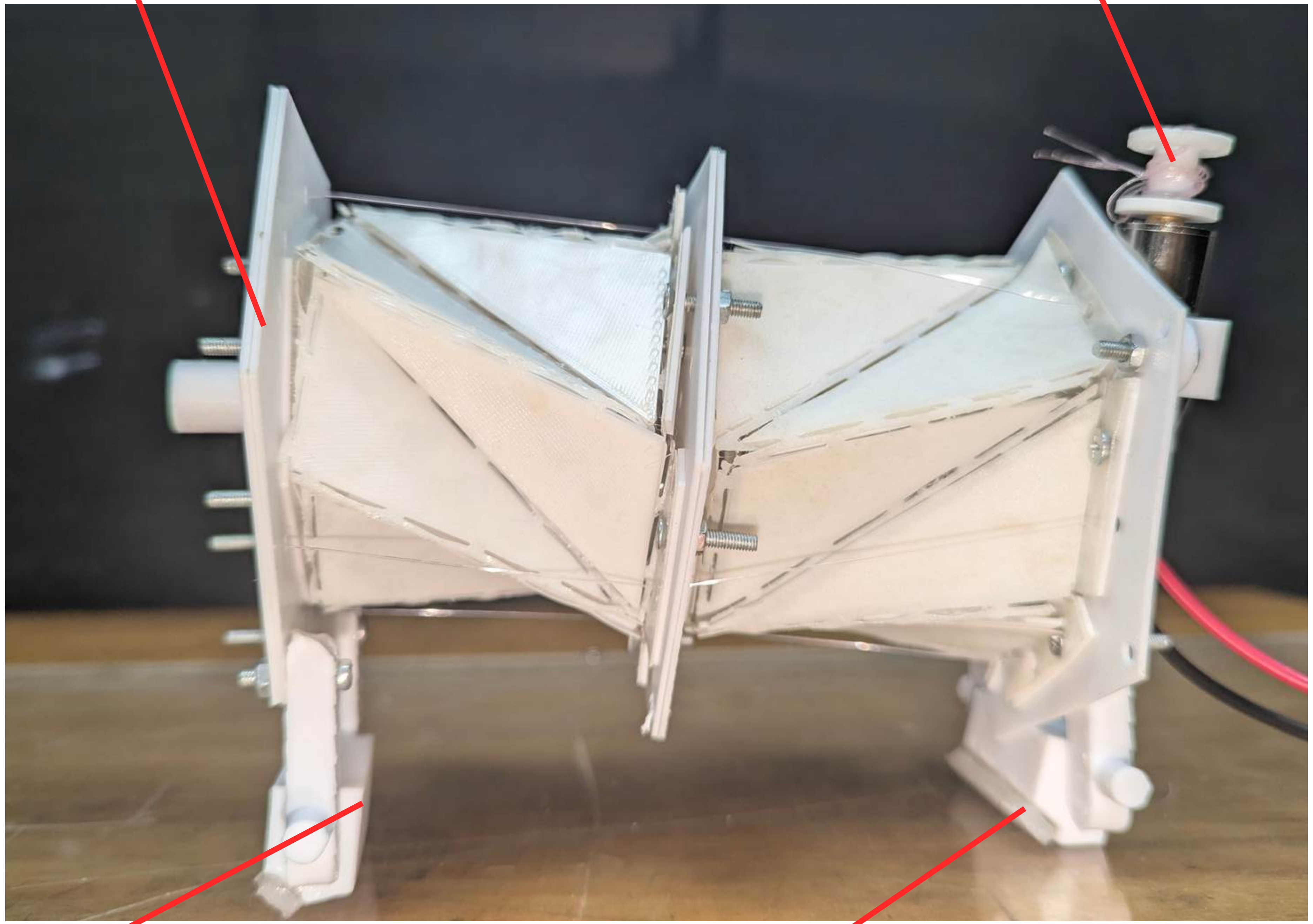
Prototype 3

12 V, 60 RPM 16GA-050 DC gear motor

Motor holder attached to the front plate

Spool rotates CW and CCW to actuate locomotion



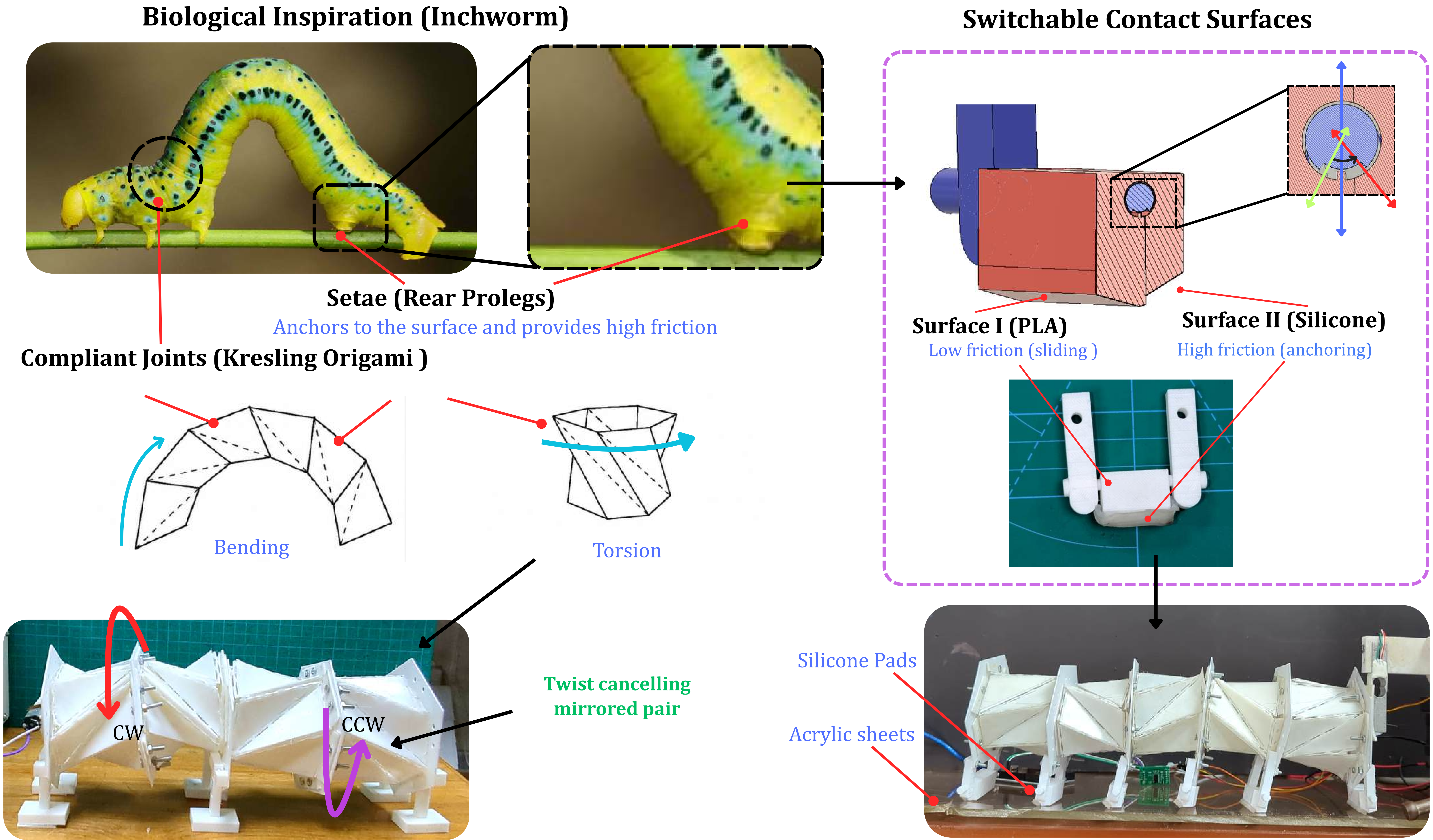


surface I (PLA)

surface II (Silicone)

Current functional prototype

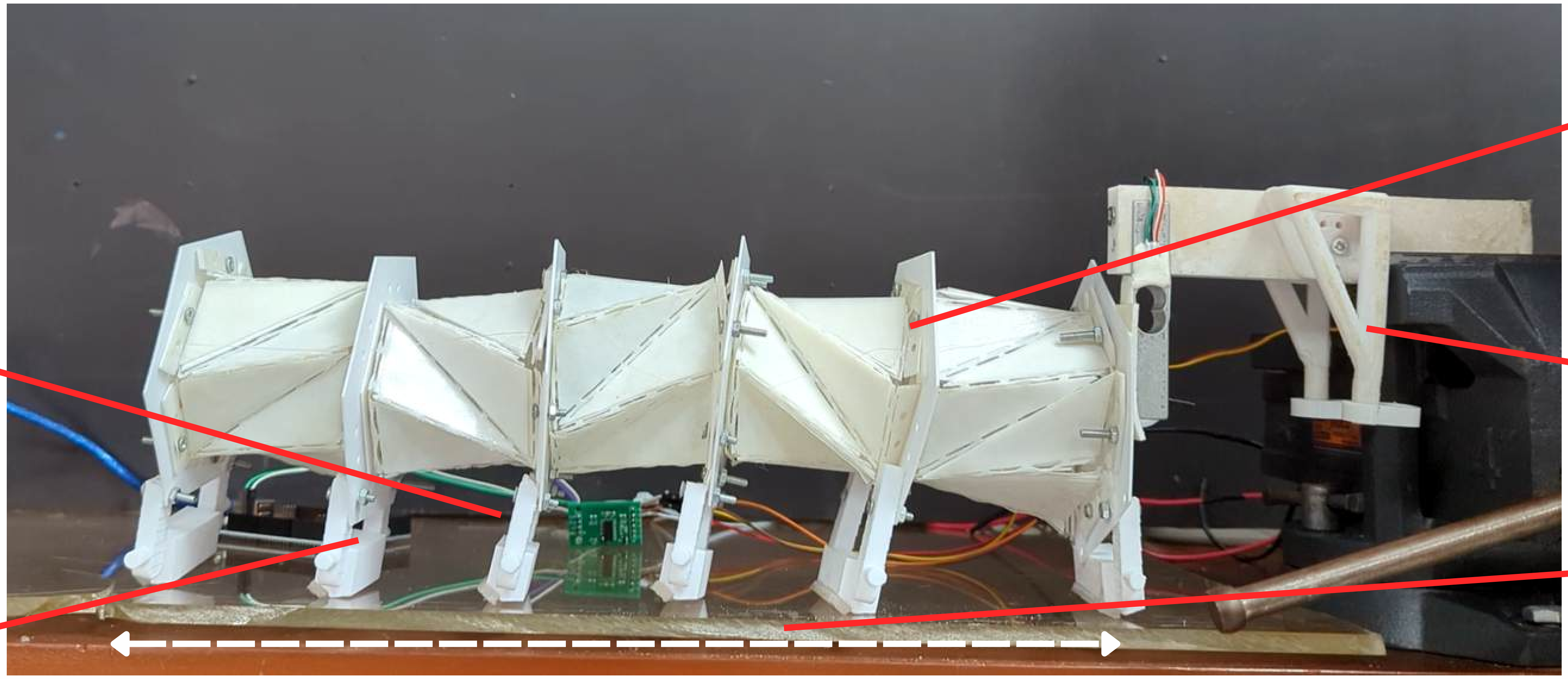
How is the chracteristics of an inchworm implemented in the integrated system ?



Does this system actually work?

Silicone Pads (2mm)

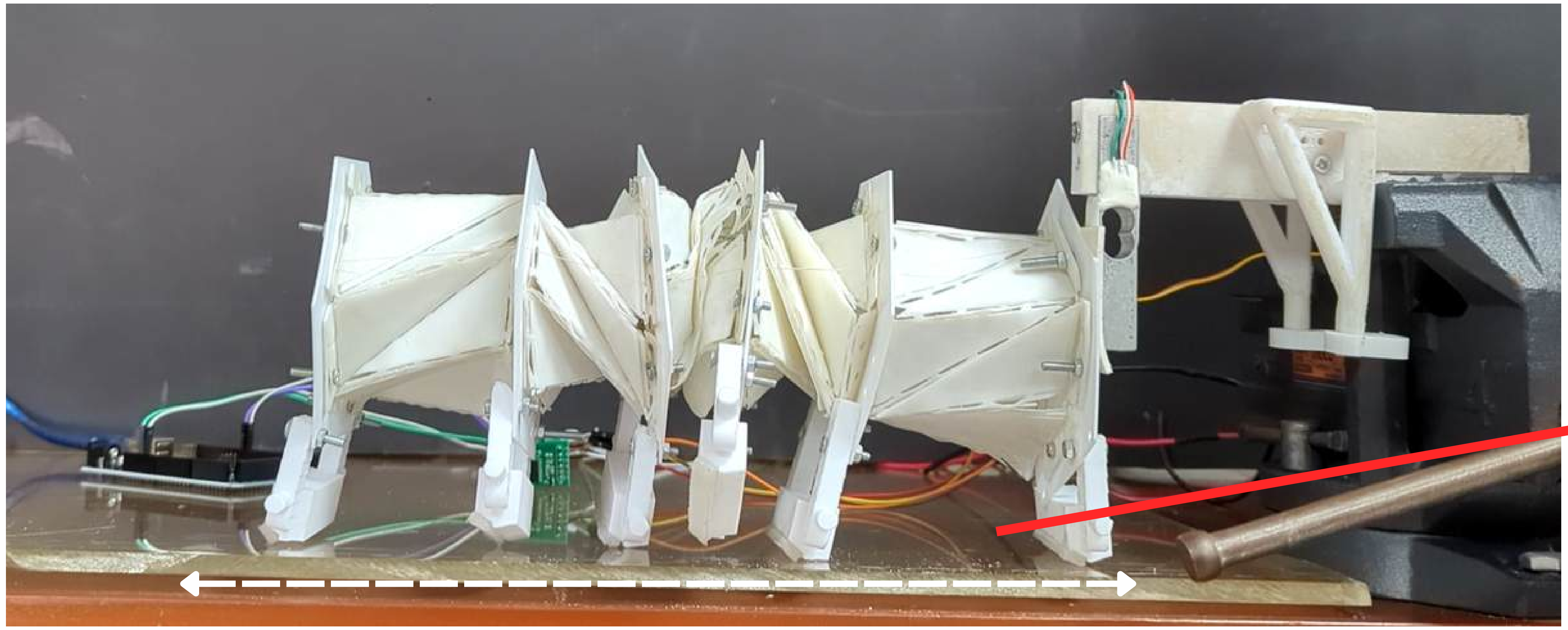
Acrylic Sheets (3mm)



Passive module (TPU) that allows lateral movement

Load Cell (5kg)

Fully Expanded Robot (33cm)



Fully Contracted Robot (18.5cm)

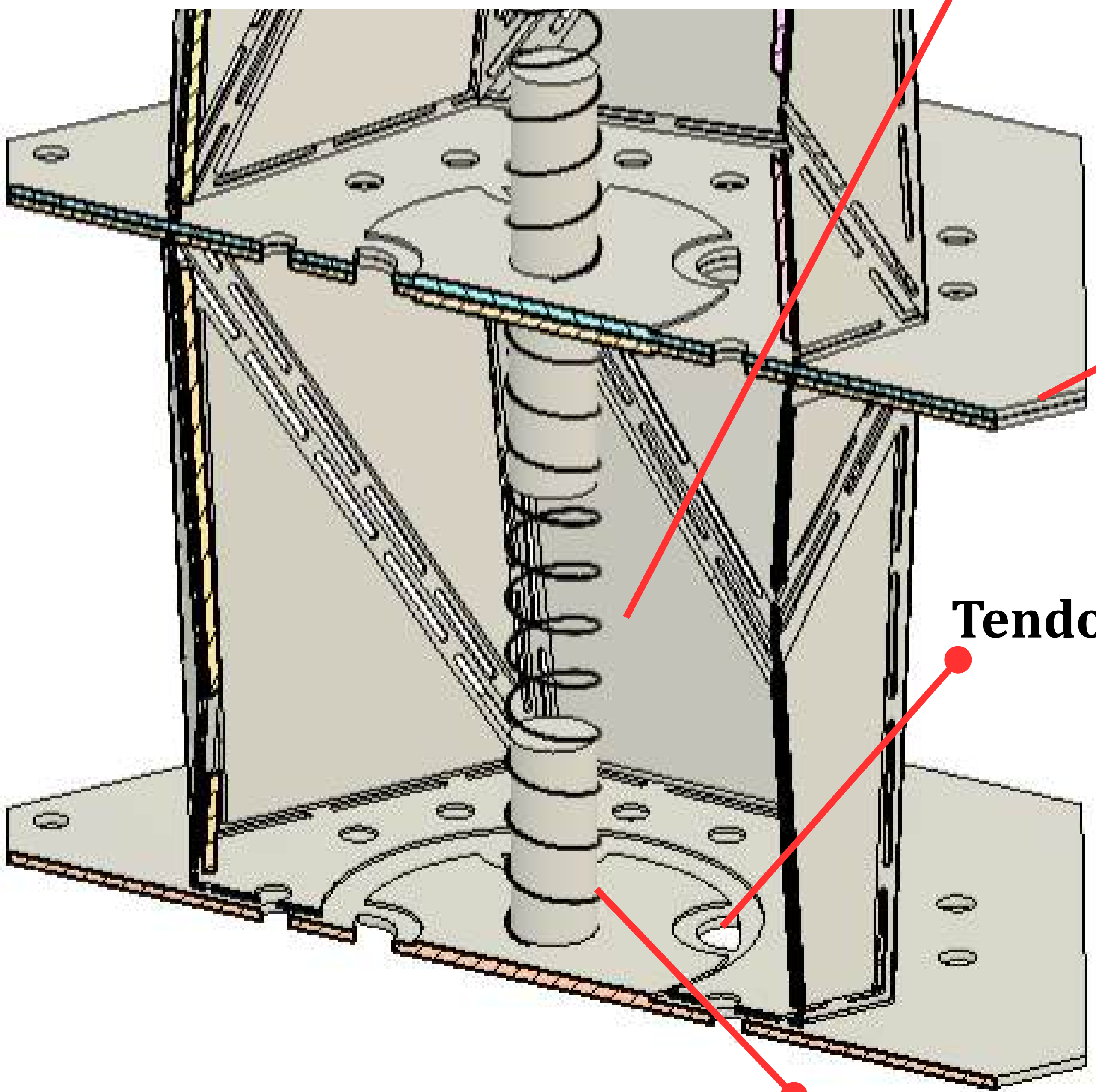
Unable to move forward, due to **lack of restoration force**

How can the restoration force of the chassis material be increased with the design constraints ?

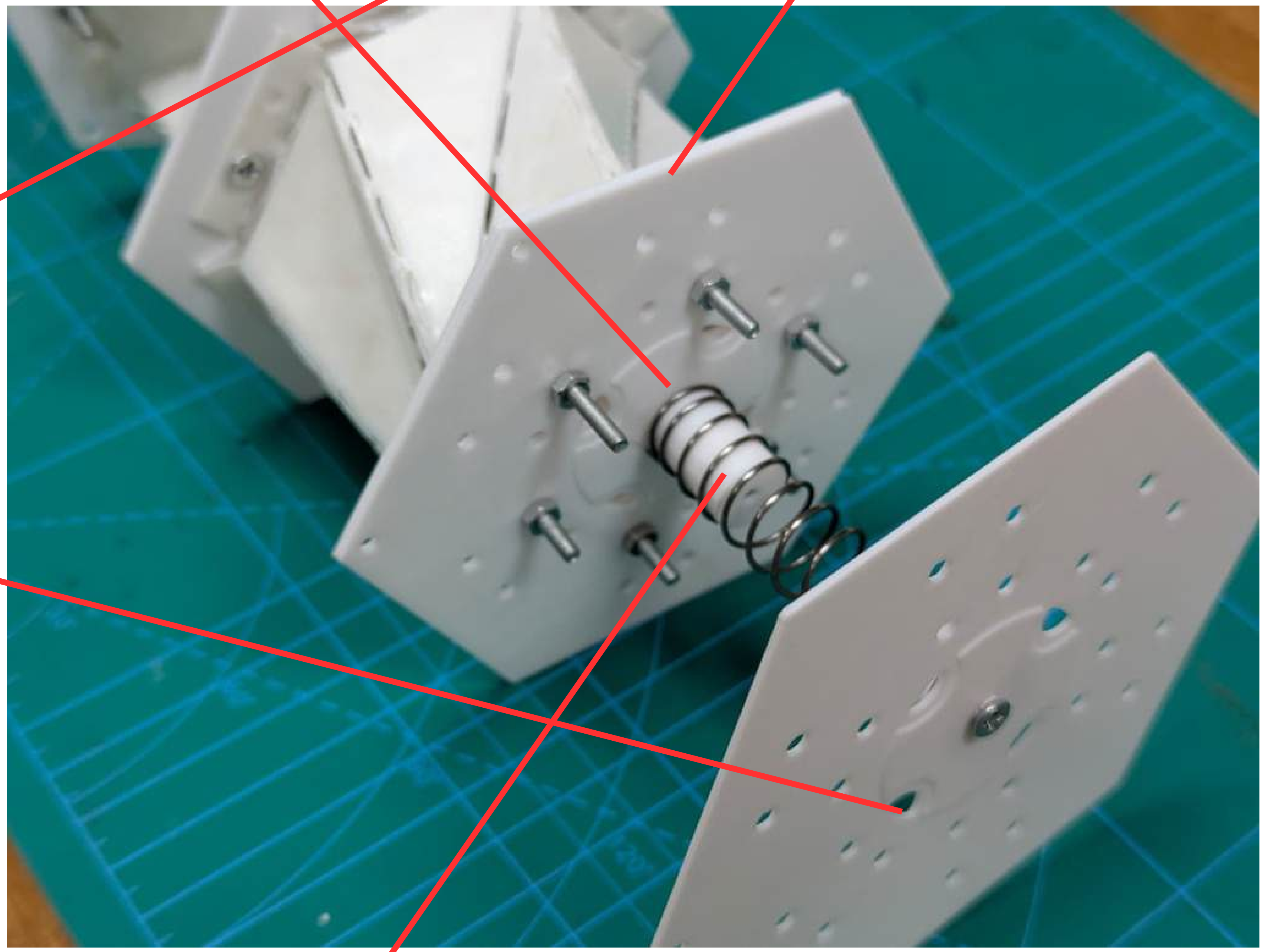
Compression spring between successive plates

Spring Steel For Higher Flexibility

Plates for attaching mirrored modules

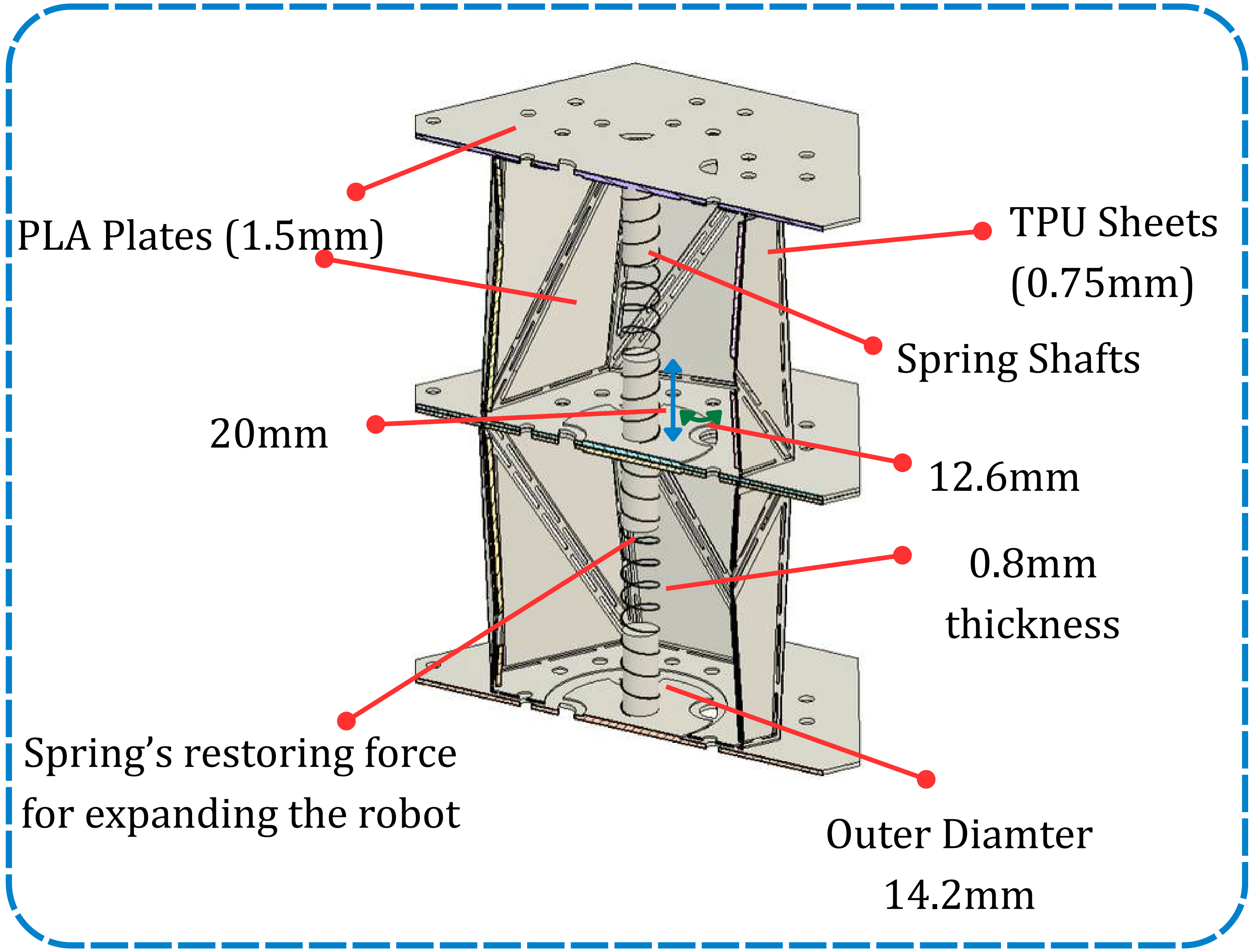


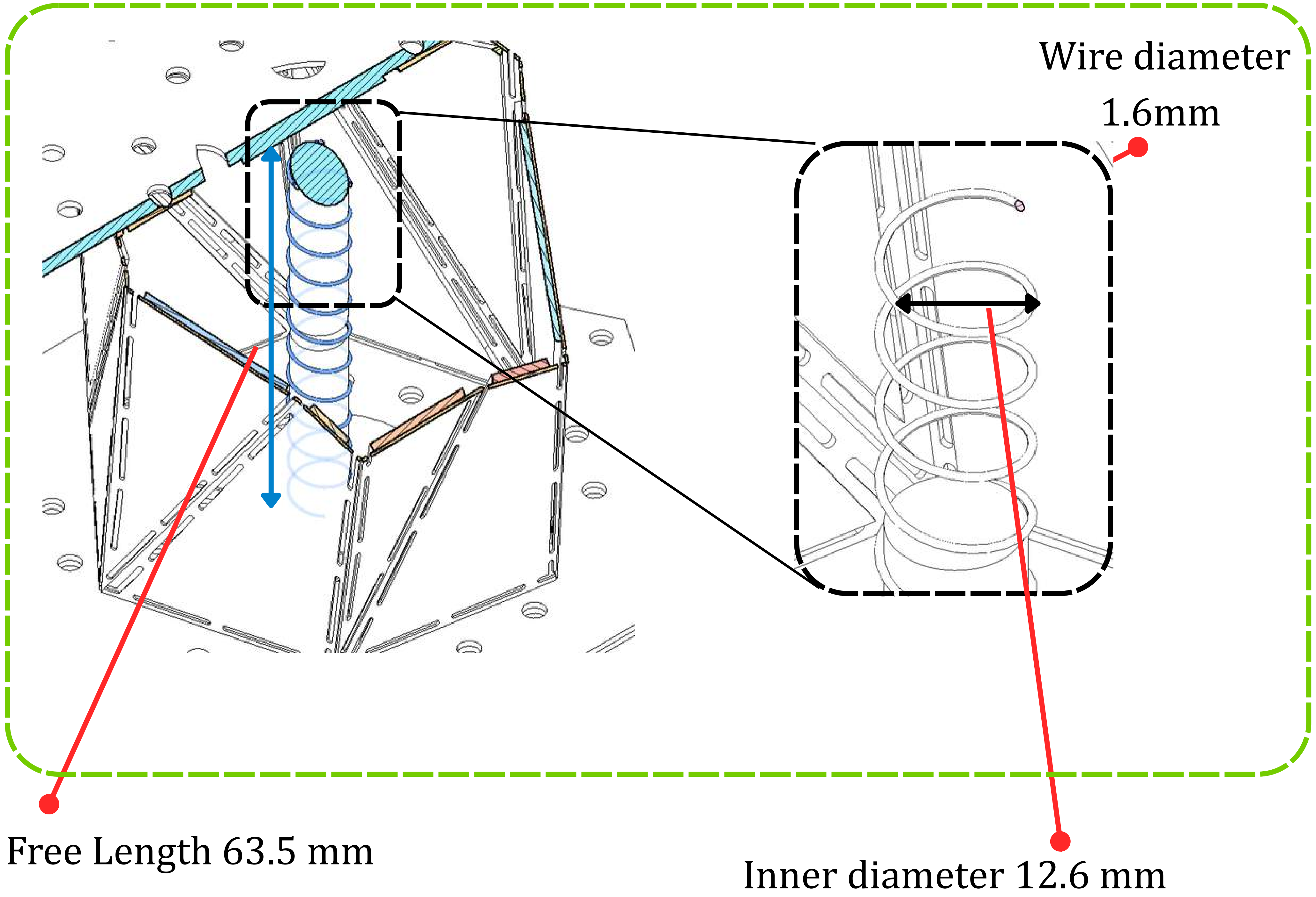
Tendon channels



Shafts to attach the spring to the shared plates and prevent buckling

Design constraints of the compression springs





Spring index → **$4 \leq C \leq 12$** → **8.875**

Pre compressed springs in each module **supplies the necessary restoration force**

Mechanical and Kinematic Analysis

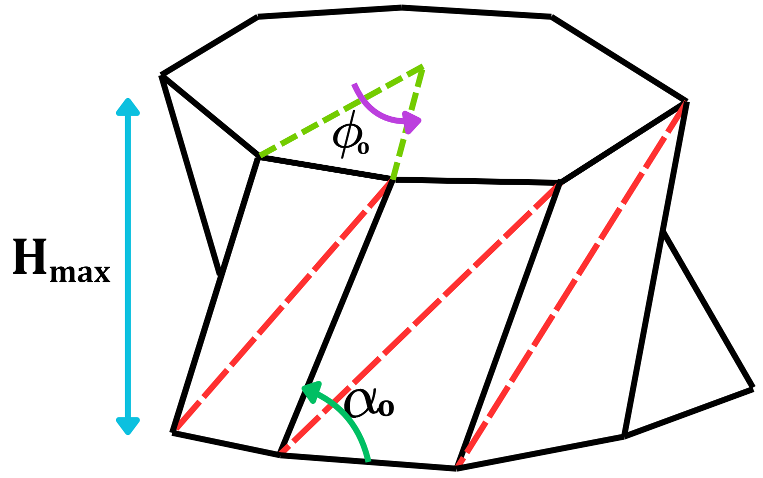
For contraction, the actuator has to overcome the restoration force of the Kresling chassis, compression spring, and frictional resistance simultaneously

$$F_{total} = F_{origami} + F_{spring} + F_{friction}$$

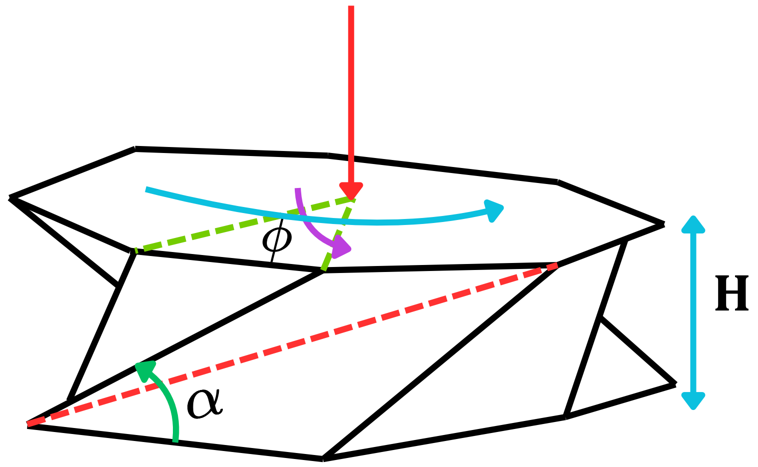
$$F_{origami} = n \cdot k_i \cdot (\alpha - \alpha_o) / (L \cdot \cos \alpha)$$

$$F_{spring} = k_{spring} \cdot (\delta_{pre} + \delta_{act})$$

$$F_{friction} = \mu \cdot F_N$$



Open State



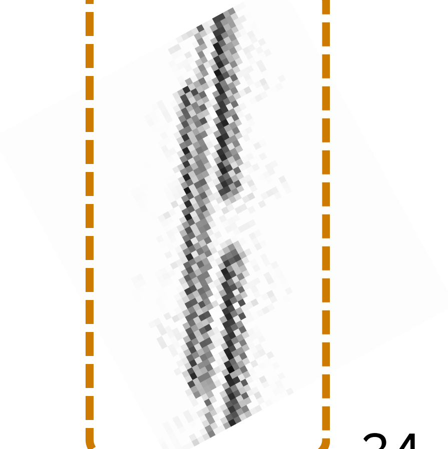
Contracted State

Bending stiffness

$$K_i = E \times I_{reduced}$$

Reduced second moment of area due to elliptical slits in the hinges

$$I_{reduced} = (b \times h^3) / 12 - \Sigma[(\pi \times a_j \times b_j^3) / 64]$$



Mechanical and Kinematic Analysis : Motor Capability & Operating Margin

$$F_{spring} = k_{spring} \cdot (\delta_{pre} + \delta_{act})$$

preload compression

$$\delta_{pre} = L_o - L_{pre\text{-}load}$$

compression due to axial force

$$\delta_{act} = L_{pre\text{-}load} - L_{current}$$

Springs are arranged in series

$$k_{equivalent} = k_{spring} / N_{module}$$

Torque produced by the system

$$T_{actuator} = (F_{total} \times r_p)$$

$\rightarrow$ 1.954E−03 Nm

F_{total}: 0.352 N

Stall to operating ratios

using a 12 V, 60 RPM 16GA-050 DC gear motor with I_{stall} 1.1A and T_{stall} 0.6276 Nm

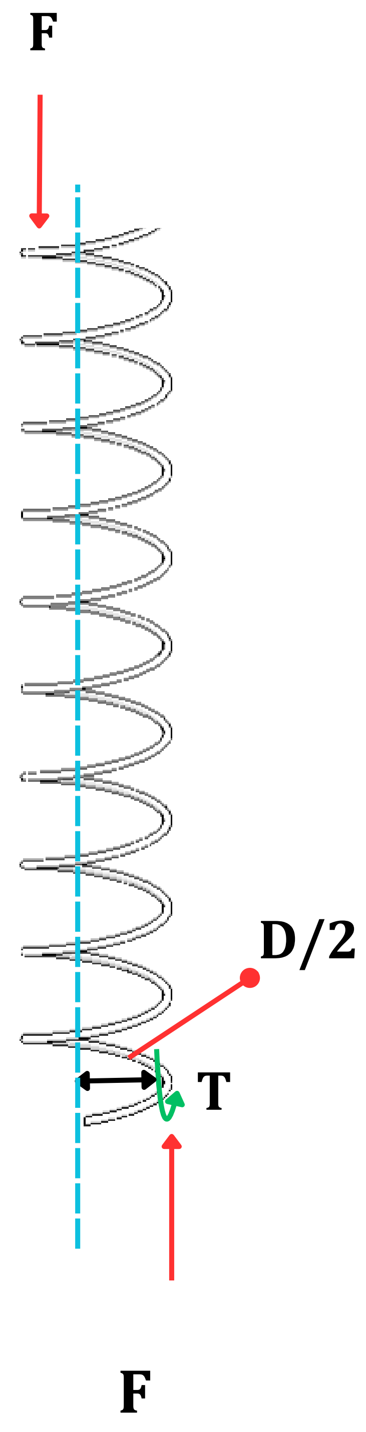
$$SF_{current} = I_{stall} / I_{max}$$

Stall current ~**5.39**× the maximum current experimentally observed during locomotion

$$SF_{Torque} = T_{stall} / T_{actuator}$$

321× stall-to-required-torque ratio

Axial Force



Dynamic Traversal Load and Motor Safety Factor Analysis

Load Cell Empirical Peak 0.352 N

Required pulley torque for compression spring

Treq ≈ 1.95 mN m

Imax = 0.204 A on an acrylic sheet

16GA-050 DC Gear Motor

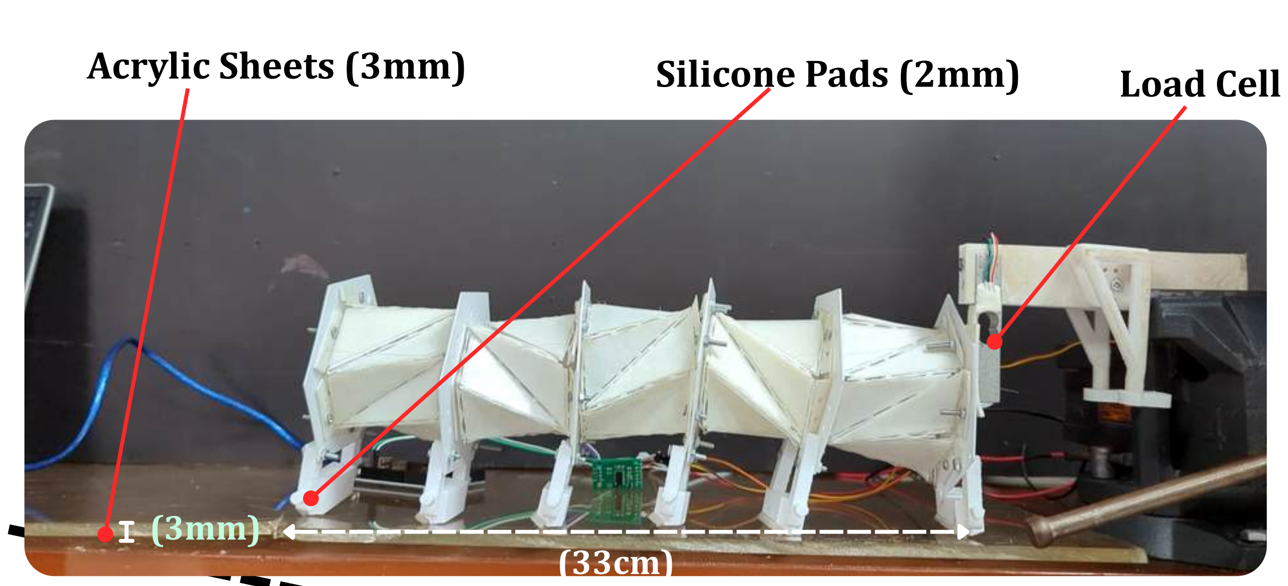
Stall Torque 0.6276 N·m

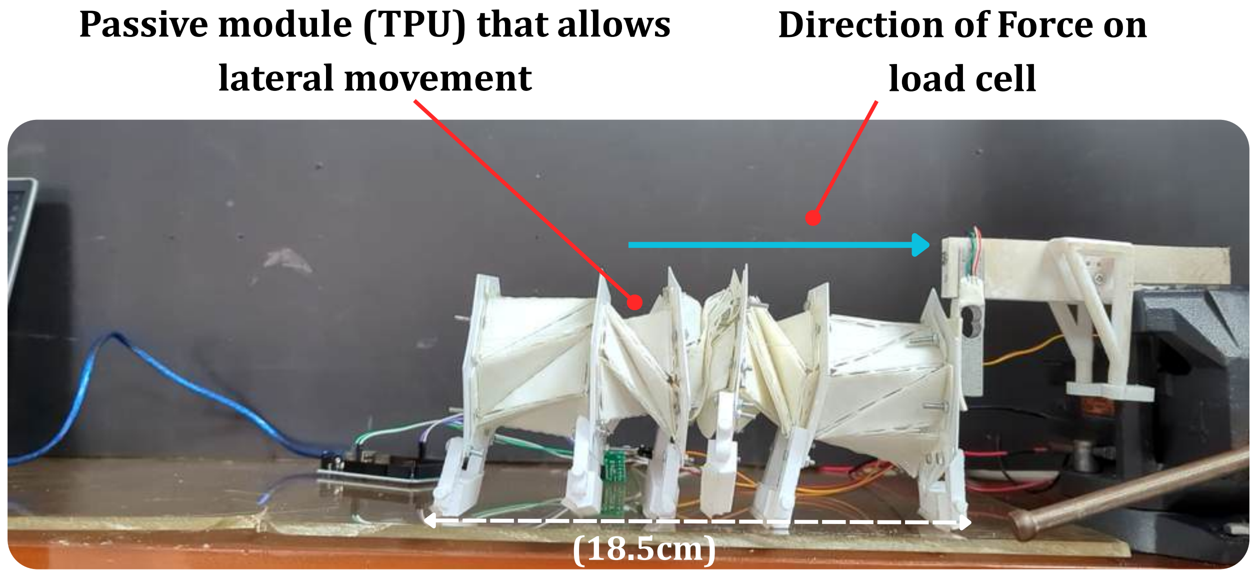
Stall Current 1.10 A

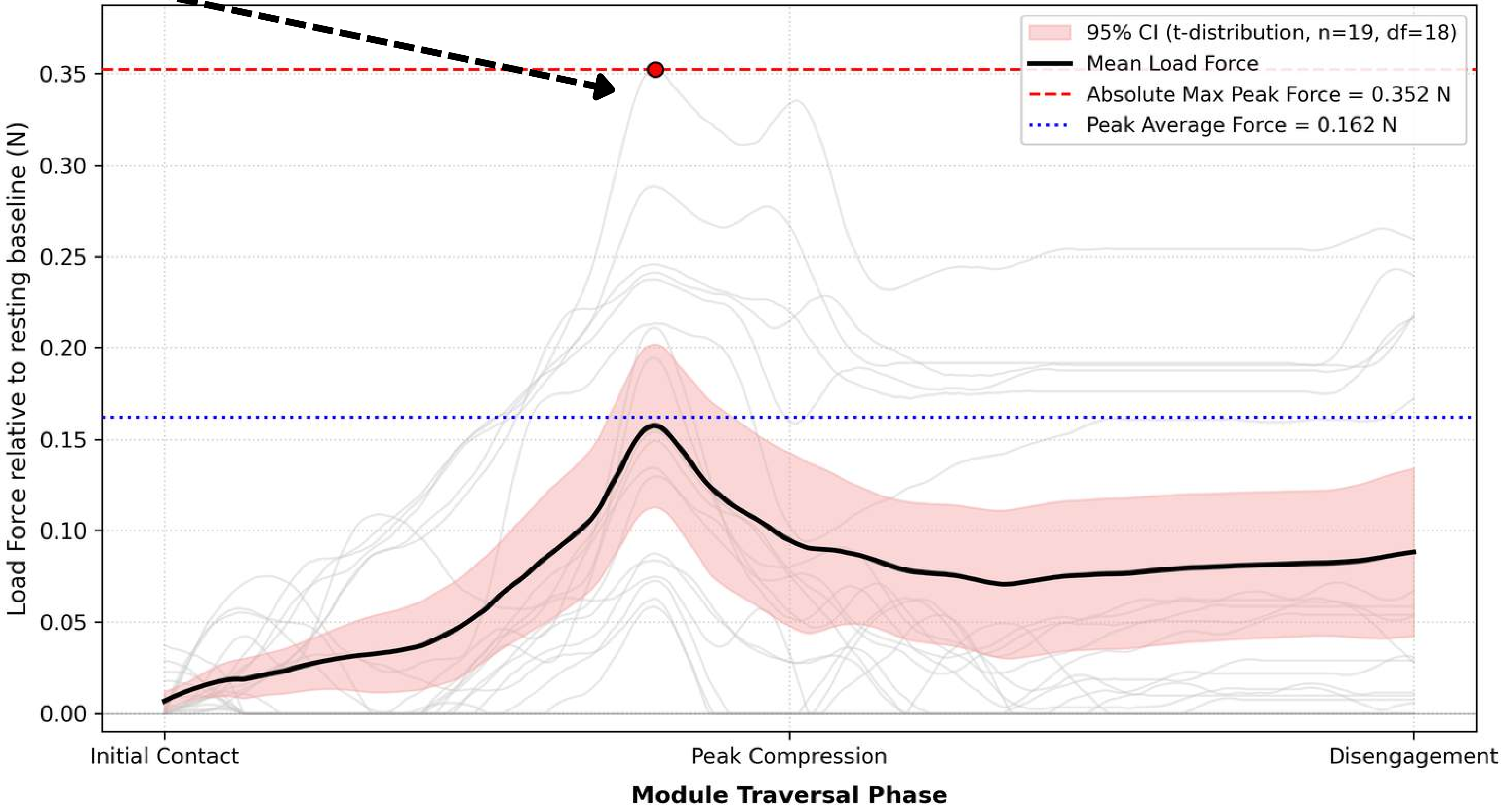
Safety Factor Analysis

Torque Safety Margin (SF_T) 321.3×

Current Safety Margin (SF_I) 5.39×

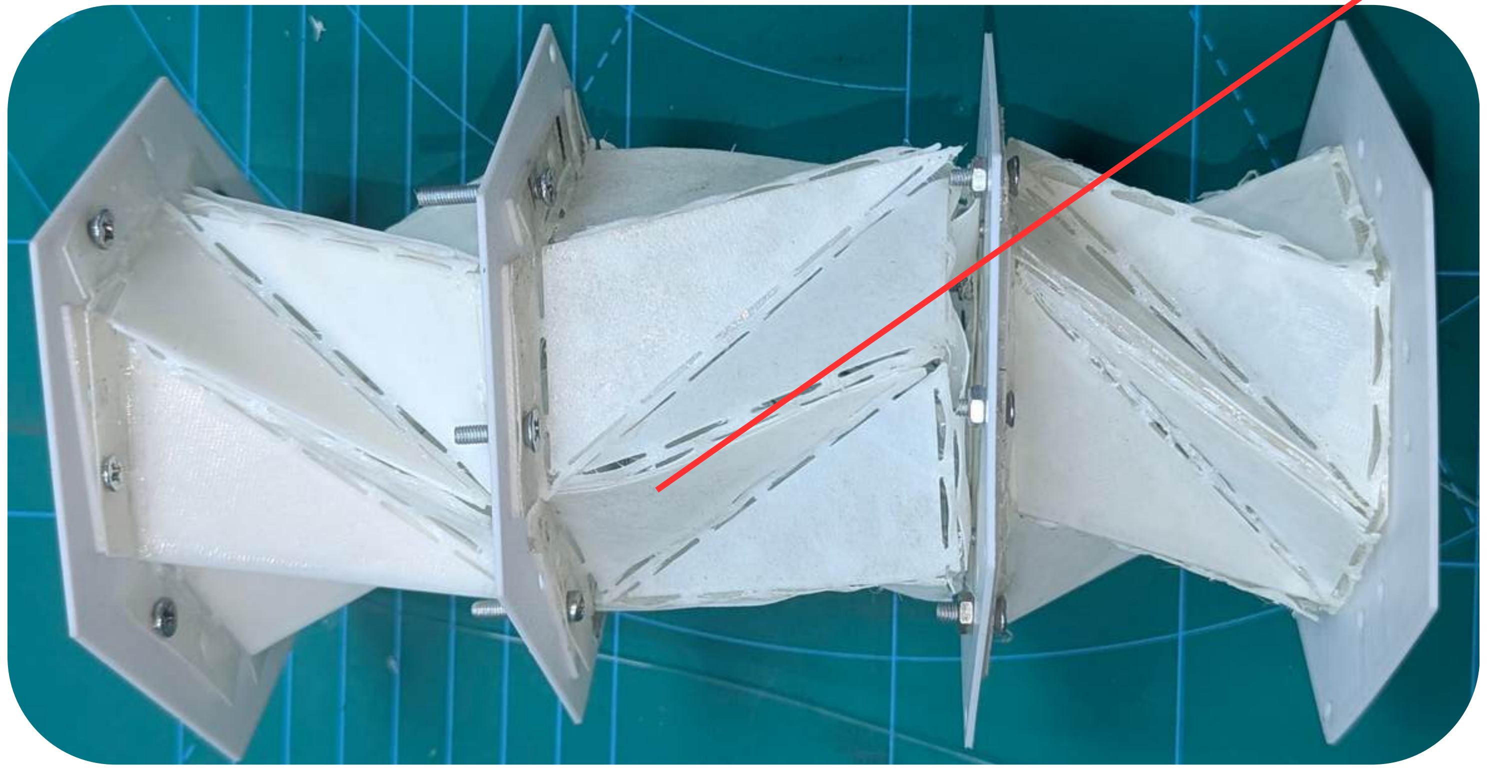






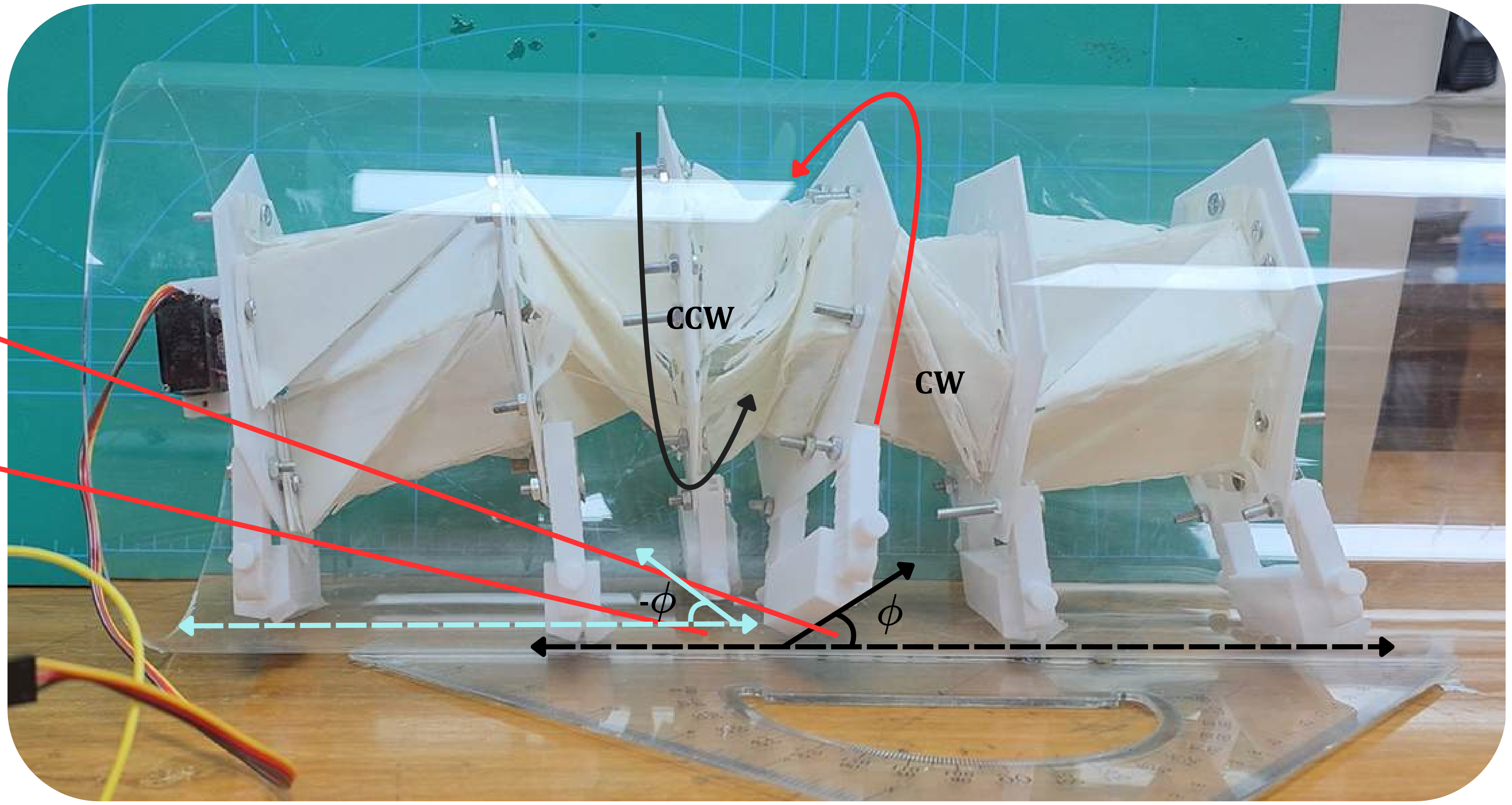
Integration of compliant passive module for yaw and pitch

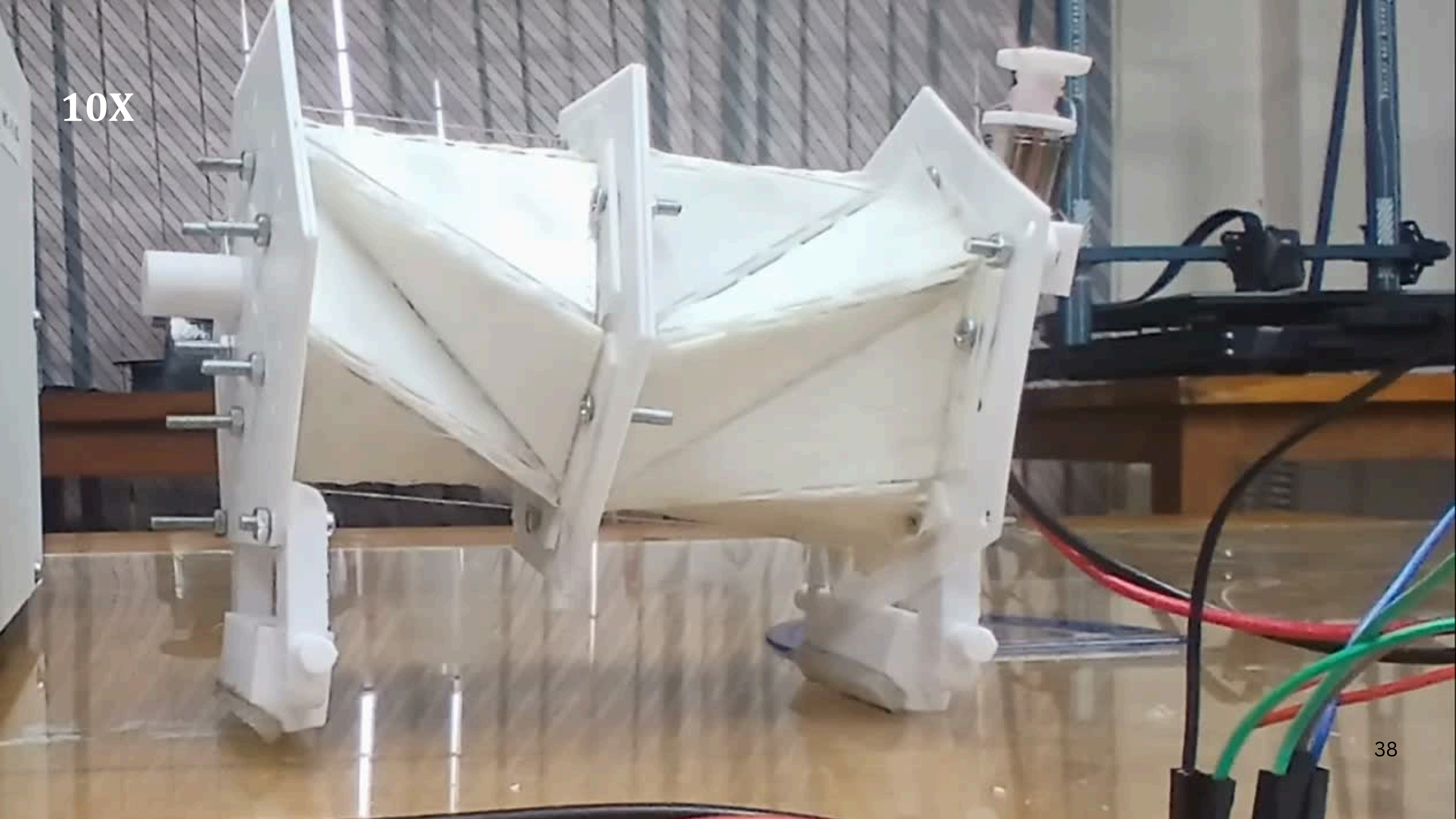
Passive module between two **active modules**

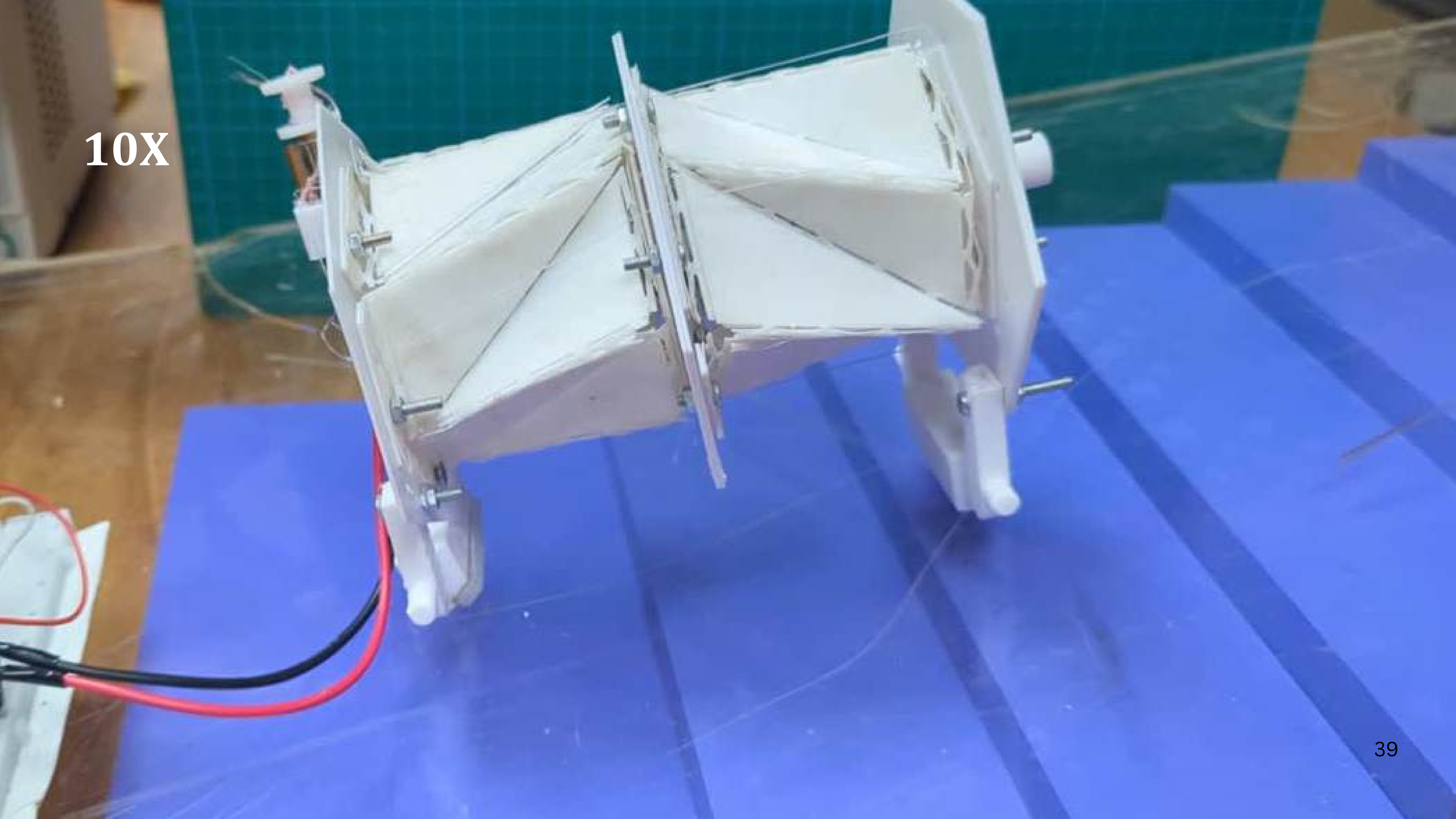


Allows Yaw and pitch

Peristalsis twisting led to the
misalignment of the foot

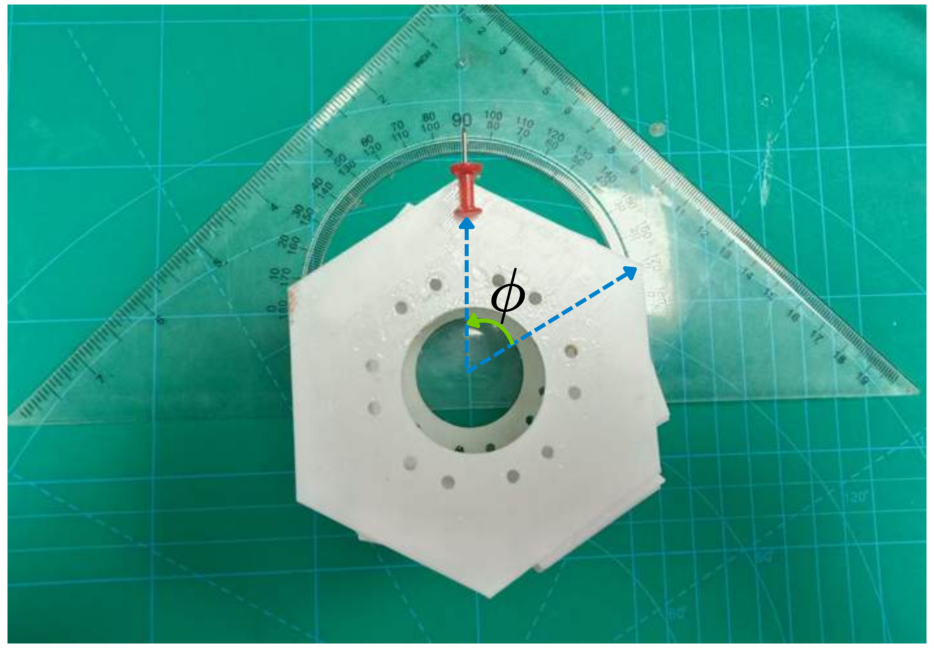



10X


10X

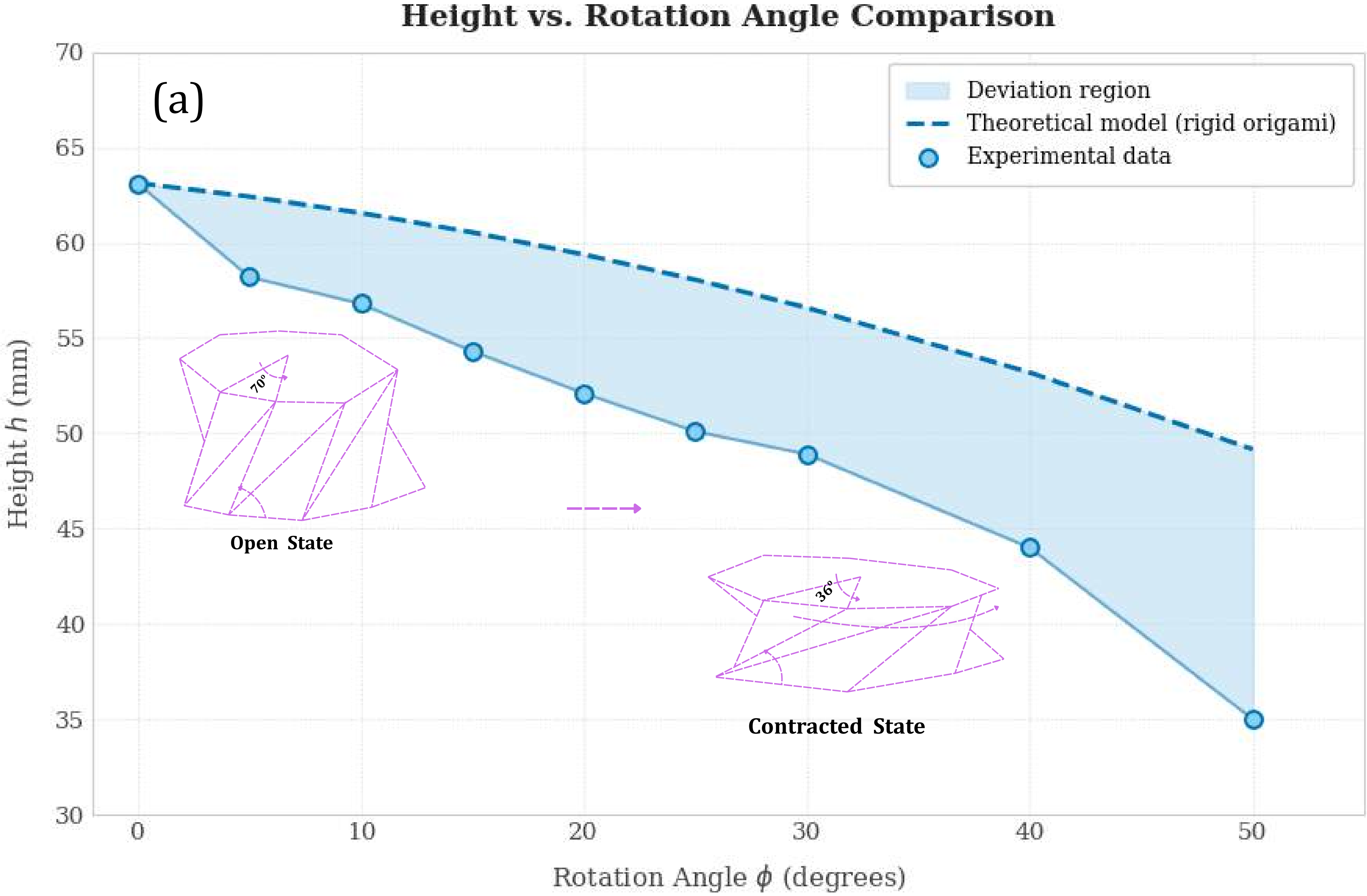
Kinematic Model Validation of the Kresling Module

$$h(\phi) = \sqrt{a^2 - 4R^2 \sin^2\left(\frac{\gamma_0 + \phi}{2} - \frac{\pi}{12}\right)}$$

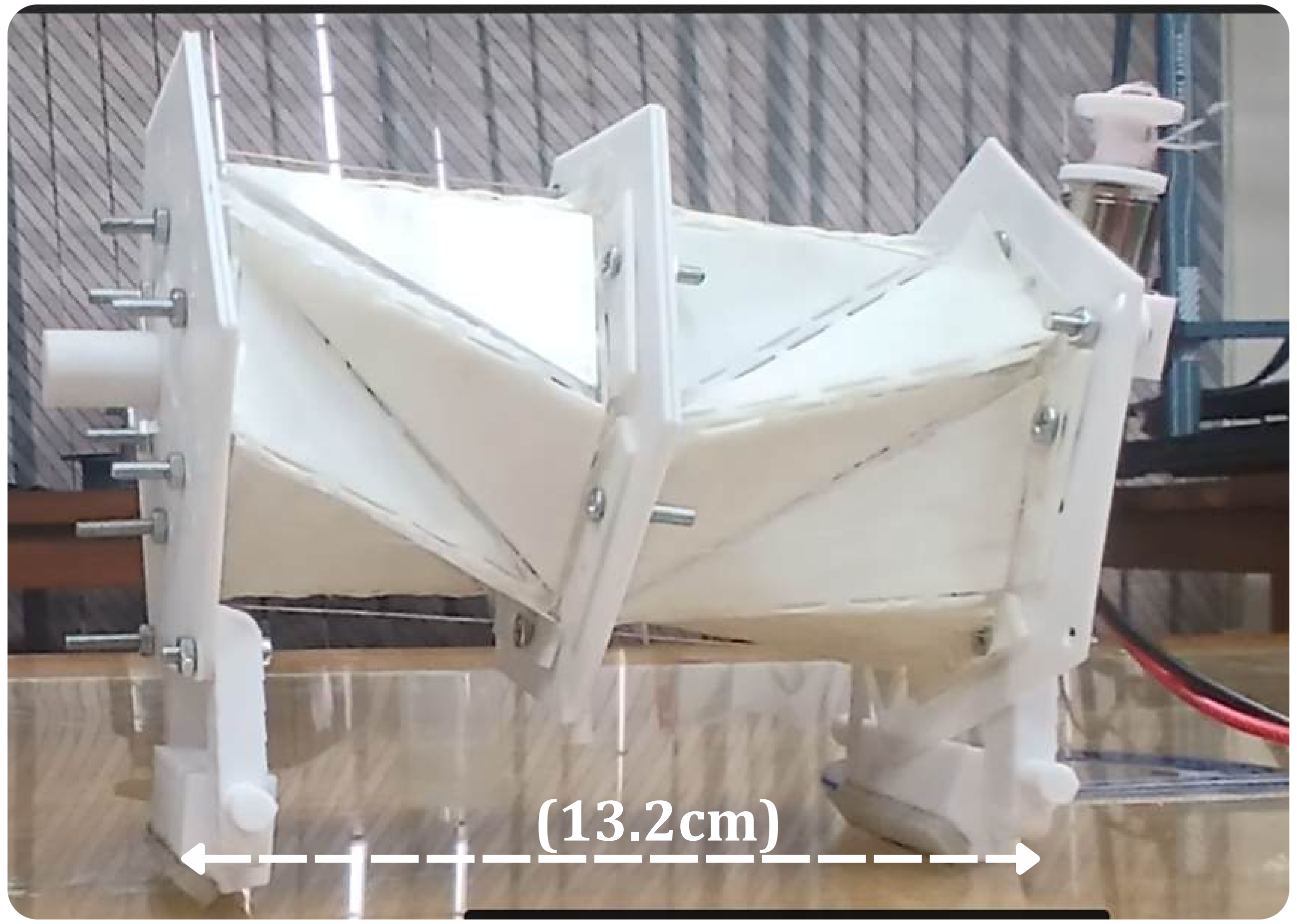


SIngle Unit

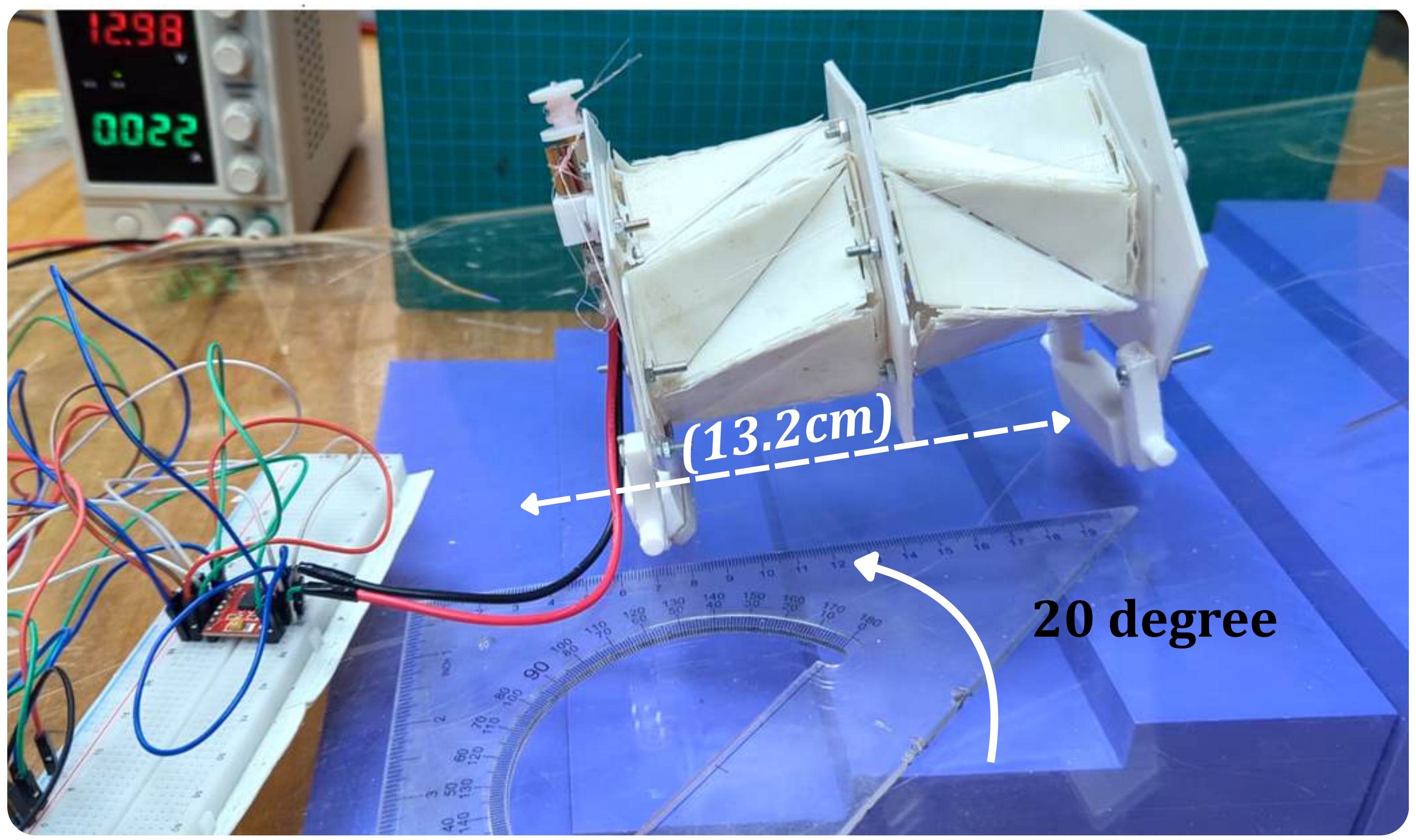
The **derived height–rotation angle relationship** closely follows the **experimental trend**, with **increasing Error(deviation) at higher rotation angles.**



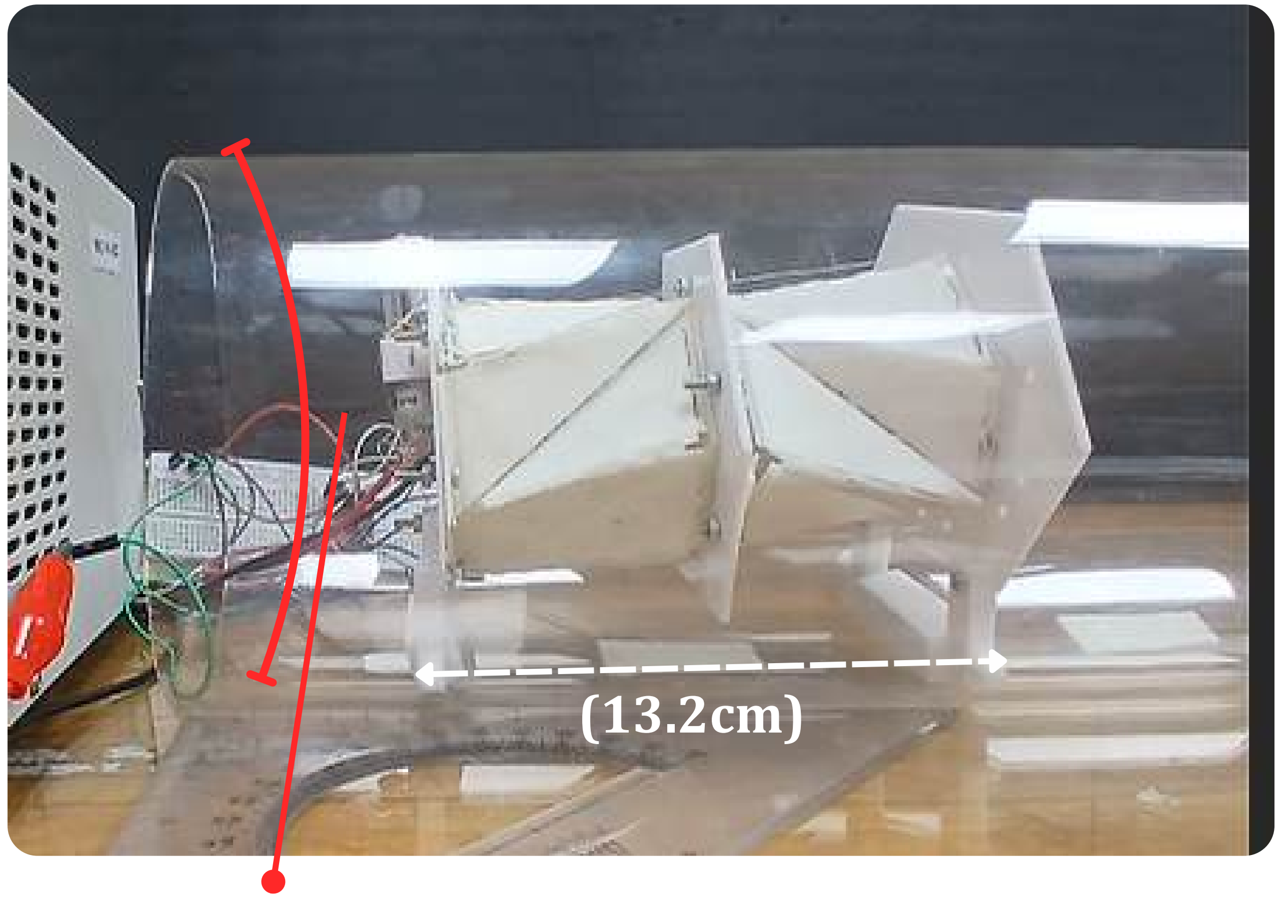
Experimental Setup



Horizontal Acrylic Sheet



Inclined Acrylic Sheet(20 degree)



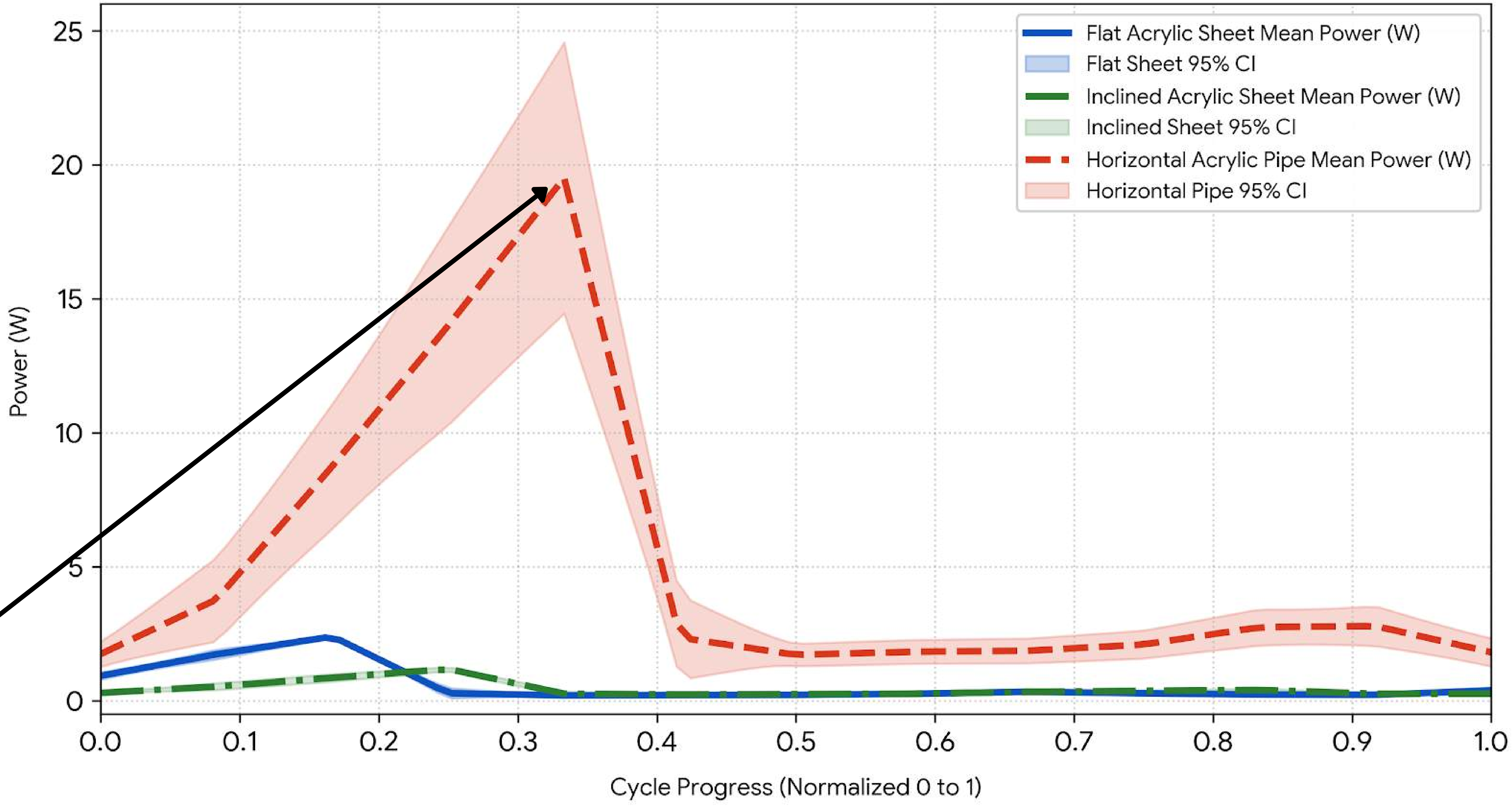
Horizontal Acrylic Pipe(5.6 inch)

Evaluates locomotion feasibility across flat, inclined, and constrained internal pipe geometries.

Gait Cycle Power Comparison Across multiple environment(Flat acrylic sheet, inclined acrylic sheet(20 degree), Horizontal Acrylic Pipe)

Successful Multi-Terrain Locomotion: Achieves reliable crawling across all 3 media (Flat, 20° Incline, and 5.6-inch ID Pipe) with low planar power < 2.5W.

High In-Pipe Power Spike (approx 19.5W): **Curved pipe profile limits** optimal **foot contact** and elevates radial normal force, driving power demand up by **approx 8.5X** during **peak contraction** of cycle.



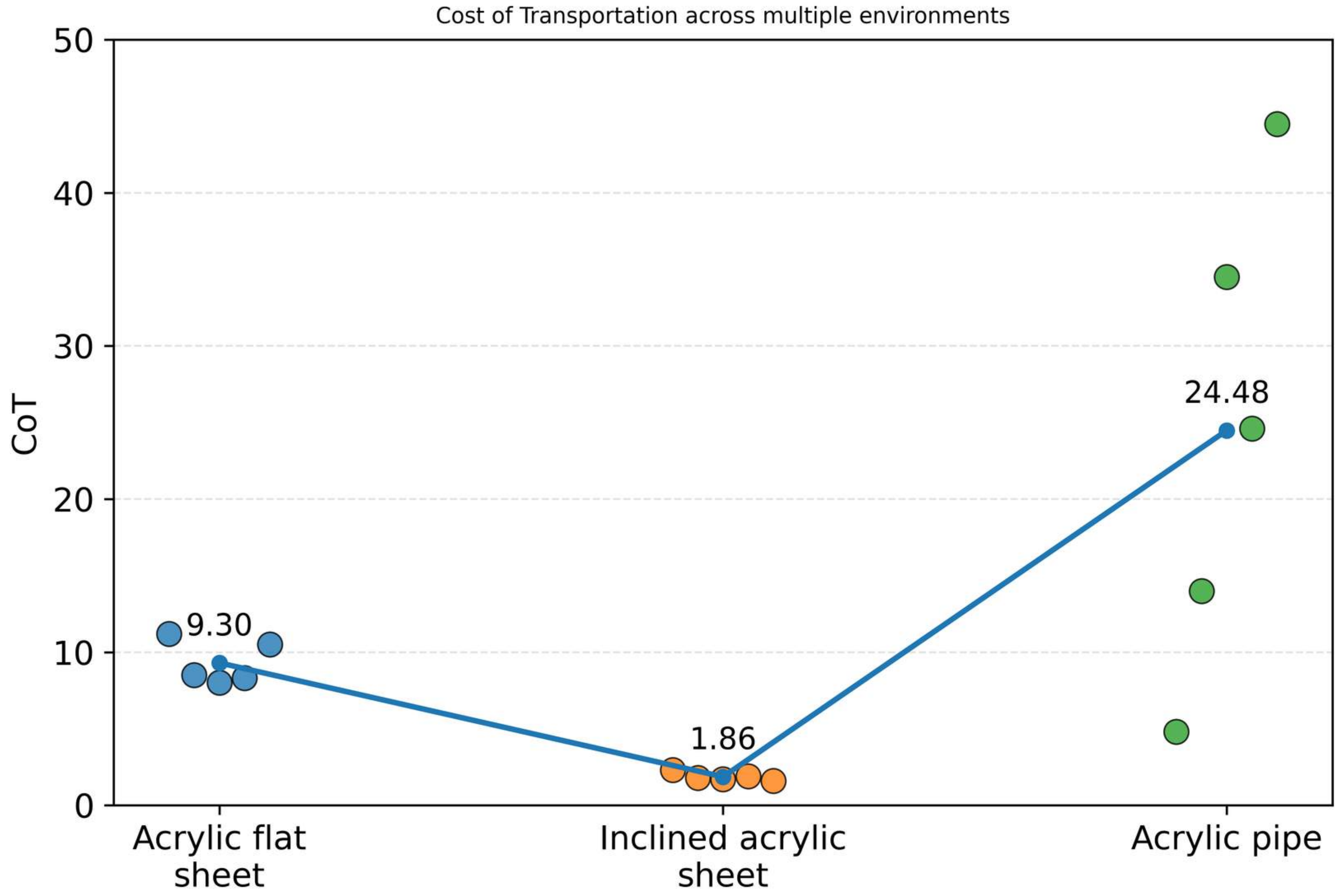
Result & Discussion

Cost of Transport (CoT)

$$CoT = \frac{P}{m \cdot g \cdot v}$$

CoT Trend: Inclined (1.86) < Flat (9.30) < Pipe (24.48)

Steeper inclines yield **superior energetic efficiency,**
Pipe geometry imposes a severe energetic penalty.



Environment	Time (s)	Distance (cm)	Velocity (mm/s)
Flat Acrylic Sheet	217	6.2	0.27
Inclined Acrylic Sheet	59	5	0.85
Horizontal Pipe (5.6")	210	4.5	0.21

Low velocity (0.21 mm/s) increases travel time, amplifying continuous power draw and radial wall friction.